Bridging a Correlated-Noise Likelihood and Evidence-Based Inference onto the Next-Generation GIGA-Lens Scene API: a Certification, Benchmark, and Mechanism Campaign*

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Technical report: campaign ***`claude-giga-lens-linus`****, experimental program complete 2026-07-21 (53.51 of 100 A100 h); companion to Benson and Huang (2026)*

Abstract

Our companion campaign (Benson and Huang, 2026) closed with a two-sided verdict on the real *HST*/WFC3-IR lens DESI–165.4754−06.0423: a drizzle-anchored correlated-noise likelihood is *necessary but not sufficient* — it flips the binned-product bimodality (a ∼191-nat per-basin evidence swing, exposing the steep basin as a noise-covariance artifact) yet over-corrects the slope to $\gamma = 1.103 \pm 0.008$, $\sim 17\sigma$ below the 1.433 native-diagonal anchor — and with the finding that real ill-conditioned and bimodal lens posteriors need a two-stage preconditioned recipe plus per-basin SMC evidence. Meanwhile the GIGA-Lens team's next-generation rewrite ("the upstream development branches") rebuilt the forward model around a unified multi-plane, multi-band *scene API* whose validation suite was, at the commit we vendored, not yet run, and which contains neither a correlated-noise likelihood nor an evidence layer. This report bridges the two lineages under a pre-registered, budget-fenced, proposer≠grader protocol (53.51 of a 100 A100 h cap). The outcome is deliberately two-sided. *Delivered:* (i) the first external certification of the scene API forward model for the EPL+shear + Sérsic/shapelet configuration class — machine-precision cross-stack parity (forward image 5.7×10^{-15}; gradients 1.5×10^{-11}) on both machines, with three documented convention reconciliations, including a real $\sim 2 \times 10^{-3}$ Sérsic-b_n approximant difference, and an fp128-referenced restatement of the Occam log-determinant gate after the original cross-algorithm reference proved noisier than the quantity it gated; (ii) a ported `CorrelatedImageData` likelihood term, exact in its analytic limits, that *reproduces the money number cross-stack*: $\gamma = 1.1005$ (offset 0.0027) with $\log Z$ agreement across stacks and machines to 0.11 nats in the low basin and 0.37 nats in the steep — $\gamma = 1.103$ now stands on two independent implementations; (iii) a sampler decision matrix from the first microcanonical-SMC benchmark on lens posteriors: prior-seeded MAMS-SMC is the only vehicle in the campaign that produced evidence at all (analytic targets, a double-source-plane ridge at λ=1, both basins of a 74-dim bimodal target with retention 1.000, and the correlated real-data refit), and where

*All work reported here — code, experiments, analysis, and text — was performed by Claude Code (Anthropic), guided by Greg Benson. This document is a technical report. Results obtained on the scene-API substrate are labeled *proposed (UNCERTIFIED — external)* throughout: we are the producer; the grader is the upstream GIGA-Lens development team (§5). Co-authorship on any publication whose results run through that substrate is offered to the team; their sign-off gates all such material.

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it fails it fails *by cost, never by health* — every non-completion is a wall-cap on a healthy sampler; on the 33-dim multi-plane real-data cell (MUSE cutouts provided by the team, used with permission) *neither* cold-start SMC nor warm MAMS converges at matched budget, while the classic single-stage MAP→SVI→HMC recipe produced no accepted posterior anywhere in the campaign, reproducing the known single-stage pathology on the new substrate. Unadjusted-MCLMC mutations are demoted for evidence work: a 20.1σ bias screen and a measured $\times 56$ minor-mode evidence inflation that SMC resampling does not launder at point estimate. Two cross-cutting negatives bind future evidence claims: bootstrap $\log Z$ errors understate true errors (measured $+3.06 \pm 0.35$ nats where the truth is ≈ 0), and a frozen MCMC-derived reference was itself indicted by an 11σ gate failure whose decomposition shows both basins with *more* evidence and γ-disjoint support — frozen references need coverage certificates. *Bounded, not decided:* the anchor arbitration (does 1.433 reproduce end-to-end on the new substrate?) closed PARTIAL-BY-VEHICLE-EXHAUSTION with no quotable γ. *Rejected:* stationarity of the noise-model class behind $\gamma = 1.103$ (calibrated $p = 0.010$; cross-block kernel spread $16.4\times$ the drizzle-registration envelope) — the standing working hypothesis for the over-correction, unconfirmed but unrefuted after the injection-recovery test was confounded by scene-subtraction residue. Every quantitative claim traces to a JSON artifact, a job id, and a gate row in `CAMPAIGN.md`.

1 Summary for Lens Modelers

This section collects, up front, the five things a lens modeler checks first. Every number is quoted from its on-disk artifact (the campaign spine, `research/synthesis_tables.md`, binds); results on the scene API substrate carry the report's *proposed (UNCERTIFIED — external)* label. The system throughout is the real *HST*/WFC3-IR lens DESI–165.4754−06.0423; the three competing mass models are the native-diagonal **anchor** ($\gamma = 1.433 \pm 0.034$), the correlated-likelihood binned-low **money number** ($\gamma = 1.1032 \pm 0.0086$), and the same-product **diagonal-low** solution ($\gamma = 1.293 \pm 0.012$), the first and third inherited from the companion campaign (Benson and Huang, 2026). A post-campaign epilogue (E1, 2026-07-22) adds a fourth, PI-requested fit: **MCLMC** on the `v2d` native diagonal target, $\gamma = 1.4683\,[1.4343, 1.5048]$ — the first fully convergence-certified posterior on that target, consistent with the anchor at $0.72\sigma_{\mathrm{comb}}$ (item 4).[1]

(1) Data, model, residual — and the caustic structure. Figure 11 is the key diagnostic: data, model, and residual/σ for all three posterior-median models rendered on the *same* `v3b` binned pixels. The one-look read: the correlated optimum ($\gamma = 1.103$) leaves a strong coherent lens-center residual that its own whitened metric prices as "probably correlated noise," while in real space the binned data prefer the $\gamma \approx 1.29$ diagonal solution (§7.4.3) — our diagnosis is that 1.103 is a whitened-metric optimum standing on a stationarity assumption the field measurably violates (§7.4.1). Figure 1 completes the geometric view with the lens-plane critical curves and source-plane caustics of the same three models plus the E1 `v2d`-native MCLMC fit (money-model $\theta_{\mathrm{E}} = 2.624 \pm 0.005$; MCLMC $\theta_{\mathrm{E}} = 2.654$), making the mass-model disagreement visible as caustic structure rather than only as a slope number.

(2) Reduced χ^2 of the best-fit models. Table 1 gives both currencies. Under its own production metric the money fit is formally excellent: whitened $\chi^2/\mathrm{dof} = 8946.8/9273 = 0.965$ (0.973 after subtracting the 46 nonlinear + 28 marginalized-amplitude parameters).[2] Two honest qualifiers frame

[1] Source artifact: `data/e1_harvest.json`.

[2] Source artifact: `data/summary_rchi2_whitened_corrlow.json` — recomputed for this section at the identical posterior-median model point on the validated stack; it bit-matches the head-to-head artifact's γ and whitened $\log L$ before being quoted.

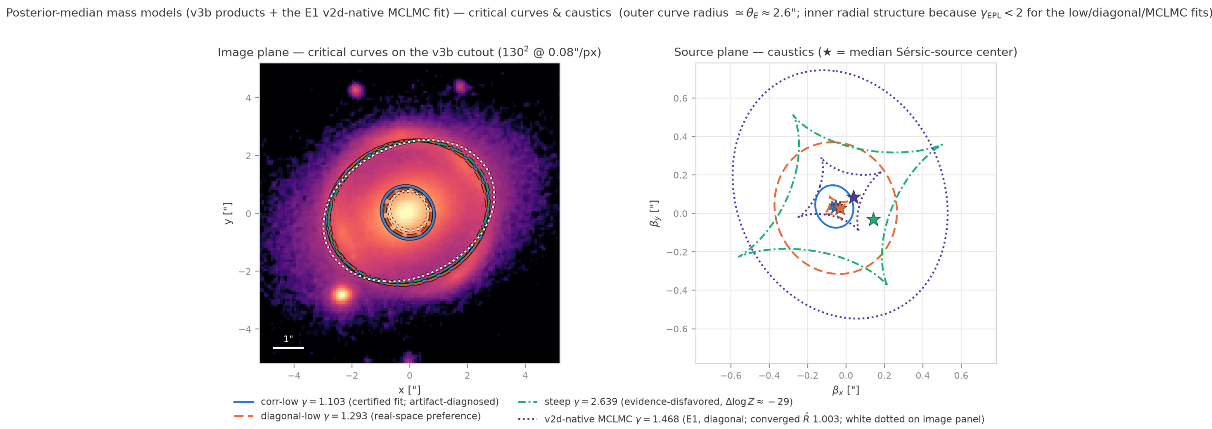

Figure 1: Critical curves (lens plane, over the `v3b` binned cutout, asinh stretch) and caustics (source plane; stars mark the median Sérsic-source centers) for the four posterior-median mass models on disk, following the critical-curve/caustic display conventions of the team's GIGA-Lens discovery paper (Gu et al., 2022): corr-low $\gamma = 1.103$ (the certified but artifact-diagnosed correlated fit), diagonal-low $\gamma = 1.293$ (the real-space preference), steep $\gamma = 2.639$ (evidence-disfavored, $\Delta \log Z \approx -29$), and the E1 `v2d`-native MCLMC fit $\gamma = 1.468$ (violet on the source panel; neutral white dotted on the image panel, where no fourth hue clears the dark-background palette floors). Scale sanity: the outer critical curve's median radius matches each model's $\theta_{\rm E}$ to $< 5\%$ for all four (money model: $2.632''$ vs. $\theta_{\rm E} = 2.624''$; MCLMC $2.652''$ vs. $2.654''$), and the $\gamma < 2$ models show the expected inner radial critical curve and caustic. Deflections from the vendored scene API EPL(50)+shear at the certified conventions; source artifacts `data/rfig_modeler_params.npz`, `data/mclmc_diag_v2d.npz`, `data/rfig_modeler_checks.json`.

the real-space column. First, *this field's own-model floor is* ≈ 1.6*, not* 1.0: the injection-build gate showed that even a ground-truth model of an injected scene scores $\chi^2/\mathrm{px} = 1.598$ full-keep versus 0.952 on the sky subset — essentially all of the excess is bright-object scene-subtraction residue (bright-object subset 2.71, lens-center $r < 1.2''$ subset 5.46)[3] — so the diagonal-low value 1.578 *is* the attainable field level, while 7.4–8.3 for the anchor and money models is genuine real-space misfit (the anchor row also carries a cross-product resolution handicap: the same parameters render $\chi^2/\mathrm{px} = 1.251$ on their home native product). Second, the whitened 0.965 inherits the whitened metric's known caveat: its stationary kernel class is *rejected* by the field at calibrated $p = 0.010$ (§7.4.1), and the same metric prefers the money model over diagonal-low by $+63$ nats in $\log L$ ($+501$ in log-posterior), inverting the real-space ordering — a reduced χ^2 of 0.965 here certifies internal consistency, not noise-model correctness. The E1 MCLMC row is evaluated by the same machinery on its *home* native product: $\chi^2/\mathrm{px} = 1.228$ full-keep (1.484 arc band) at its posterior median — marginally better than the anchor point's home 1.251 on the same pixels (the evaluator was cross-checked against the old-stack head-to-head value to 0.01% at the anchor point).[4]

(3) Sampling method. In HMC-vs-MCLMC vocabulary: every headline fit in this report was sampled by *adaptive tempered SMC* (Del Moral et al., 2006) ($\lambda : 0 \to 1$, per-stage adaptive step targeting a weighted ESS of $0.7N$, per-stage bootstrap $\log Z$) whose mutation kernel is *MAMS — the Metropolis-adjusted variant of MCLMC* (Robnik and Seljak, 2024a,b): microcanonical (MCLMC) dynamics made exact by a Metropolis accept/reject step, tuned to the MAMS paper's target

[3] Source artifact: `data/t11_injection_build_report.json`; `data/t11b_residue_mask_report.json`; `data/t04_realspace_headtohead.json`.

[4] Source artifact: `data/e1_mclmc_home_chi2.json`.

Table 1: Goodness of fit for the three campaign best-fit models on the same `v3b` binned pixels: real-space χ^2/px (diagonal-solved amplitudes; 16,653 keep px full field, 7,825 px arc band $r \in [1.2, 4.2]''$) and, for the production correlated fit, the whitened reduced χ^2 over the 9,273 whitened dof; plus the E1 MCLMC fit at its posterior median on its home `v2d` native product (marked [a]; same evaluator convention). Whitened $\log L$ for the three `v3b` models is in Table 8. Sources: `data/t04_realspace_headtohead.json`, `data/summary_rchi2_whitened_corrlow.json`, `data/t11_injection_build_report.json`, `data/e1_mclmc_home_chi2.json`.

model (γ)	χ^2/px full	χ^2/px arc band	whitened χ^2/dof
diag-low 1.293 (home reference)	**1.578**	**1.929**	—
anchor 1.433 (from `v2d`; home 1.251)	7.436	3.622	—
corr-low 1.103 (money)	8.321	2.756	**0.965**
MCLMC 1.468 (E1; home `v2d` native px[a])	1.228[a]	1.484[a]	—

[a]E1 row only: evaluated on its home `v2d` *native* product (5,865 keep px; 2,937 arc-band px at 0.13″), not the shared `v3b` binned pixels of the other rows — the direct analogue of the anchor's home 1.251; `data/e1_mclmc_home_chi2.json`.

field's own-model floor (truth model of injected scene): full-keep 1.598; sky subset 0.952 —
the excess over 1.0 is bright-object scene-subtraction residue, faced by every fit on this product.

acceptance of 0.90. It is therefore neither plain HMC nor raw (unadjusted) MCLMC. The distinction is load-bearing: unadjusted-MCLMC mutations were demoted at the correctness gate (20.1σ evidence bias on the ill-conditioned analytic target; a measured $\times 56$ minor-mode evidence inflation on the 74-d bimodal target), and everywhere the classic single-stage MAP$\to$SVI$\to$HMC recipe ran in this campaign it failed its own acceptance rule (Table 2; full decision matrix Table 9, §9.5, §9.6) — consistent with the companion finding that these real lens posteriors need the two-stage re-preconditioned recipe (Benson and Huang, 2026). The E1 epilogue adds **MCLMC proper** — the PI-requested vehicle; the team's own vendored implementation (`full_mclmc_with_adapt_sharded`: windowed mass-matrix adaptation, run with their Stan-style mass-matrix regularization option enabled) — used as the direct posterior sampler on both diagonal targets from warm starts in the physical basin. On the native product it delivers the first fully convergence-certified posterior of the program (item 4); tempered SMC remains the evidence/model-comparison layer — the two are complementary, not competing.[5]

(4) Convergence metrics. The E1 MCLMC fits supply the campaign's first $\hat{R}$/ESS-certified direct-sampler rows, and the `v2d` native fit leads Table 2: $\hat{R}_{\mathrm{worst}} = 1.0031$, $\mathrm{ESS}_{\mathrm{min}} = 30{,}334$ against its frozen gate $\hat{R} < 1.01 \wedge \mathrm{ESS} \geq 1000$ over the pooled 64 chains (Fig. 2); its `v3b` twin fails both frozen gates and is reported as a NO-QUOTE migration finding (below). For the SMC fits, which have no $\hat{R}$, the honest analogues are independent-seed repeats, the per-stage weighted-ESS floor maintained by the adaptive λ-schedule, unique-particle retention after resampling, MAMS acceptance against its 0.90 target, and — strongest of all — the cross-stack, cross-machine reproduction. Table 2 is the consolidated convergence ledger for every fit quoted in this report, with $\hat{R}$/ESS given where MCMC arms ran (the classic-recipe arms are both rejections, reported as such). For the two inherited diagonal fits (anchor 1.433, diag-low 1.293) the per-fit $\hat{R}$/ESS live in the companion report's campaign ledger (Benson and Huang, 2026) and are cited, not re-derived here. Table 2 covers the fits *quoted* in this report; Appendix A extends it to the complete per-fit ledger — one row for every fit the campaign ever ran, including the failed, rejected, collapsed, and budget-blocked arms.

[5]Source artifact: `data/mclmc_diag_v2d.json; data/e1_harvest.json`.

Table 2: Consolidated convergence ledger. wESS = final-stage weighted ESS; "unique" = unique particles after resampling; the adaptive λ-schedule holds per-stage wESS at $0.7N$ by construction. The two E1 MCLMC rows are gated at their own frozen thresholds ($\hat{R} < 1.01 \wedge \mathrm{ESS} \geq 1000$, pooled 64 chains; containment band $\gamma \in [1.15, 2.0]$, report-never-drop). Sources: `data/mclmc_diag_{v2d,v3b}.json` + `data/e1_harvest.json`, `data/t02_t03_gate_eval.json` (+ per-seed run JSONs), `data/results-perlmutter/l0g2_v3b_scene_seed2.json`, `data/b1r_decision_matrix.json` + `research/checkpoint_b1r_close.md`, `data/results-perlmutter/l0_arbitration_classic.json`, `data/b3_run_ref_sys{0..3}_l4-8.json`, Benson and Huang (2026).

fit / arm	vehicle, n	diagnostics	verdict
`v2d` native diag, **MCLMC** (E1, PI-requested) — $\gamma = 1.4683\,[1.4343, 1.5048]$	vendored windowed-adaptation MCLMC; 2 seed groups $\times$ 32 chains $\times$ 4000 draws	$\hat{R}_{\text{worst}} = 1.0031$ vs gate < 1.01; $\mathrm{ESS}_{\min} = 30{,}334$ vs ≥ 1000; γ: $\hat{R}$ 1.0017, ESS 39,794; seed-group medians agree to 0.0002; 0 containment violations (all 256k draws in $[1.324, 1.637]$); 1.49 h on one L4	**PASS both frozen E1 gates** — γ quotable, not escalated; $+0.0353 = 0.72\sigma_{\text{comb}}$ from the 1.433 anchor (Fig. 2)
`v3b` binned diag, MCLMC (E1)	same; 4 seed groups $\times$ 16 chains	$\hat{R}_{\text{worst}} = 1.379$, $\mathrm{ESS}_{\min} = 139$ (both lens-light I_e); 46/46 scene params over the $\hat{R}$ gate; 64/64 chains migrate below the $[1.15, 2.0]$ band during burn-in (Fig. 4)	**NO-QUOTE + FINDING**: structural migration, not sampling depth — the one allowed escalation declined
`v3b` corr **low** — $\gamma = 1.1032 \pm 0.0086$ (**money**)	tempered SMC + MAMS; 3 seeds $\times$ 128	λ=1 @28 stages every seed; wESS 118–128/128; unique 77–82/128; seeds $\{1.1032, 1.0967, 1.1005\} \Rightarrow \sigma_{\text{seed}}(\gamma) = 0.00325$, $\sigma_{\text{seed}}(\log Z) = 1.22$ nats	**converged + seed-certified** (T0.2, §7.1); n=3 caveat: ±46% on σ_{seed} itself
`v3b` corr steep ($\gamma \approx 2.64$)	same; 3 seeds $\times$ 96	λ=1 @21–22 stages; wESS 36–96/96 (seed-2 final resample low); unique 50–59/96; $\sigma_{\text{seed}}(\gamma) = 0.00664$	converged; $\Delta \log Z$(steep − low) $= -28.9/-32.3/-31.2$ seed-stable
same fit, scene API port, Perlmutter A100 (L0-G2)	same; 1 seed	λ=1 @27/22 stages; wESS 123.5/128 (low), 47.6/96 (steep); unique 81/128, 54/96	**PASS**: $\Delta\gamma = 0.0027$; $\log Z$ agreement 0.11 (low) / 0.37 (steep) nats *across stacks and machines* (§7.7)
carousel 33-d multi-plane REAL (B1r; hardest cell)	same; 1 seed $\times$ 128	$\lambda = 0.151$ @36 stages; unique 92–104/128 every stage; MAMS accept 0.871–0.934 (mean 0.899 vs 0.90 target); stage wESS pinned at $0.7N$	**no posterior**: sampler healthy, killed by wall-cap — a cost verdict, not a convergence failure (§9.2)
classic MAP→SVI→HMC, warm, `v2d` (arbitration arm)	50 chains, 250 burn-in/750 draws; escalated 500/1500	$\hat{R}_{\text{worst}} = 44.3$ (escalated 11.7) vs rule < 1.05; $\mathrm{ESS}_{\min} = 53.6$ (63.0 escalated); all chains railed $\gamma \approx 1.987\,[1.983, 1.991]$ at the prior edge	**NO-VERDICT**: γ not quotable; reproduces the known T2 single-stage pathology on the scene API (§9.5)
classic reference chains, hs2 scenes ×4 (B3)	frozen 50-chain recipe	$\hat{R}_{\text{worst}} = 4.36$–$7.87$; $\mathrm{ESS}_{\min} = 63.7$–$83.9$ vs rule $\hat{R} < 1.05 \wedge \mathrm{ESS} \geq 200$	all 4 **rejected** by the preregistered acceptance rule (§9.6)
anchor 1.433 (`v2d` diag) and diag-low 1.293 (`v3b` diag)	companion campaign (two-stage re-preconditioned recipe)	provenance row: per-fit $\hat{R}$/ESS ledgered in the companion report	converged there; cited, not re-derived (Benson and Huang, 2026)

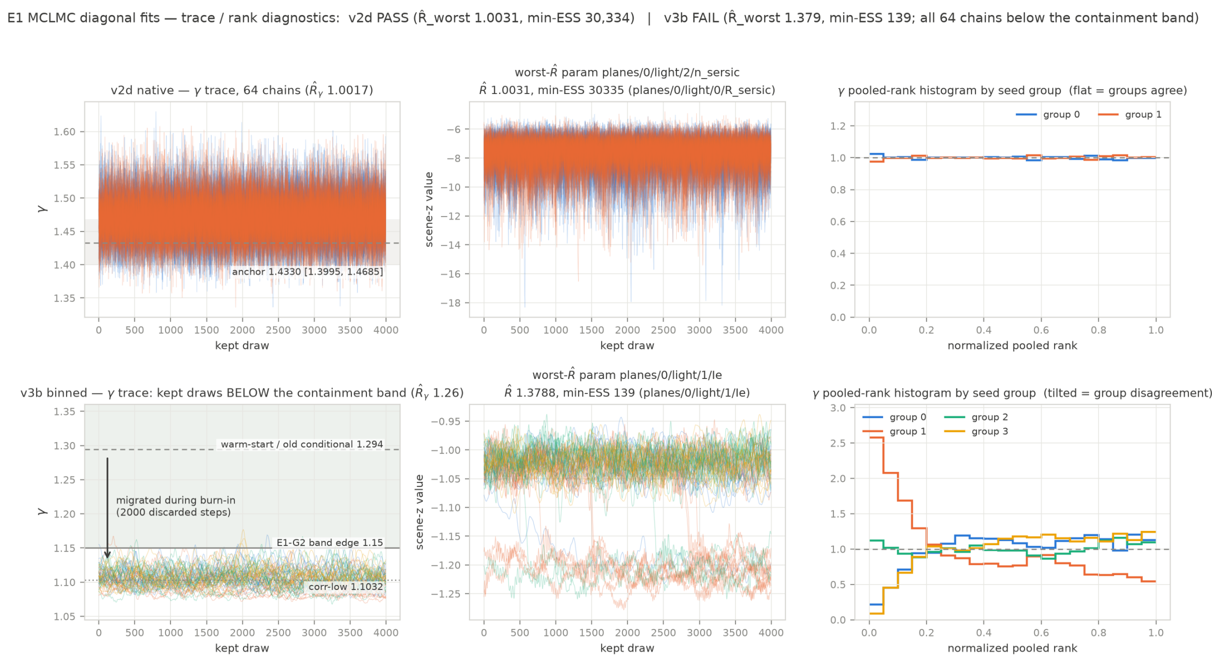

Figure 2: E1 MCLMC trace/rank diagnostics, both diagonal fits (plotted before any gate math per house rule). Top row, `v2d` native: the γ trace of all 64 chains sits on the anchor band $1.4330\,[1.3995, 1.4685]$ ($\hat{R}_\gamma$ 1.0017), the worst-$\hat{R}$ parameter (a lens-light $n_{\text{sérsic}}$, $\hat{R}$ 1.0031; the fit's min-ESS, 30,334, sits on a lens-light $R_{\text{sérsic}}$) mixes cleanly, and the per-seed-group pooled-rank histograms are flat (groups agree). Bottom row, `v3b` binned: the kept draws sit almost entirely *below* the E1-G2 containment-band edge (1.15; per-chain fraction of kept draws outside the band: min 0.94, median 1.00) after the chains migrated away from the 1.294 warm start during the 2000-step burn-in; the worst-$\hat{R}$ parameter (lens-light I_e, $\hat{R}$ 1.379, min-ESS 139) splits between shelves, and the tilted rank histograms localize the disagreement both within and between seed groups. Source artifacts: `data/mclmc_diag_{v2d,v3b}.{npz,json}`; `data/e1_harvest.json`.

(5) The posterior. Figure 3 shows the posterior corner for the production correlated `v3b` fit with the three seed repeats overlaid. The read-out: the Einstein radius is tight and stable ($\theta_{\rm E} = 2.6239 \pm 0.0049$, a 0.2% constraint) while the two basins remain cleanly separated in γ (low 1.103 vs. steep 2.64) with a seed-stable evidence preference for the low basin of $\approx$ 29–32 nats; the seed overlays coincide within $\sigma_{\rm seed}(\gamma) = 0.00325$, which is what licenses quoting $\sigma_{\rm tot} = 0.0086$ on the money number (§7.1).[6] The regenerated corner now also overlays the E1 `v2d`-native MCLMC posterior (violet; 256,000 pooled draws, scene-native keyed extraction; its equal-weight draw median reproduces the run artifact's pooled $q_{50} = 1.4683$ exactly), so the native-vs-binned displacement in γ is visible in the same axes as the seed-repeat spread.[7] The number the correlated posterior constrains, and the standing caveat on interpreting it, are the subject of the rest of this report: $\gamma = 1.1032 \pm 0.0086$ is certified, cross-stack reproduced — and diagnosed as an artifact of a stationary noise-model class the field rejects.

The E1 caveat on the binned product — reported, not smoothed. The `v3b` arm of E1 is the honest counterweight to the clean `v2d` pass, and we state it plainly: *on the binned product, even MCLMC under the plain diagonal likelihood migrates toward the unphysical low-γ region.* Warm-started inside the physical basin (the $\gamma = 1.2941$ conditional cloud — itself only a conditional slice of an $\hat{R} = 843$ bimodal chain), all 64 chains in all four independent seed groups left the containment band during the 2000-step burn-in and spent essentially all of their kept draws at $\gamma \approx 1.08$–1.15 (Fig. 4); the kept-draw pooled median, 1.1047, is descriptive only (the ensemble is unconverged, $\hat{R}_\gamma = 1.261$) but sits 0.0015 from the certified correlated-fit 1.1032. The one escalation the frozen protocol allowed was declined, with the reasoning on the ledger: the failure mode is structural migration out of the band, not sampling depth — doubling draws cannot un-migrate chains, and conditioning post hoc on the band would be a goalpost move — so the verdict is NO-QUOTE plus a finding. The finding is the sampling-side face of the campaign's binned-product pathology (§7.4.1, §7.4.3): the binned pixels themselves prefer the low-γ shelf under likelihoods that discount what the native product resolves — seen with tempered SMC under both the correlated likelihood (the money fit) and the diagonal one (the B5 low basin at $\gamma = 1.092$, §9.3), and now with the PI's own MCLMC under the plain diagonal likelihood. The finding's honest scope is what this run tested: one product, one likelihood, one warm-start cloud, at a fixed $2000 + 4000$-step budget — longer runs or other initializations could yet behave differently; what is established is the migration itself, not the shelf's posterior weight. It further undermines the old 1.2941 conditional as a diagonal reference for this product — and it is why the `v2d`-*native* MCLMC fit, $\gamma = 1.4683\,[1.4343, 1.5048]$, is the quotable E1 deliverable.[8]

2 Introduction

2.1 Two lineages

The companion campaign (Benson and Huang, 2026) attacked the two ingredients that the code-comparison literature flags as under-addressed in fast GPU lens modeling (Galan et al., 2024): correlated image noise and honest posterior multimodality. It delivered a validated drizzle-anchored correlated-noise likelihood ($C = D^{1/2} K D^{1/2}$, convolutional whitening gated at operator-norm flatness $e_{\rm op} \le 0.02$, generalized ridge marginalization with the Gaussian Occam term) and the first systematic sampler benchmark on strong-lens posteriors. On the real system DESI–165.4754−06.0423

[6]Source artifacts: companion `data/results/e2_v3b_low_smc_canary_fix.json`; `data/t02_t03_gate_eval.json`.
[7]Source artifact: `data/rfig_modeler_checks.json`.
[8]Source artifact: `data/e1_harvest.json; data/mclmc_diag_v3b.json`.

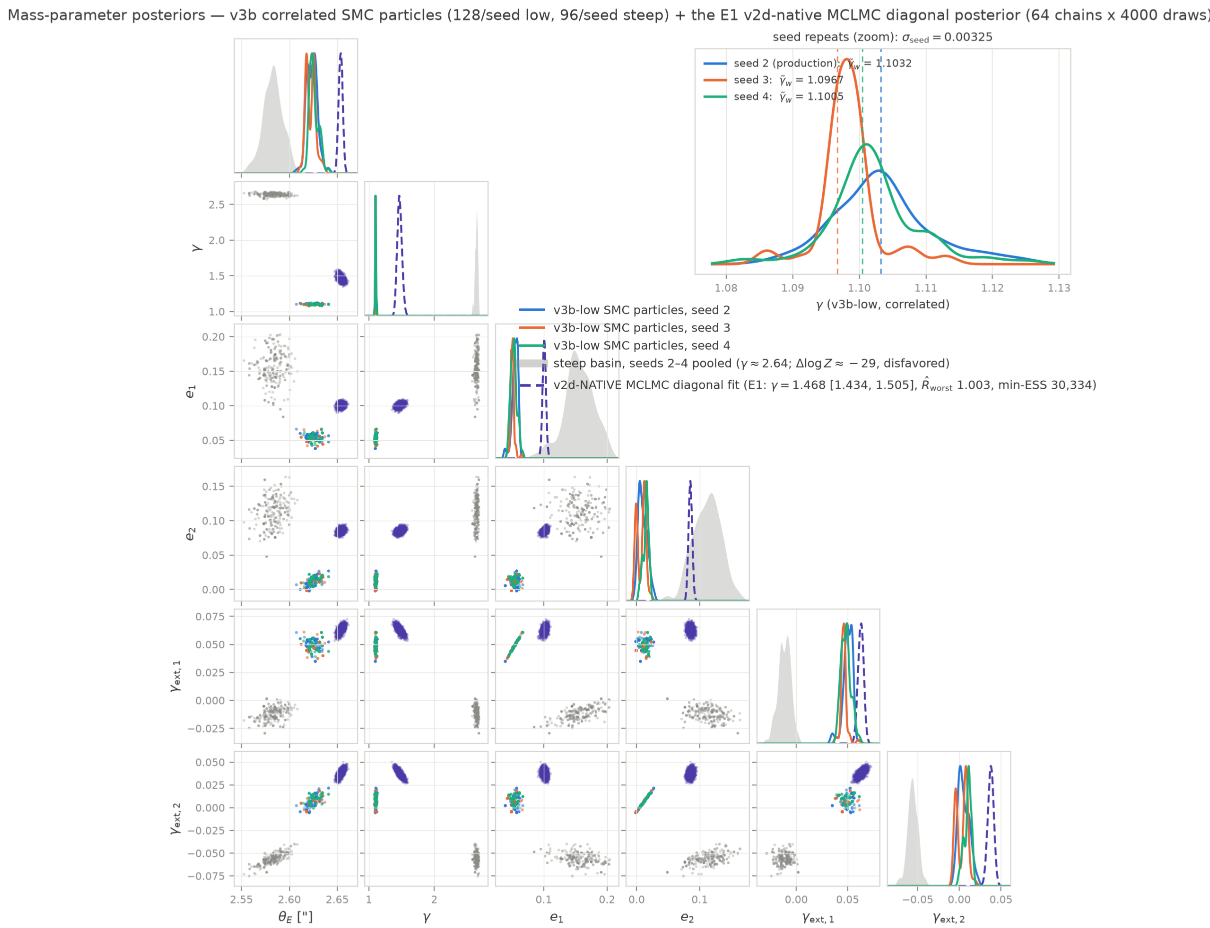

Figure 3: Posterior corner of the six mass parameters ($\theta_{\rm E}$, γ, e_1, e_2, $\gamma_{\rm ext,1}$, $\gamma_{\rm ext,2}$) from the stored equal-weight `v3b`-low correlated SMC particles, seeds $\{2, 3, 4\}$ overlaid — the SMC convergence visual (SMC has no $\hat{R}$; the seed-repeat spread $\sigma_{\rm seed} = 0.00325$ plus the per-stage ESS/acceptance diagnostics of Table 2 are the analogues). Grey: the steep basin (seeds 2–4 pooled, $\gamma \approx 2.64$; evidence-disfavored by $\approx$ 29–32 nats). Inset: γ zoom with the per-seed weighted medians of record $\{1.1032, 1.0967, 1.1005\}$; the equal-weight particle medians reproduce them to ≤ 0.0027, well inside $\sigma_{\rm stat} = 0.008$. Violet: the E1 `v2d`-native MCLMC diagonal posterior (256,000 pooled draws; equal-weight draw median asserted equal to the run json's pooled γ $q_{50} = 1.4683$; it lies outside the γ zoom inset by construction). Source artifacts: companion `data/results/e2_v3b_low_smc_canary_fix.json` (+ seed-3/4 repeats), `data/t02_t03_gate_eval.json`, `data/rfig_modeler_params.npz`, `data/mclmc_diag_v2d.npz`, `data/rfig_modeler_checks.json`.

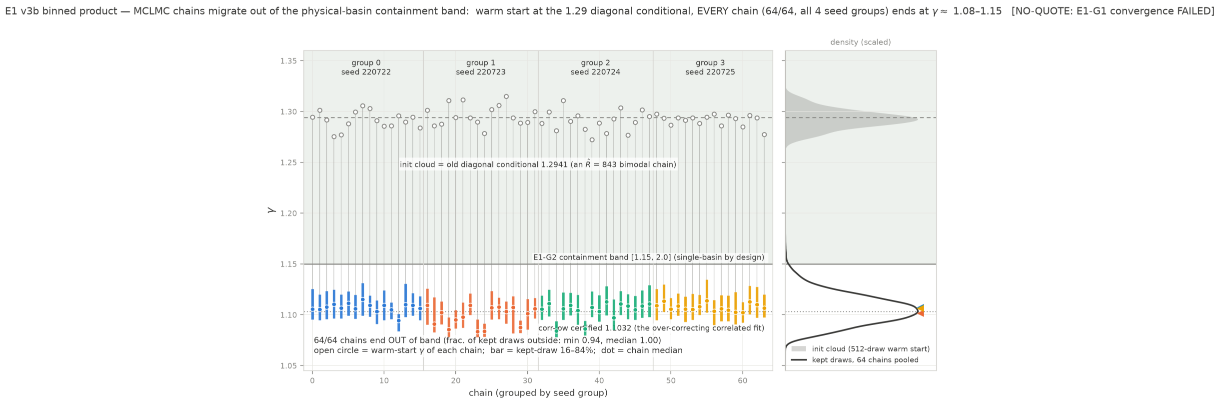

Figure 4: The E1 `v3b` migration finding (the run's money visual, plotted before gate math). Left: for each of the 64 chains (grouped and colored by seed group), the warm-start γ (open circle, drawn from the 1.2941 diagonal-conditional init cloud) against its kept-draw 16–84% interval (bar) and median (dot): every chain exits the E1-G2 containment band $[1.15, 2.0]$ during burn-in and spends its kept draws at $\gamma \approx 1.08$–1.15 under the *diagonal* likelihood (fraction of kept draws outside the band: min 0.94, median 1.00; all 64 violations reported, none dropped, per the frozen report-never-drop rule). Right: init-cloud vs. pooled kept-draw marginals; the pooled kept-draw median 1.1047 sits 0.0015 from the corr-low certified 1.1032 — descriptive only, as the ensemble fails every convergence gate. Source artifacts: `data/mclmc_diag_v3b.json` (`gates.E1_G2`); `data/e1_harvest.json`.

(Huang et al., 2025) the verdict was *necessary but not sufficient*: the correlated likelihood flips the binned-product bimodality — per-basin evidence swings ∼191 nats, from $\Delta \log Z = +162$ (diagonal) to -29 (correlated), showing the upsampled steep basin to be a noise-covariance artifact — but over-corrects the slope, restoring a low basin at $\gamma = 1.103 \pm 0.008$ that sits $\sim 17\sigma$ *below* the 1.433 native-diagonal anchor, so that the correlated slopes across pixel scales *bracket* the anchor rather than unifying onto it. Explaining that bracket — and certifying the machinery on which any explanation would run — is unfinished business this campaign inherits. On the inference axis, the companion campaign found that the two hardest real posteriors (a 46-dim condition-10^{14} marginalized target and a 74-dim bimodal target) defeat off-the-shelf samplers, and that per-basin tempered SMC evidence is the only trustworthy mode-weight instrument.

Independently, the GIGA-Lens team — authors of the original GPU recipe (Gu et al., 2022) and its multi-node successor (Huang et al., 2026) — has been rebuilding the library on what we refer to throughout as *the upstream development branches*: a JAX-only rewrite (Bradbury et al., 2018) around a unified "scene API" (multi-plane, multi-band scene composition; a documented `Dataset/LikelihoodTerm` seam; a physicality layer; experimental microcanonical sampler kernels). Three facts about the commit we vendored (§3) motivate the bridge. First, its validation suite was present but *unrun* — the forward model had no external certification. Second, it contains no correlated-noise likelihood, no analytic source marginalization with an Occam term, and no tempering/SMC evidence layer — precisely the assets the companion campaign validated on the old stack. Third, its likelihood seam is documented and stable enough to carry an upstream-shaped port. Per the campaign's collision rules (§5) we do not characterize any unpublished results of those branches; where campaign cells ran on data or targets provided by the team (the 33-dim multi-plane "carousel" cell on real MUSE cutouts), the data were *provided by the team and used with permission, with any comparison against their own runs deferred pending their sign-off.*

2.2 Campaign goals

The campaign (branch `claude-giga-lens-linus`; plan frozen before any GPU spend) set five goals:

1. **Certify the substrate.** Cross-stack parity of the scene API forward model against the validated companion-campaign stack, at machine precision, on both campaign machines (gate battery F1–F8, §3).

2. **Port the payload.** An upstream-shaped `CorrelatedImageData` likelihood term on the documented likelihood seam, exact in its analytic limits, plus a mutation-kernel-generalized tempered-SMC evidence layer (MC-SMC, §4); then the acid test: does the new substrate *reproduce* the companion campaign's money number $\gamma = 1.103$ and its evidence at posterior level?

3. **Benchmark the bet.** A pre-registered decision matrix for prior-seeded MAMS-SMC against warm MAMS-alone, MCLMC, and the classic MAP→SVI→HMC recipe, across analytic, synthetic-scene, double-source-plane, high-condition, bimodal, and real multi-plane target classes — with two-sided honest predictions written before the runs.

4. **Close mechanism suspects.** Why does the correlated likelihood over-correct? Free (0 GPU-h) and cheap discriminators for the profile-curvature, companion-galaxy, spectrum-flooring, and noise-stationarity suspects, then a confirmatory injection-recovery test on the real noise field.

5. **Arbitrate the anchor, and hand off.** A two-stack end-to-end check of the 1.433 premise itself; and a claims-register handoff to the team (certification evidence, port, a first formal simulation-based-calibration (Talts et al., 2018) run of the pipeline class, ops lessons), every scene-substrate verdict labeled *proposed (UNCERTIFIED — external)*.

2.3 Findings in brief (two-sided by design)

House style holds honest negatives to be first-class results; we summarize both signs here and expand in the results sections that follow.

Certification delivered. F1–F8 pass at machine precision on both machines (forward image 5.7×10^{-15} relative; constrained-space gradients 1.5×10^{-11}), *given* three documented convention reconciliations applied on our side of the fence with the vendor unpatched — the largest a real $\sim 2 \times 10^{-3}$ model-level difference between the two stacks' Sérsic b_n approximants. One gate (F6, the Occam log-determinant) required a signed, ledgered restatement against fp128 truth after the original cross-algorithm reference was measured to be noisier than the tolerance it enforced (§6.3).

The port reproduces cross-stack. The ported correlated term is exact in its analytic limits (delta-kernel identity to 5.8×10^{-11}; dense-Cholesky reference to 5.5×10^{-12} nats) and, run end-to-end on the new substrate on an independent machine, reproduces $\gamma_{\rm low} = 1.1005$ (offset 0.0027 against the certified 1.1032 ± 0.0086), $\log Z$ to 0.11 nats in the low basin and 0.37 nats in the steep, and the low-basin evidence preference at seed-family magnitude ($\Delta \log Z = -29.36$ against $-28.9/-32.3/-31.2$). The companion campaign's headline number now stands on two independent stacks.

The benchmark maps capability boundaries. Prior-seeded MAMS-SMC is the only vehicle that produced $\log Z$ at all in this campaign, and its failures are uniformly *cost walls on a healthy sampler*, never divergence or collapse. The classic single-stage recipe, at frozen settings, produced zero accepted posteriors campaign-wide — including a faithful reproduction, on the new substrate, of the known condition-10^{14} single-stage pathology ($\hat{R}_{\text{worst}} = 44.3$, all chains prior-railed). The carousel-class real-data cell defeats *both* vehicles at matched budget (0 posterior draws each). MCLMC mutations are demoted for evidence: a 20.1σ bias screen and a $\times 56$ minor-mode inflation that resampling does not launder at point estimate. Two cross-cutting findings bind all future evidence gates: bootstrap $\log Z$ uncertainties understate true errors (measured $+3.06 \pm 0.35$ nats across an exact reparameterization where the truth is ≈ 0), and one of our own frozen references was indicted by the very gate it anchored — an 11σ failure decomposing into *more* evidence in both basins with γ-disjoint ensemble support, i.e. a within-basin coverage defect in the reference, not (only) the vehicle.

Stationarity rejected; mechanism standing but unconfirmed. The stationary noise-kernel class behind $\gamma = 1.103$ is provably violated by the real field (calibrated $p = 0.010$, replicated with the arc excluded; cross-block kernel spread $16.4\times$ the drizzle-registration envelope), while the cheap alternatives died: profile curvature is structurally dead at zero GPU cost, the companion galaxy is exonerated (-0.0021 shift), and spectrum flooring is rejected by thousands of nats. The confirmatory injection-recovery test returned NO-CONFIRM / NO-EXONERATE: positive-signed biases outside both pre-registered zones with a control that failed high *and* sick, implicating the injection construction (scene-subtraction residue) under the pre-registered honesty clause. Noise-model-class misspecification remains the working hypothesis — unconfirmed, unrefuted.

Arbitration bounded, not decided. The two-stack anchor check exhausted four vehicles without producing a quotable γ on the 46-dim diagonal target; it closed PARTIAL-BY-VEHICLE-EXHAUSTION, with the premise carried indirectly by machine-precision likelihood parity (F1–F6) plus the *converged* correlated-target reproduction above. "1.433 reproduces at posterior level on the new substrate" is therefore *untested*, and is flagged as such wherever the anchor is used.

The remainder of this report: §3 (substrate and certification battery), §4 (correlated likelihood port and the MC-SMC evidence layer), §5 (protocol, labeling, and budget fences), followed by the results sections (benchmark decision matrix; mechanism program; cross-stack reproduction and arbitration; gifts and handoff), the discussion of open items, and the reproducibility appendix.

3 Methods I: Substrate, Environment, and Cross-Stack Certification

3.1 Vendored substrate and pinned environment

The campaign runs against a frozen snapshot of the upstream development branch, vendored at commit `80916d2` (full SHA in `vendor/gigalens-linus/VENDORED_REF.txt`) via `git archive` — *unpatched*, installed `--no-deps`. Every campaign process asserts the vendor identity at import (`require_vendor_ref`), and a ledgered re-pin procedure (motivation → re-archive → diff review of the three subclassed surfaces → full gate-battery re-run green) guards against silent drift of the moving branch; it was never needed — the pin held for the whole campaign. All campaign-side adaptation lives in the shim package `cgl2/` (parameter mapping, guards, scene factories, the correlated term, and the samplers); the vendor tree was never edited.

Table 3 lists the pinned environment. The single deviation from the upstream declared pins — numpy 2.4.6 versus their 2.1.3 — is recorded as a known deviation and covered by the gate battery. The old validated stack (commit `58ec9a7`; Benson and Huang, 2026) is imported by path, never copied; whitener bundles and Foundry-I data products likewise. Production GPU cells ran single-GPU shared-QOS on NERSC Perlmutter A100s; certification and free checks ran on the local phoenix node (L4/A16).

Table 3: Pinned software environment (venv `cgl2`, mirrored as `cgl2-pm` on Perlmutter). Pins are seeded from the companion campaign's `constraints.txt`; all installs are constrained.

Component	Version	Note
python	3.13	
jax / jaxlib	0.6.2	upstream's declared pin; identical on both machines
BlackJAX	1.3	(Cabezas et al., 2024)
tensorflow-probability	0.25.0	(Dillon et al., 2017)
numpy	2.4.6	*known deviation* vs. upstream 2.1.3; gate-covered
tensorflow	—	absent; the vendored dependency is verified inert
vendor	`80916d2`	UNPATCHED; `--no-deps`; identity-guarded
old stack	`58ec9a7`	validated companion-campaign stack, by path

3.2 Reference-artifact parity design

Both stacks ship as a Python package named `gigalens`, so the originally planned single-process dual import is impossible (a name collision the upstream documentation itself warns about). The enacted design (ledgered decision D3) is *reference-artifact parity*: the old validated stack, running in its own certified virtual environment, dumps reference arrays (forward images, design-matrix columns, log-likelihoods, constrained-space gradients) to `data/parity_refs.npz` for the foundry v2d and v3b inputs at a reference point z_{ref} plus three seeded perturbations; the scene API stack then recomputes the identical quantities in the campaign environment and compares. Both sides run the same jax 0.6.2, which preserves 10^{-12}-class comparability across the process boundary. Comparison is *in constrained parameter space* through the campaign's name-keyed parameter map (never positional ordering), so any flat-vector convention bug fails parity by construction.

3.3 The F1–F8 certification battery

Table 4 gives the battery and its frozen thresholds (fixed at P0, before any science ran); the measured results, on both machines, are reported with the certification results in §6 (Table 5). F1–F4 constitute, to our knowledge, the first external certification evidence for the scene API forward model in the EPL+shear + 4×Sérsic / Sérsic+shapelets ($n_{\max} = 6$) configuration class — the vendored validation suite was unrun at the pin. F8 repeats the entire battery under the Perlmutter-native pinned environment on an A100, so that the stack is certified on both machines before any production cell. Parity at these thresholds holds *given* three documented convention reconciliations, applied on our side of the seam with the vendor unpatched; they are themselves findings and are reported with the results (§6.2).

The whitener bundles imported from the companion campaign were revalidated in place (gate W): operator-norm flatness e_{op} reproduced exactly, erosion masks reproduced from first principles, kernel hashes pinned. The three production bundles (v2d, v3b, v3) are admissible under the strict $e_{\mathrm{op}} \le 0.02$ gate; the ledgered relaxed arm `v2d_relaxed` ($e_{\mathrm{op}} = 0.0312$ against its own design target 0.05) is correctly *inadmissible-by-design* and fenced from production use.[9]

[9] Source artifact: `data/whitener_manifest.json`.

Table 4: Cross-stack certification battery design: old validated stack (`58ec9a7`) versus the scene API (`80916d2`), same foundry `v2d/v3b` inputs, $z_{\text{ref}} + 3$ seeded perturbations, constrained-space comparison, scored on the worst case over products and perturbations. Thresholds frozen at P0; F6 is quoted under its restated form (§6.3). Measured results: Table 5 and §6–§6.4.

Gate	Statement	Threshold
F1	forward image	$\leq 10^{-12}$ rel
F2	design-matrix columns	$\leq 10^{-12}$ rel
F3	diagonal masked loglik + χ^2	$\leq 10^{-8}$
F4	grad of marg. loglik (constrained)	$\leq 10^{-8}$ rel-L_2
F5	delta-kernel corr. term $\equiv$ stock	$\leq 10^{-10}$
F6	Occam $-\frac{1}{2}\log\det A$ vs fp128 truth	cond-scaled (§6.3)
F7	flat-z round-trip + name audit	exact (info)
F8	full battery, Perlmutter-native env	report-only
03-A	corr. term vs dense-Cholesky (32^2 toy)	≤ 0.1 nat
03-B	delta identities (fwd; conv$\equiv$multiply)	$\leq 10^{-10}$
W	whitener-bundle revalidation	$e_{\text{op}} \leq 0.02$ strict

4 Methods II: The Correlated-Noise Likelihood Term and the MC-SMC Evidence Layer

4.1 `CorrelatedImageData`: an upstream-shaped port

The correlated-noise payload is ported as a dataset class `CorrelatedImageData` with its companion likelihood term `CorrelatedImageLikelihoodTerm` (`cgl2/correlated.py`), subclassing the scene API's documented `Dataset` and `LikelihoodTerm` seam — shaped so that upstream adoption is a move, not a rewrite. Design points, each carrying a validation gate:

- **Frozen whitener bundles as data.** The constructor takes a frozen whitener bundle (convolution taps, $D^{-1/2}$ diagonal, eroded keep-mask, e_{op}, kernel hash) and validates it raise-never-default; `event_size` is the kept whitened degrees of freedom. The whitener seam is deliberately pluggable: the stationarity rejection (§2) means a locally-stationary (per-region) kernel class is the anticipated successor, and the term is agnostic to which bundle it is handed.
- **Single-render contract.** One basis render per likelihood evaluation, via `lstsq_simulate` with `return_stacked=True`, honoring the upstream single-forward-eval contract and reusing the fused convolution/pooling path verbatim; residual and design columns are whitened by a grouped depthwise convolution wrapped in `jax.checkpoint` (the memory fix that keeps SMC particles under the $\leq$200 MB budget). Masked operations use `jnp.where`, never multiplication by a mask.
- **Ridge marginalization with the Occam term.** The linear (shapelet) amplitudes are analytically marginalized under a generalized ridge prior, *including* the $-\frac{1}{2}\log\det A$ Gaussian-evidence term that a plain linear solve omits — the term that makes downstream Bayes factors meaningful. Whitened χ^2 is exposed through the upstream `reports_chi2` channel.
- **Exactness in analytic limits.** With a delta kernel the term reduces exactly to the stock `ImageLikelihoodTerm` (F5: 5.8×10^{-11}; conv$\equiv$multiply identity 0.0); on a 32^2 toy with a dense covariance it matches an exact Cholesky reference to 5.5×10^{-12} nats against a 0.1-nat gate (03-A).[10] Production wiring adds an old-vs-new whitened marginalized log-likelihood/gradient

[10]Source artifact: `data/correlated_term_validation.json`.

cross-stack check at z_{ref} plus three seeded perturbations (worst $|\Delta \log L_{\text{marg}}|$: 2.4×10^{-8} on `v2d`, 7.2×10^{-7} on `v3b`)[11] and a remat-exactness gate (checkpointed vs. un-checkpointed whitening, difference exactly 0.0).[12] Separately, B0's target-adapter parity gate compares old and new log-densities at 64 prior draws (max $|\Delta|$ exactly 0.0).[13]

For SMC the likelihood is evaluated in f64: a f32 forward path carries a ∼0.3-nat evaluation noise floor that would pollute Metropolis–Hastings acceptance in the mutation kernels.

4.2 MC-SMC: tempered SMC with microcanonical mutation kernels

The evidence layer (`cgl2/samplers/smc_micro.py`) generalizes the companion campaign's validated adaptive tempered-SMC driver (Del Moral et al., 2006) — the scene API's `ProbModel` log-prior / log-likelihood split maps exactly onto the BlackJAX convention — with the mutation kernel abstracted over three choices:

- **HMC** — the proven baseline from the companion campaign.
- **MAMS** (Robnik and Seljak, 2024b) — the *primary* and the campaign's bet: Metropolis-adjusted microcanonical dynamics, whose exact π_λ-invariance at every tempering stage is what makes the SMC evidence estimate unbiased.
- **MCLMC** (Robnik and Seljak, 2024a) — unadjusted microcanonical dynamics, admitted *diagnostic-only* behind a pre-registered bias gate (below), never as an evidence kernel.

Kernel logic is copied with attribution from the upstream experimental implementations, never imported (those modules reach into private BlackJAX internals and the pre-rewrite simulator). Per-λ tuning was frozen at P0, before any science cell: mass matrix from the regularized weighted particle covariance; step size from a short dual-averaging pilot per stage (10 iterations, acceptance target 0.90 for MAMS per the MAMS integrator recommendation, 0.80 for HMC; an energy-variance-per-dimension pilot for MCLMC); 4 kernel draws per stage; adaptive λ-schedule by bisection on the incremental-weight ESS at target $0.7N$ with systematic resampling; $\log Z$ accumulated per stage with a per-stage bootstrap error σ_{boot}. Gradient evaluations are ledgered exactly (pilot draws billed identically), which is what makes the budget-matched benchmark arms of the results sections well-defined.

B0 correctness gates (frozen at P0). On analytic targets the layer passes where it must and fails where the class fails, and both are recorded:[14] target adapters build 11/11; on the 2-dim bimodal `mix2`, $|\log Z \text{ error}| = 0.073 \le 3\sigma_{\text{boot}}$ with minor-mode weight 0.219 against truth 0.2; on the 46-dim ill-conditioned target the worst-parameter $|z| = 1.83$ passes while the $\log Z$ error is 2.40 nats at $\sigma_{\text{boot}} = 0.37$ — i.e. $6.6\sigma_{\text{boot}}$, the first measurement of the campaign-wide finding that *bootstrap* $\log Z$ *errors understate true errors on ill-conditioned targets*; on `funnel10` the layer *fails* (-0.66-nat $\log Z$ deficit) — a pre-existing limitation of the adaptive tempered-SMC class, reproduced in the old validated stack and carried as a caveat, not a port defect. The pre-registered MCLMC bias screen fired: MCLMC-vs-MAMS $\Delta \log Z = +10.79 \pm 0.54$ on the ill-conditioned target, i.e. 20.1σ,[15] demoting MCLMC to diagnostic/cost-frontier-only for the entire campaign —

[11] Source artifact: `data/parity_report_scene.json`.
[12] Source artifact: `data/l0_remat_gate.json`.
[13] Source artifact: `data/smc_b0_report.json`.
[14] Source artifact: `data/smc_b0_report.json`.
[15] The ledger prose records "30σ"; the artifact of record `data/smc_b0_report.json` gives $10.79/0.536 = 20.1$. Per the campaign's discrepancy-flag rule (§5) the artifact value is quoted; the demotion ($> 3\sigma$ rule) is unaffected.

a demotion whose independent confirmation (×56 minor-mode evidence inflation on the 74-dim bimodal target) appears in the results.

4.3 The checkpoint/resume retrofit

The benchmark's first production wave lost ∼17.6 of its 18.7 A100 h to zero-artifact timeouts (only 1.12 h produced readable science): the frozen SMC driver wrote results only at $\lambda = 1$ and printed no stage progress, so wall-capped jobs burned silently. The response — kept out of the frozen driver, which was never edited — is a checkpointing retrofit (`cgl2/samplers/ckpt.py`) with three properties:

1. **Per-stage full state.** A delegating `CheckpointKernel` (build/tune/mutate verbatim) writes a full-state checkpoint per λ-stage (post-mutation particles, λ, step size, trajectory length, acceptance, and the exact per-stage gradient/log-probability ledger) plus one progress line; a pre-registered in-job wall cap raises *after* the checkpoint, with an exit-3 PARTIAL protocol — a wall hit preserves progress by construction, and no gate math is ever run on a non-λ=1 ensemble.

2. **Evidence-complete resume.** Weight-stage log-likelihoods are recorded through the driver's existing `loglik_batch_fn` seam (delegation only), so a resumed run retains the *full* $\log Z$ accumulation and per-stage bootstrap — not a partial-from-resume approximation.

3. **Bit-identity.** On `--resume`, the RNG chain is fast-forwarded by replay of the recorded log-likelihoods; the retrofit gate proves a fresh run and an interrupted-then-resumed run *bit-identical* including $\log Z$ and its bootstrap, and that resume-after-completion rebuilds the identical result with zero new forward evaluations. (Documented caveat: under strict-f32 execution the checkpointed ensemble is an f64 reconstruction of the f32 state, so resume there is protocol-identical up to an f32-rounding-level perturbation; everywhere else it is bit-exact.)

The retrofit ships with a 58-test suite, green on both machines. Its operational effect is visible throughout the results: the recovery lane turned the first wave's zero-artifact failure mode into readable recoveries (3 of 4 cells), and the 12-hour, three-leg chained real-data carousel run — and every partial-λ cost readout quoted in the decision matrix — exists only because interrupted evidence runs became resumable and readable at all.

5 Methods III: Protocol

5.1 Pre-registration and frozen gates

Every experimental cell was preceded by a *design checkpoint* logged to the campaign record before submission: hypothesis, predicted direction and magnitude, falsifier, and derived threshold. Certification gates (F1–F8, B0) were frozen at P0 in the repository README before any science ran. Thresholds parameterized by then-unknown quantities carried declared placeholders that were *finalized, not moved*, when the parameterizing measurement landed: the seed-repeatability study fixed $\sigma_{\text{seed}}(\gamma) = 0.00325$ (low basin), 0.00664 (steep), and $\sigma_{\text{seed}}(\Delta \log Z) = 1.79$ nats (with the n=3 caveat — ±46% χ-distribution sampling error — quoted at every use), and the downstream gates inherited these by a ledgered finalization row.[16] Where a pre-registered gate proved undecidable as written, the record shows the branch taken rather than a rescored verdict: outcomes in this report

[16] Source artifact: `data/t02_t03_gate_eval.json`.

include FAIL-as-written, NOT-DECIDABLE-AS-REGISTERED, PARTIAL-BY-BUDGET, NOT-EVALUABLE-WITHIN-BUDGET, and AMBIGUOUS-AT-FROZEN-THRESHOLDS — honest negatives are first-class results, and no threshold was moved after data. Two further house rules: diagnostic figures are produced and inspected *before* gate text is written (a rule adopted from the upstream team's research culture), and amendments (a re-registered gate, a descoped cell) enter the record as explicit ledgered decisions with the original always preserved.

5.2 Proposer ≠ grader: the UNCERTIFIED-external label

The campaign separates producing a result from certifying it. For every claim that runs through the scene API substrate, we are the *proposer*; the *grader* is the upstream GIGA-Lens team (and Greg Benson for our-side scope). Accordingly, every scene-substrate verdict in this report carries the label *proposed (UNCERTIFIED — external)*, and the deliverable of record is a claims register in the team's own lab-notebook format (`papers/handoff/CLAIMS.md`: fourteen claims, each with pre-registered criterion, measured numbers, named artifacts, and a mandatory doubt report). "Validated" is reserved, throughout this report, for the old certified stack; scene-API results are "reproduced/measured, UNCERTIFIED (external)." Claims derived from data or targets provided by the team are additionally marked SIGN-OFF GATED in the register: internal-memo material, publication-gated on the team's explicit sign-off. In this report the carousel real-data cell appears under exactly that regime — *data provided by the team, used with permission; any comparison against their own runs is deferred pending sign-off* — and the upstream repositories are referenced only as the upstream development branches, without characterization of their unpublished results.

Four bright lines, frozen in the plan before the campaign began, bind the published text: (i) no result is presented so as to be readable as an MCLMC-vs-HMC efficiency comparison on a unimodal target — every sampler number in this report attaches to a multimodality, evidence, cold-start, or noise-model question; (ii) nothing derived from the upstream branches' unpublished code or results appears externally without the team's sign-off; (iii) the team's ongoing staged campaigns are untouched — our prototypes are offers, not applications; (iv) co-authorship is offered on any publication whose results run through their substrate.

5.3 Budget fences and operational discipline

The campaign ran under a 100 A100 h hard stop with per-phase caps and a ledger-before-results rule: the estimated cost of every job is appended to the campaign ledger *before* any result is read, and actuals are harvested from `sacct`. Realized spend was 53.51 A100 h — certification slice 2.21, injection-recovery 7.80, benchmark 22.45 of a 24-h cap, the user-funded real-data extension 15.85 of 18, and the payload/arbitration phase 5.21 of 17 (component sums differ from the total by 0.01 h of per-row rounding).[17] Sub-fences were binding, not advisory: chained runs carried per-leg wall caps with pre-declared stop conditions (the carousel chain stopped at its fence with $\lambda = 0.15$, and the cell was closed by an explicit ledgered decision, D9, rather than extended), budget-matched comparison arms were capped at their partner's *realized* gradient ledger, and cells whose measured cost exceeded remaining headroom closed as NOT-EVALUABLE-WITHIN-BUDGET rather than overrun. Unused programmed items are closed explicitly in a final gate-ledger sweep (six programmed items proposed NOT-RUN with reasons — among them the S7 flow-preconditioned-MAMS arm, whose prerequisite B1 λ=1 ensemble never came to exist, and the X3 evidence-scored Vela-ladder prototype, never mobilized — plus B4 proposed NOT-EVALUABLE-WITHIN-BUDGET) rather

[17] Source artifact: `CAMPAIGN.md (A100-hour ledger)`.

than silently dropped.[18]

Operational rules that each cost real budget to learn once, then became protocol: production SMC jobs pin high-memory GPUs (`-C gpu&hbm80g`; memory sizings measured under rematerialization do not transfer to un-rematerialized runs); the submitted batch-script snapshot is verified by checksum via `scontrol write batch_script` (SLURM snapshots the script at submission — a "resubmitted fix" can silently run the old code); every campaign file is checksum-audited on the remote before production (single-writer manifest, union-merge on conflict); a dead-man watchdog with per-job expected artifacts and stall alerts armed every wave; and the lensing likelihood is never evaluated on CPU. A full ops-lessons table, with the job ids that motivated each rule, accompanies the reproducibility appendix.

5.4 Provenance rule

Every headline number in this report is traced to an on-disk artifact (JSON/NPZ path), and where it was produced on Perlmutter, to a SLURM job id; the campaign ledger `CAMPAIGN.md` records the gate row and the decision context. Where the ledger prose and the artifact of record disagree, the synthesis that feeds this report flags the discrepancy rather than smoothing it — seven such flags are on record, and in every case this report quotes the *artifact* value.[19] The experimental record — plan, ledger, design checkpoints, gate evaluations, and the claims register — ships with the repository.

6 Results I — Certification: the Scene API Reproduces the Validated Stack

The campaign's foundation result: the next-generation scene API — the team's upstream development branch, vendored unpatched at commit `80916d2` — evaluates the same forward model, likelihood, and gradients as the previous campaign's validated stack to within machine precision, on both campaign machines, for the configuration class of the real lens. Everything downstream (the mechanism program of §7, the cross-stack reproduction of §7.7) stands on this section. All scene-substrate numbers are proposed (UNCERTIFIED—external): we are the producer; the upstream team is the grader.

The certification battery F1–F8 compares the scene API's simulator and likelihood surfaces against the validated `58ec9a7` stack of the preceding campaign on the same DESI–165.4754−06.0423 inputs (`v2d` native and `v3b` binned products), at the reference point plus three seeded perturbations, compared in constrained parameter space. The battery ran at zero A100 cost (phoenix L4 free tier); its Perlmutter repetition (F8) took 58 s of A100 time.

6.1 The F1–F8 parity battery, both machines

All hard gates pass on both machines (Table 5).[20] The forward image and design columns agree at the 10^{-15} level — the two stacks are evaluating the *same function*, not merely similar ones — and the constrained-space gradient, which compounds every convention in the chain (profiles, grids, PSF, masking, bijectors), agrees to 1.5×10^{-11} relative L_2. The Perlmutter-native re-run under the

[18] Source artifact: `research/gate_ledger_final.md`.

[19] Source artifact: `research/synthesis_tables.md (§4, discrepancy flags)`.

[20] Source artifact: `data/parity_report_scene.json; data/results-perlmutter/parity_report_scene_pmnative_55980038.json`.

production pinned environment (jax 0.6.2, x86 A100; job 55980038) passes every hard gate with the F1 statistic *bit-consistent* with the phoenix value (5.682×10^{-15}), certifying the stack on both machines and unblocking every scene-API production cell of the campaign.

Table 5: Cross-stack certification battery: scene API (vendored upstream development branch, unpatched) vs. the validated `58ec9a7` stack, same DESI–165.4754−06.0423 inputs, constrained space, $z_{\rm ref}+3$ seeded perturbations. F6 is the restated conditioning-scaled gate (§6.3); the F8 column's F1 value is bit-consistent with phoenix. Phoenix column: `data/parity_report_scene.json`; Perlmutter column (F8, job 55980038, 58 s): `data/results-perlmutter/parity_report_scene_pmnative_55980038.json`.

gate (quantity)	tolerance	phoenix L4	Perlmutter A100 (F8)
F1 forward image (rel)	10^{-12}	5.7×10^{-15}	5.682×10^{-15}
F2 design columns (rel)	10^{-12}	3.9×10^{-15}	pass
F3 diagonal masked $\log L + \chi^2$	10^{-8}	7.5×10^{-9}	pass
F4 constrained gradient (rel L_2)	10^{-8}	1.5×10^{-11}	1.314×10^{-11}
F5/F5b delta-kernel $\equiv$ stock	10^{-10}	5.8×10^{-11} / 0.0	pass
F6 Occam vs fp128 truth (§6.3)	cond-scaled	v2d 1.19×10^{-11}; v3b 4.48×10^{-10}	3.562×10^{-10}
F7 flat-z round-trip (46 names)	exact	0.0	pass

One scoping note, recorded for honesty: the pre-registered F1 statement (forward image, old vs. `SceneSimulator.simulate`) is gated as written; the *full* model image with each stack's own solved source amplitudes is reported informationally (worst 1.1×10^{-12}), because it compounds F1×F2 through the $\mathrm{cond}(A) \approx 7 \times 10^7$ amplitude solve — a path the likelihood itself never takes (the quadratic form that enters $\log L$ is gated via F3/F4).

6.2 Three convention reconciliations

Parity at the level of Table 5 holds *given three documented convention reconciliations*, each measured before being reconciled. They are findings in their own right — stock-vs-stock, the two stacks differ at exactly these scales:

1. **Sérsic b_n approximant (a real $\sim 2 \times 10^{-3}$ model-level difference).** The old stack uses $b_n = 1.9992\,n - 0.3271$; the scene API uses $b_n = \exp(0.6950 + \ln n - 0.1789/n)$. Both are literature approximants to the exact b_n equation; we did not adjudicate which is closer to exact for the use range. Measured via an informational `native_profiles` arm: $\sim 2 \times 10^{-3}$ relative model-image difference for the tested class.[21] Any cross-stack comparison at the 10^{-3} level needs this reconciliation.

2. **f32 shapelet Hermite prefactor** in the old stack ($\sim 10^{-8}$ basis delta).

3. **f32 coordinate grids + un-renormalized subgrid PSF kernel** in the old stack ($\sim 10^{-4}$ / 2×10^{-6} image deltas at `v2d/v3b`).

The reconciliations are implemented on our side, as subclasses and instance overrides in `cgl2/scene_build.py` (docstring items 1–3); **the vendored branch is unpatched**. Declared environment deviation: our numpy is 2.4.6 vs. the branch's 2.1.3 pin, covered by the gate battery. The certification covers one configuration class — EPL+Shear with 4×Sérsic lens light and Sérsic+shapelets ($n_{\rm max} = 6$) source — and makes no statement about other profile or multi-plane paths.

[21] Source artifact: `data/parity_report_scene.json (native_profiles arm)`.

6.3 F6: an honest gate failure at the f64 noise floor, and its signed restatement

F6 — parity of the Occam term $-\frac{1}{2}\log\det A$ of the generalized ridge marginalization — **failed as originally written**, and the failure is worth more than the pass that replaced it. The original gate compared the scene-API jax-Cholesky value against a numpy `slogdet` reference at $\leq 10^{-10}$; on `v3b` it failed at 1.31×10^{-10}.[22] The noise-floor analysis showed why the gate, not the code, was wrong: at $\mathrm{cond}(A) = 7.0 \times 10^7$, pure-numpy Cholesky and `slogdet` on the *identical* matrix already differ by 7.2×10^{-10}, and against an fp128 ground truth the jax-Cholesky value errs by 4.5×10^{-10} — *more accurate* than the numpy `slogdet` gate reference itself (7.1×10^{-10}) and than the old validated stack's own stored value (1.3×10^{-9}). The defensible statement is that the gate *reference* errs at $\sim 7 \times 10^{-10}$ there, so demanding 10^{-10} cross-algorithm agreement is not meaningful at that conditioning (not that 10^{-10} is unreachable by any f64 implementation).

The gate was therefore restated — by signed exception (Benson sign-off, 2026-07-15), not silently moved — to compare against fp128 truth with a conditioning-scaled tolerance:

$$\left|\log\det A_{\text{jax-chol}} - \log\det A_{\text{fp128}}\right| \;\leq\; \max\left(10^{-10},\; 5\,\epsilon\,\mathrm{cond}(A)\cdot 10^{-2}\right),$$

under which F6 passes: `v2d` 1.19×10^{-11} (tol 10^{-10}, floor governs at cond $= 7.9 \times 10^6$); `v3b` $4.48 \times 10^{-10} \leq 7.76 \times 10^{-10}$ (cond $= 7.0 \times 10^7$). The original recommendation and the full fp128 evidence are preserved in the artifact.[23] The transferable lesson, offered to anyone writing $\log\det$ gates for marginalized likelihoods: at $\mathrm{cond}(A) \gtrsim 10^7$ a cross-algorithm f64 tolerance below $\sim 10^{-9}$ tests the reference, not the code.

6.4 The correlated-noise likelihood port is exact in its analytic limits

The drizzle-anchored correlated-noise likelihood of the previous campaign was ported onto the scene API's documented `Dataset`/`LikelihoodTerm` seam, as `CorrelatedImageData` plus its matching likelihood term: one `lstsq_simulate` render per evaluation, grouped-depthwise convolutional whitening, and generalized ridge marginalization with the Occam term. Its validation gates, all passed:[24]

- **Delta-kernel identity (F5/F5b):** with a delta whitening kernel the correlated term reproduces the stock `ImageLikelihoodTerm` to 5.8×10^{-11} (χ^2 branch exact, 0.0).
- **Dense-Cholesky exact reference (03-A):** on a 32^2 toy with a full dense covariance, the ported term agrees with the exact reference to 5.5×10^{-12} nat (gate ≤ 0.1 nat); delta identities (03-B) hold to 9.1×10^{-13} / 0.0.
- **Whitener bundles re-validated on import:** operator-norm whiteness $e_{\mathrm{op}} \leq 0.02$ (strict) holds for the three production bundles (`v2d`, `v3b`, `v3`); the `v2d_relaxed` bundle is correctly flagged inadmissible-by-design under the strict gate (it was built as the ledgered relaxed arm).

The $-\frac{1}{2}\log\det A$ term — which a plain least-squares implementation drops — is included, and matters for any downstream Bayes factor. This ported term is the engine of the cross-stack reproduction in §7.7.

[22] Source artifact: `data/parity_report_scene.json (products.v3b.F6.chol_minus_numpy)`.

[23] Source artifact: `data/parity_report_scene.json (products.*.F6)`.

[24] Source artifact: `data/correlated_term_validation.json; data/parity_report_scene.json; data/whitener_manifest.json`.

7 Results II — The Bracket Mechanism on the Real Lens

The previous campaign ended with a verdict — the correlated likelihood is necessary but not sufficient — and a puzzle: on DESI–165.4754–06.0423 the three data products bracket the power-law slope (fine-correlated $\gamma = 1.816$, native-diagonal anchor 1.433, binned-correlated 1.103) instead of unifying. This campaign ran a pre-registered elimination program against that bracket. The outcome: two suspects are dead (radial profile curvature, the companion galaxy), one proposed cure is falsified (spectrum flooring), and the noise-model class *— a single stationary convolution kernel per product — is directly rejected on the field that produced $\gamma = 1.103$, making noise-model-class misspecification the standing working hypothesis. The confirmatory injection experiment came back confounded, honestly reported below, so the mechanism is indicted, not convicted. Source/PSF representation remains alive and untested.*

Table 6 is the final scoreboard; the subsections that follow give each row's evidence. Except where noted, the mechanism program ran on the *validated old stack* (these are our-stack results, not scene-substrate results); X1-G0 and the T0.4 trilogy cost zero GPU hours.

Table 6: Final suspect scoreboard for the γ bracket (fine 1.816 / anchor 1.433 / binned 1.103). Every verdict is pre-registered; artifacts as cited in the subsections.

suspect	verdict	discriminating measurement
profile curvature (radial mass structure)	**DEAD** (X1-G0, §7.2)	no monotone r_{eff} ordering in 24/24 variants; would require $\lvert d\gamma_{\mathrm{loc}}/d\ln r\rvert \approx 226$ vs. $O(1)$ physical
companion galaxy (LL2/LL3 misfit)	**EXONERATED** (T0.3, §7.3)	whiten-then-drop shift -0.0021 ($0.26\,\sigma_{\mathrm{stat}}$; band was $\geq +0.024$)
spectrum flooring (information discard at spectral zeros)	**FALSIFIED** (T0.4-2, §7.4.2)	data reject floored kernels by $\sim$3.2k–33k nats per point
stationarity of the noise-model class	**REJECTED — standing mechanism** (T0.4-1, §7.4.1)	calibrated $p = 0.010$; cross-block kernel spread 16.4× the drizzle-registration envelope
injection methodology (T1.1 construction)	**CONFOUNDED** → data-side redesign (§7.5–7.6)	control fails high *and* sick; fit-side rescue loses 85.8% of whitened dof
source model / PSF representation	**ALIVE (untested)**	inherits the bracket question: the driver acts at fixed radius (X1-G0)

7.1 The number under interrogation: γ certified at 1.1032 ± 0.0086, with a seed-stable basin preference

Before interrogating the binned-correlated slope, the campaign certified it (T0.2: three matched-seed repeats of the production 128-particle tempered-SMC fit on `v3b`, seeds $\{2, 3, 4\}$; 2.21 A100-h with T0.3).[25] The low-basin medians are $\{1.1032, 1.0967, 1.1005\}$, giving $\sigma_{\mathrm{seed}}(\gamma) = 0.00325$ — the pre-registered kill ($\sigma_{\mathrm{seed}} > 0.024$) is not tripped — and

$$\gamma^{\mathrm{binned}}(\mathrm{corr},\ \mathrm{low}) \;=\; 1.1032 \pm 0.0086,$$
$$\sigma_{\mathrm{tot}} \;=\; \sqrt{\sigma^2_{\mathrm{stat}} + \sigma^2_{\mathrm{seed}}} \;=\; \sqrt{0.0080^2 + 0.00325^2} \;=\; 0.00864 \;\approx\; 1.08\,\sigma_{\mathrm{stat}}.$$

[25]Source artifact: `data/t02_t03_gate_eval.json`.

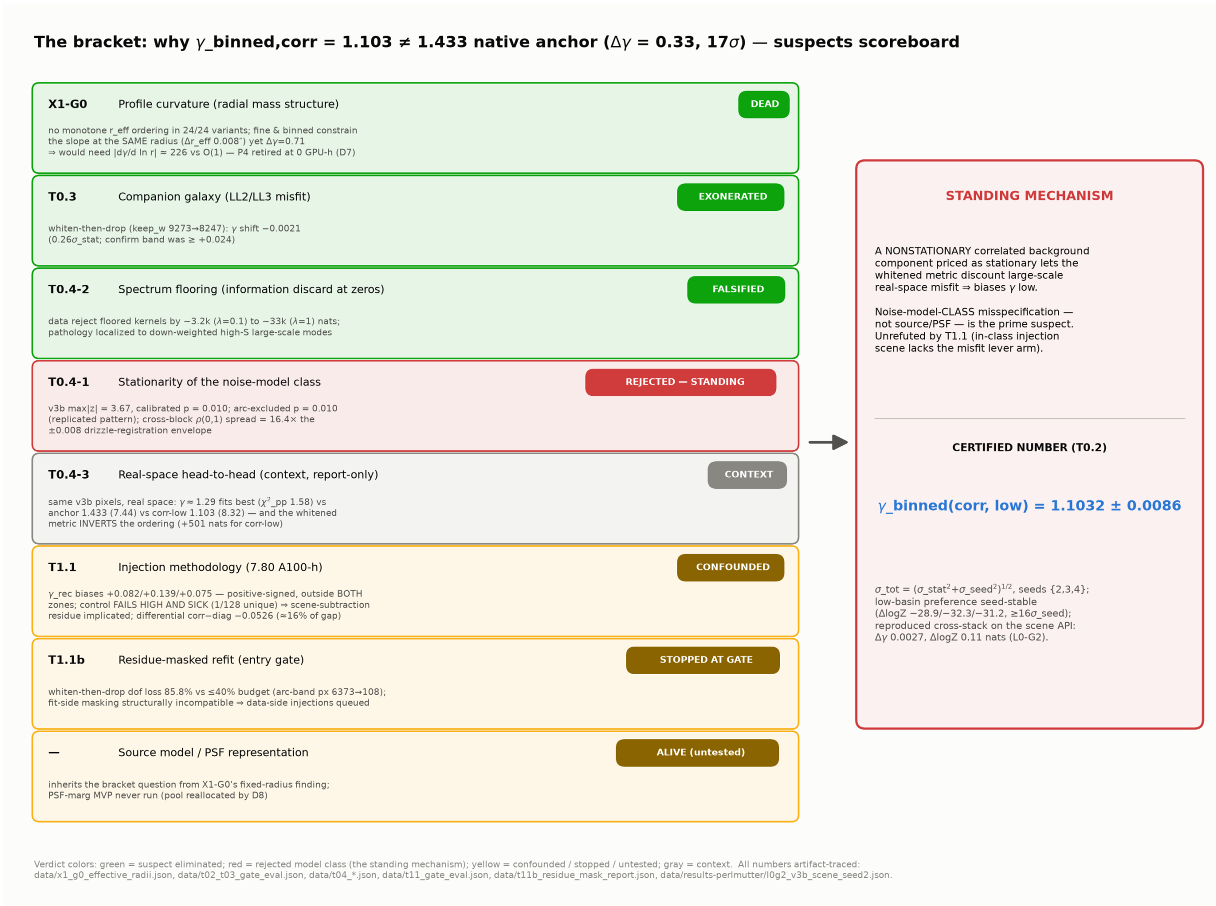

Figure 5: The bracket suspects scoreboard (the graphical form of Table 6): one card per pre-registered discriminating measurement, in program order. Profile curvature DEAD (X1-G0 entry-gate kill executed as written; P4 retired at zero GPU cost, decision D7); companion EXONERATED (T0.3); spectrum flooring FALSIFIED by 3.2k–33k nats (T0.4-2); stationarity of the noise-model class REJECTED at calibrated $p = 0.010$ — the standing mechanism (T0.4-1); the real-space head-to-head as context (T0.4-3; the anchor row carries a cross-product resolution handicap — honest caveat); injection–recovery CONFOUNDED by scene-subtraction residue under its pre-registered honesty clause (T1.1, n=3, no coverage claims); the residue-masked refit stopped at its entry gate (T1.1b, 0 A100-h); source/PSF representation ALIVE (untested). $\gamma = 1.1032 \pm 0.0086$ is certified (T0.2) and reproduced cross-stack (L0-G2); the E3 kernel-scan deferral remains the standing caveat on absolute-γ claims. Sources: the per-suspect artifacts cited in the subsections of this section.

The steep basin repeats at $\{2.6393,\ 2.6522,\ 2.6485\}$ ($\sigma_{\text{seed}} = 0.00664$), and the per-matched-seed basin evidence gap $\Delta \log Z(\text{steep} - \text{low}) = -28.88/-32.27/-31.19$ nats is **seed-stable in sign and magnitude** — a low-basin preference $\geq 16\,\sigma_{\text{seed}}$ from zero, with $\sigma_{\text{seed}}(\Delta \log Z) = 1.79$ nats (Fig. 6). That last number is load-bearing: the campaign's cross-cutting finding that bootstrap errors understate evidence error makes this measured seed scatter the honest unit for every downstream $\Delta \log Z$ gate. Caveat, quoted with every use: $n = 3$ seeds, so the σ estimates themselves carry $\pm 46\%$ sampling error (χ-distribution). Downstream thresholds were re-parameterized from the measured σ_{seed} at this point — a ledgered finalization, not a goalpost move.

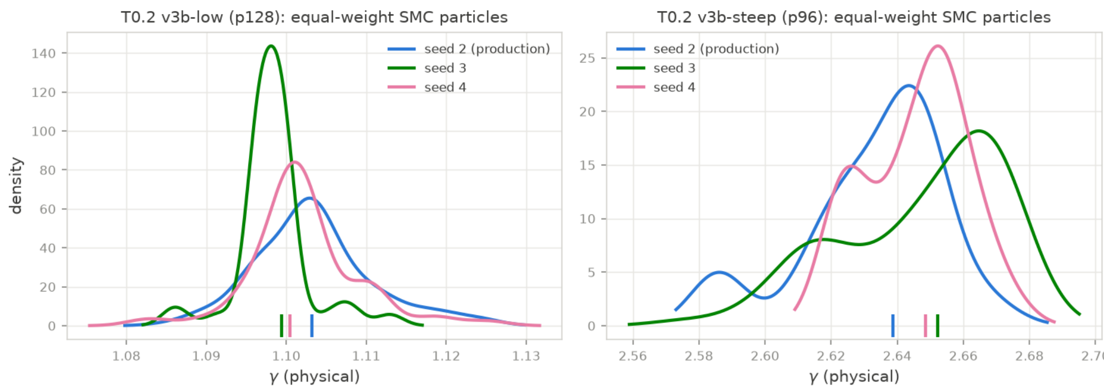

Figure 6: T0.2 seed-repeat certification: low- and steep-basin γ posteriors for seeds $\{2, 3, 4\}$ (production tempered SMC, `v3b` correlated likelihood). The money number is $\gamma = 1.1032 \pm 0.0086$ (σ_{tot}, seed scatter folded in); the basin preference $\Delta \log Z(\text{steep} - \text{low}) \approx -29$ to -32 nats is seed-stable. Source: `data/t02_t03_gate_eval.json`.

7.2 X1-G0: profile curvature is structurally dead

The P4 hypothesis held that the EPL single-power-law assumption drives the bracket: whitening and binning re-weight spatial frequencies, so each product constrains the slope at a different effective radius, and a curved $\kappa(r)$ then yields different local slopes. Its pre-registered entry gate (X1-G0, zero GPU cost) required the per-product effective radii to be monotone in the same sense as the γ ordering. **The gate failed in all 24 analysis variants** (3 statistics $\times$ primary weighting plus 7 robustness variants), and P4 was retired unspent on the strength of it (decision D7).[26]

Table 7: X1-G0: slope-information effective radii by product (arc-brightness $\times$ $(S/N)^2$ weighting; three statistics agree). The required ordering (γ: $1.816 > 1.433 > 1.103$) would need r_{eff} monotone in the same sense; observed is `v3` > `v3b` > `v2d` — the middle-γ product has the *smallest* r_{eff}. Source: `data/x1_g0_effective_radii.json`.

product (likelihood)	γ	r_{eff} mean	log-mean	median
`v3` fine $0.04''$ (correlated)	1.816 ± 0.117	$2.596''$	$2.578''$	$2.613''$
`v2d` native $0.13''$ (diagonal, anchor)	1.433 ± 0.035	$2.501''$	$2.466''$	$2.543''$
`v3b` binned $0.08''$ (correlated)	1.103 ± 0.008	$2.588''$	$2.571''$	$2.612''$

The **fixed-radius magnitude argument** is stronger than the ordering kill. The fine and binned products are the same sky, and their slope information peaks at the same radius: $\Delta r_{\text{eff}} \approx 0.008''$

[26]Source artifact: `data/x1_g0_effective_radii.json; research/x1_g0_mechanism_check.md`.

($r_{\rm mean}$ 2.5964″ vs. 2.5882″), less than a quarter of a fine pixel — yet their slopes differ by $\Delta\gamma = 0.71$. For any curved $\kappa(r)$ to carry that gap across that lever arm would require a local-slope gradient $|d\gamma_{\rm loc}/d\ln r| \approx 226$ per e-fold; physical profile curvature (NFW at r_s, dPIE break) delivers $O(1)$, and even the most charitable product pairing needs ≈ 10 (Fig. 7). The positive content of the kill: *the bracket driver differs between likelihoods at fixed radial information* — it must live in the noise/likelihood treatment or in source/PSF structure, not in the radial mass profile. What the gate does *not* kill: a BPL/dPIE fit could still win on evidence for other reasons; the pre-registered *mechanism* is what is excluded.

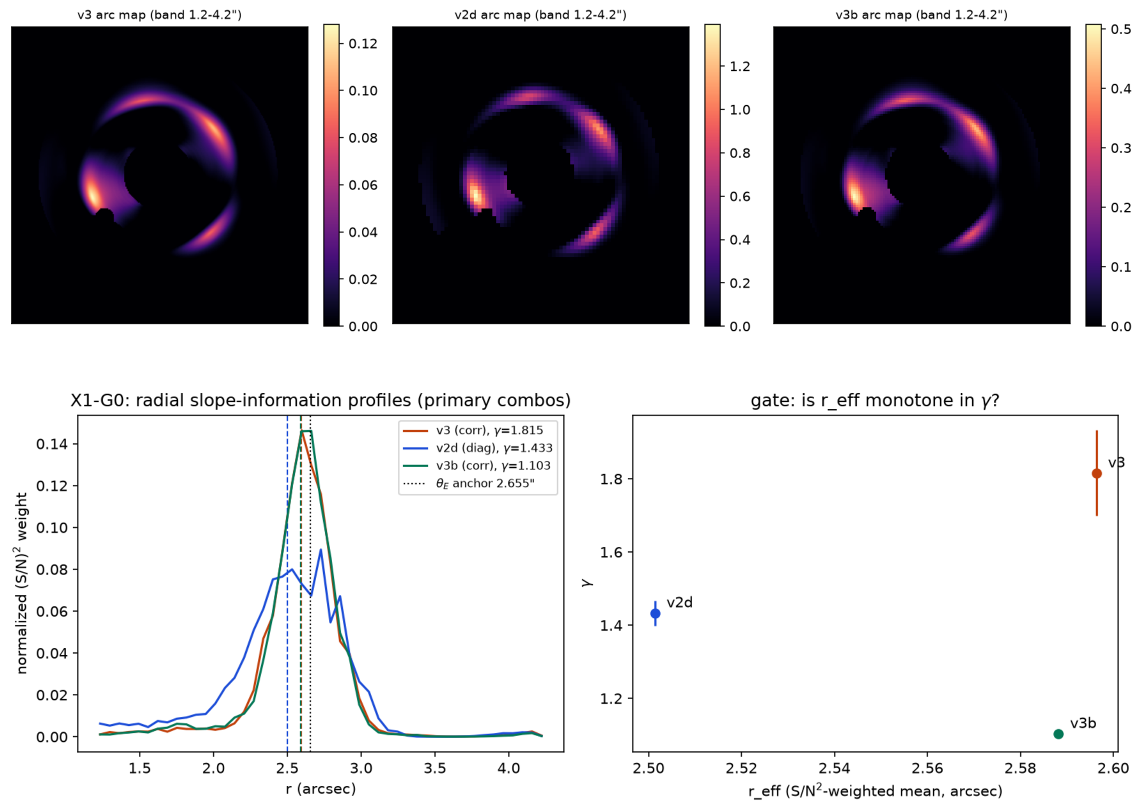

Figure 7: X1-G0 entry-gate kill. Top: arc-brightness $\times$ $(S/N)^2$ slope-information weight maps for the three products. Bottom: radial $(S/N)^2$ slope-information profiles (left) and the γ-vs-$r_{\rm eff}$ gate panel (right): no monotone ordering exists (24/24 variants), and the fine/binned pair constrains the slope at the same radius ($\Delta r_{\rm eff} \approx 0.008''$) while disagreeing by $\Delta\gamma = 0.71$. Source: `data/x1_g0_effective_radii.json`.

7.3 T0.3: the companion galaxy is exonerated

The companion galaxy (the LL2/LL3 light components, a known local misfit) was a candidate driver of the low slope. The pre-registered discriminator whitens first and then drops the companion's whitened pixels (keeping the production whitener fixed; 9273 $\to$ 8247 whitened dof), so the test isolates the companion's likelihood contribution without rebuilding the noise model. Result: $\gamma = 1.1011$ vs. production 1.1032 — a shift of -0.0021 ($0.26\,\sigma_{\rm stat}$), statistically zero against the pre-registered detection band of $\geq +0.024$, with convergence diagnostics indistinguishable from production.[27] **The companion is exonerated**; the noise-model class keeps its place as prime

[27] Source artifact: `data/t02_t03_gate_eval.json (t03)`.

suspect. (The companion run's $\log Z$ is not comparable to production — different data after the drop — and is not used.)

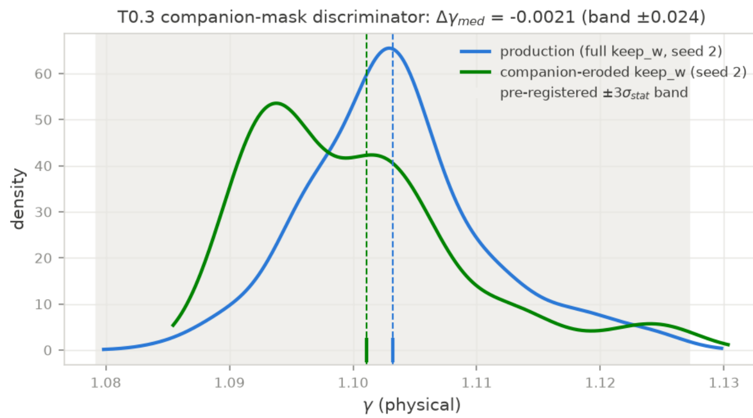

Figure 8: T0.3 companion-mask discriminator: the γ posterior under the production whitener with and without the companion's whitened pixels ($9273 \to 8247$ dof). The shift is -0.0021 ($0.26\,\sigma_{\mathrm{stat}}$) against a pre-registered detection band of $\geq +0.024$: the companion is exonerated. Source: `data/t02_t03_gate_eval.json`.

7.4 The T0.4 trilogy: the noise-model class is the standing suspect

Three zero-GPU checks on the real field, run on the validated stack with pre-registered forks, converge on one mechanism.

7.4.1 Stationarity of the noise-model class: REJECTED

The correlated likelihood behind $\gamma = 1.103$ prices noise with a *single stationary convolution kernel per product*. T0.4-1 tests that class directly: per-block kernel fits on a 3×3 grid over the field, with null bands calibrated by 200 exact FFT resamples of stationary fields under the global fitted kernel (mask geometry and finite-block scatter included), and a multiplicity-honest calibrated p from the same $\max|z|$ statistic on every null draw.[28] The verdict (Fig. 9):

- The money product `v3b` rejects the stationary class at $\max|z| = 3.67$, **calibrated** $p = 0.010$; the arc-excluded `v3` robustness arm also rejects at $p = 0.010$ ($\max|z| = 3.66$) — excluding the arc band *strengthens* the fine-product rejection (arc residuals were diluting the signal, not driving it).
- The rejection is a coherent, cross-product-replicated spatial pattern — top row of blocks over-correlated ($\rho(0,1)$ up to 0.71 vs. global 0.60), middle-left block under-correlated (z down to -3.7) — the same sign pattern in `v3` and `v3b`, with and without the arc band. That is what a real shift-variant noise field predicts and a fit fluke does not.
- Drizzle sub-pixel registration cannot explain it: the enumerated registration envelope predicts a block-to-block $\rho(0,1)$ spread of ± 0.008; the observed spread is ~ 0.28 peak-to-peak on `v3b` — $16.4\times$ **the envelope**. The nonstationarity must come from something else that varies across the skycell (coverage/weight-map structure and/or residual astrophysical background).

[28]Source artifact: `data/t04_stationarity.json`; `data/t04_stationarity_noarc.json`.

The noise-model class that produced $\gamma = 1.103$ is provably violated by the field it whitens. Honest caveats: the four test arms share pixels and are not independent — the case is the replicated pattern, not any single p-value.

7.4.2 Spectrum flooring: FALSIFIED as mechanism and as cure

A proposed benign reading of the low slope was information discard at spectral zeros: the near-singular fine whitener amplifies low-power modes, and flooring the whitener spectrum (raising $s_{\text{floor}} = \lambda$) should then cure the known "fine-low gaming" pathology, in which optimization drives $\gamma \to 1.02$ while whitened log-probability improves and real-space χ^2 collapses. T0.4-2 rebuilt the whitener across $\lambda \in \{0.05\ (\equiv \text{production, unfloored}), 0.1, 0.3, 1.0\}$ — the production build is reproduced to $\max|\Delta h| = 1.0 \times 10^{-9}$ in the taps, and the production floor is $s_{\text{floor}} = 0.05$, at which the floor does not engage (the plan's "0.1" label is corrected in the artifact) — and scored three frozen points (the v3b-low start A, the railed fine-low MAP B, the fine-steep MAP C) under exact cross-λ normalization ($\log|C(\lambda)|$ via Szegő, dense-Cholesky anchored to $\leq 7 \times 10^{-4}$ nat/px).[29]

Two results (Fig. 10):

- **Flooring never re-aligns the orderings.** The whitened-vs-real-space inversion (B beats A by ∼4200 whitened nats at production while real space prefers A by $\Delta\chi^2_{\text{pp}} = 37$) persists essentially unchanged from the unfloored whitener through $\lambda = 1.0$ (85% of modes floored): still +3832 nats. If the pathology lived at the spectral zeros, flooring them would have fixed it.
- **The data reject the floored kernels outright**: the normalized likelihood drops by ∼3.2k nats per point at $\lambda = 0.1$ ($-3187/-3328/-3304$ for A/B/C) and by ∼33k nats at $\lambda = 1.0$ ($-33{,}060/-33{,}450/-33{,}370$). "Cure by flooring" costs thousands of nats of evidence per step — more than any plausible model could repay.

The pathology is localized instead in what flooring never touches: the *down-weighting of high-power large-scale modes* ($S_{\max}/\text{mean} \approx 99$ — the fitted correlated-background component, $w_b \approx 0.27$), which prices large-scale real-space residual structure at ∼1/10 amplitude weight. Those are exactly the modes T0.4-1 shows are nonstationary: the two checks indict the same component.

7.4.3 Real space vs. the whitened metric: the orderings invert

T0.4-3 (report-only by design, a free ordering check rather than a calibrated test) asks which solution the binned pixels themselves prefer in plain real space. Forward renders of three on-disk posterior medians on the *same* `v3b` pixels, each given its ridge-solved (real-space optimal) source amplitudes:[30]

Table 8: T0.4-3 real-space head-to-head on the same `v3b` binned pixels (diag-solved source amplitudes; 16,653 keep px full field, 7,825 px arc band $r \in [1.2, 4.2]''$). The production whitened metric inverts the real-space ordering. Source: `data/t04_realspace_headtohead.json`.

model (γ)	χ^2_{pp} full	χ^2_{pp} arc band	whitened $\log L_{\text{data}}$	whitened log-posterior
anchor 1.433 (from `v2d`)	7.44	3.62	−5442.1	−5615.7
corr-low 1.103 (money)	8.32	2.76	**−4599.2**	**−4683.6**
diag-low 1.293 (home reference)	**1.58**	**1.93**	−4662.3	−5184.3

[29] Source artifact: `data/t04_lambda_arm.json`.

[30] Source artifact: `data/t04_realspace_headtohead.json`.

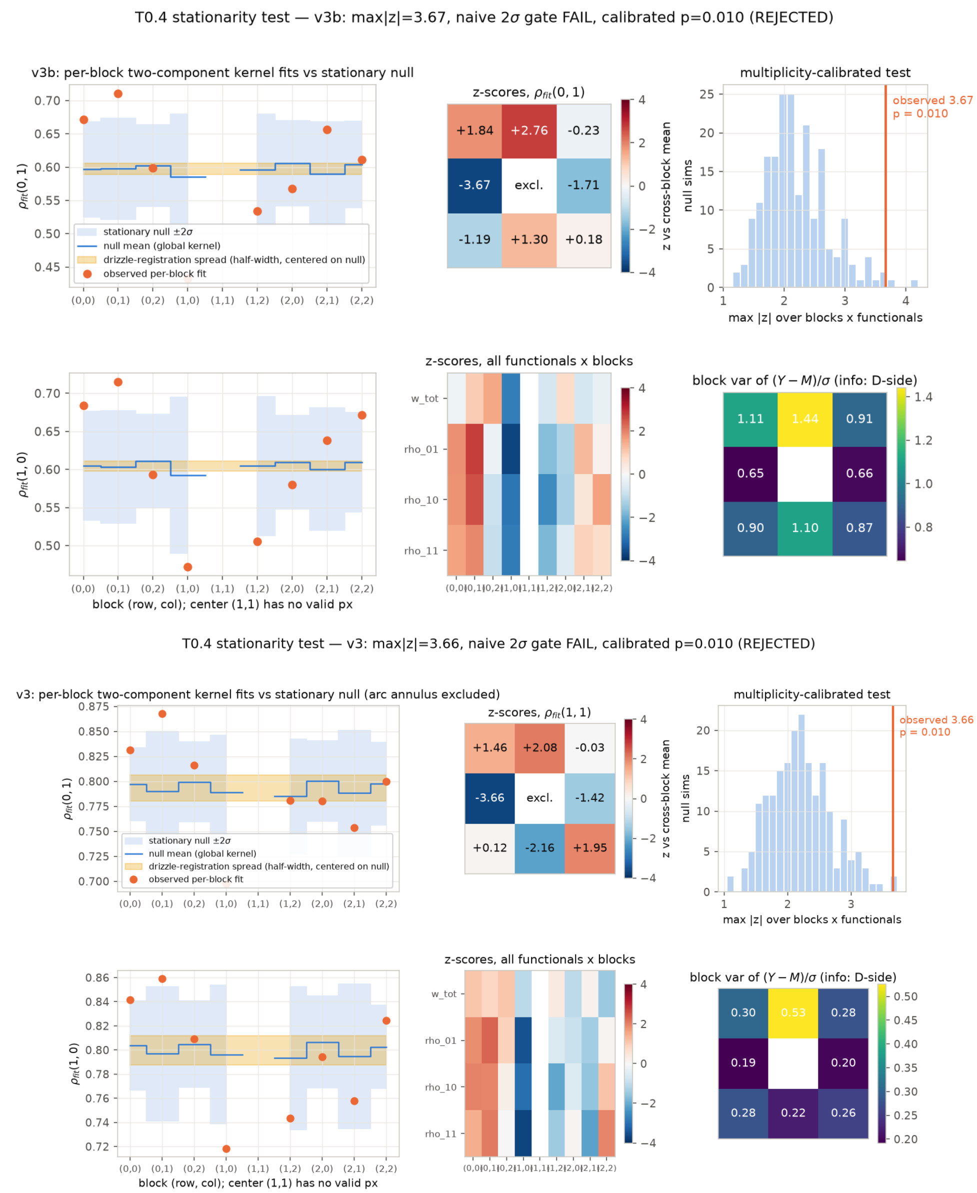

Figure 9: T0.4-1 stationarity rejection. Top: the money product `v3b` — per-block kernel functionals vs. null-calibrated bands ($\max|z| = 3.67$, calibrated $p = 0.010$). Bottom: the arc-excluded fine product `v3` ($\max|z| = 3.66$, $p = 0.010$) — excluding the arc band *strengthens* the fine-product rejection (with the arc included, `v3` gives $p = 0.17$). The spatial pattern (top row over-correlated, middle-left under-correlated) replicates across products and pixel-set arms; its amplitude is 16.4× the drizzle-registration envelope. Sources: `data/t04_stationarity.json`; `data/t04_stationarity_noarc.json`.

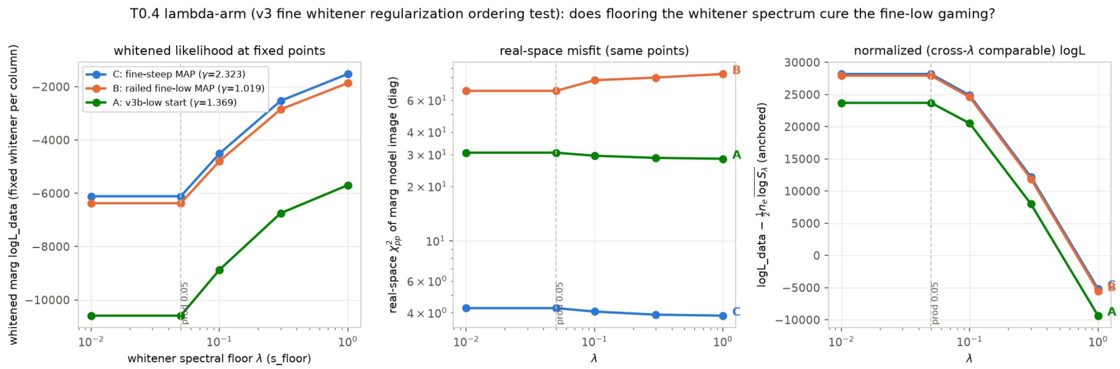

Figure 10: T0.4-2 λ-arm: whitened and normalized likelihoods of the three frozen points vs. spectrum floor λ. The gaming inversion is flat in λ (flooring does not cure it) while the normalized likelihood rejects the floored kernels by 3.2k–33k nats per point. Source: `data/t04_lambda_arm.json`.

In real space the binned data's preference is the home-product diagonal-low solution at $\gamma \approx 1.29$ ($\chi^2_{\rm pp}$ 1.58 full / 1.93 arc band); the 1.103 corr-low model is real-space poor (8.32 / 2.76), its residual dominated by smooth lens-center misfit — the same currency as the fine-low gaming. **The production whitened metric inverts this ordering**, preferring corr-low over diag-low by +63 nats in $\log L_{\rm data}$ and +501 **nats in log-posterior**. The 1.103 solution is a whitened-metric optimum, not a real-space one, *on its own product*. Honest caveat on the anchor row: its full-field number carries a cross-product resolution handicap (parameters fit at $0.13''$ rendered at $0.08''$; the same point renders $\chi^2_{\rm pp} = 1.25$ on its home product), so the defensible summary is that *neither* extreme-γ model is the binned data's real-space preference — the $\gamma \approx 1.29$ diagonal solution is (Fig. 11).

Mechanism synthesis (working hypothesis, not a conviction). One mechanism spans all of T0.4 and the X1-G0 constraint: the fitted stationary kernels contain a large correlated-background component ($w_b \approx 0.25$–0.27) estimated from sky that is provably *not* stationary across the field; the whitener therefore prices large-scale real-space misfit at $\sim 1/10$ weight everywhere, including regions where that pricing is wrong; solutions that spend exactly this currency (fine-low gaming, the 1.103 optimum) become whitened-optimal while real space disagrees — a coherent route to a low-biased γ at fixed radial information. Noise-model-*class* misspecification is thereby the prime suspect for the 1.103 over-correction; a locally-stationary (per-region kernel) class is the indicated next likelihood iteration, and the ported `CorrelatedImageData` deliberately keeps its whitener seam pluggable for it. Standing caveat: the kernel-hyperparameter scan (E3) was deferred by plan and never funded, and the alternative class was never fitted — the mechanism rests on a class-level rejection plus consistent circumstantial evidence, and the confirmatory injection test came back confounded, as the next subsection reports.

7.5 T1.1: injection–recovery is confounded — an honest negative

T1.1 was the mechanism's confirmatory experiment (7.80 A100-h; jobs 55952480/55952481/55958518 + diagonal control 55952483): inject arcs of known slope $\gamma_{\rm truth} = 1.4330$ into the *real* `v3b` drizzle residual field at three sub-pixel shifts, refit with the production correlated pipeline, and read the recovered slope against pre-registered zones — CONFIRM-low if median$(\gamma_{\rm rec} - 1.433) < -0.078$; EXONERATE if $|\text{median}| < 0.026$; a diagonal-likelihood control on the same data predicted in

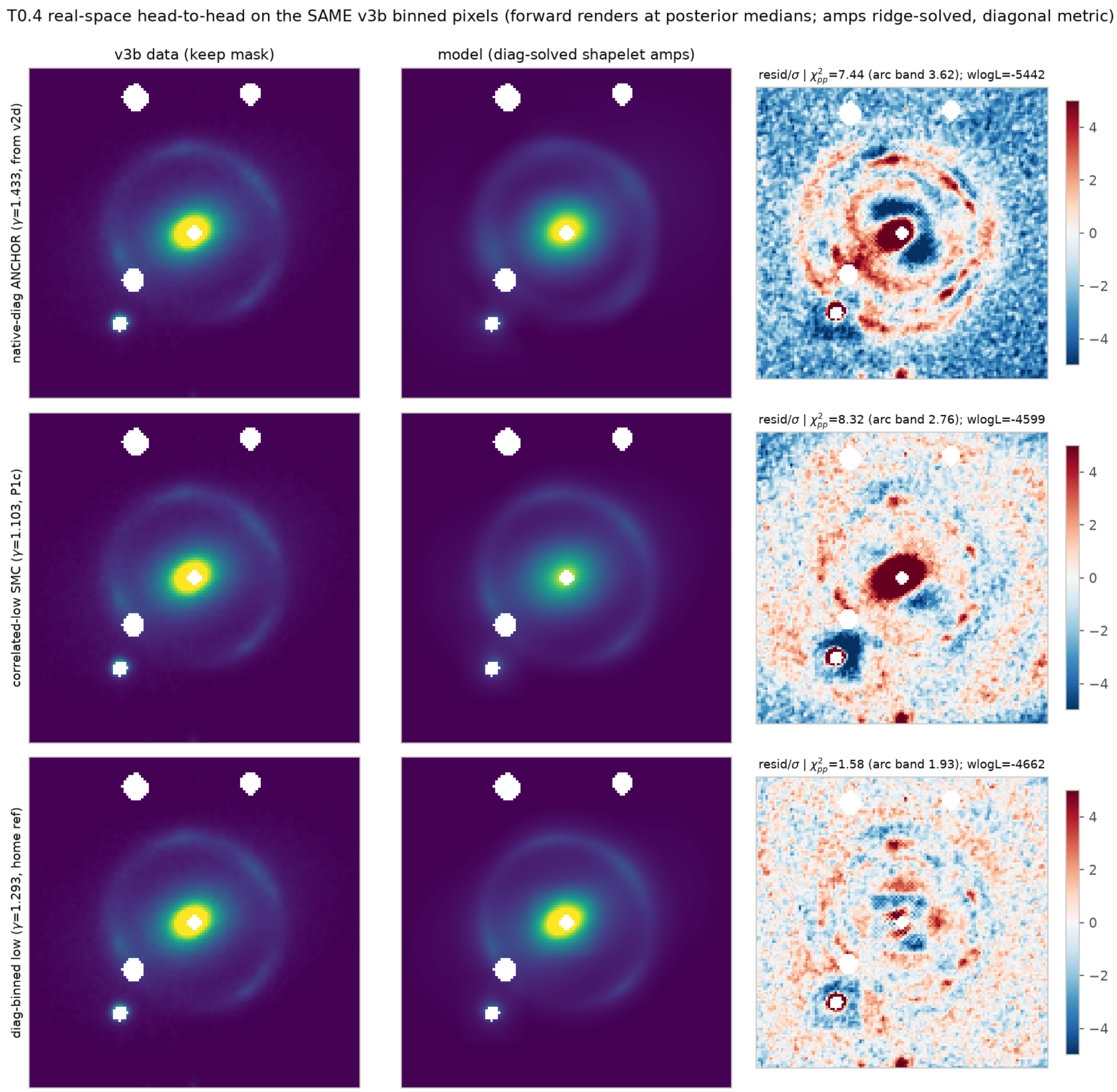

Figure 11: T0.4-3 residual maps on the same `v3b` pixels for the three posterior medians. The corr-low ($\gamma = 1.103$) render leaves strong coherent lens-center residuals that the whitened metric prices as "probably correlated noise"; real space prefers the $\gamma \approx 1.29$ diagonal solution. Source: `data/t04_realspace_headtohead.json`.

$[1.29, 1.43]$.[31]

The result lands outside both zones, on the positive side (Fig. 12): recovered medians 1.5151/1.5719/1.5076, biases $+0.0822/+0.1389/+0.0747$ (own-σ z $+2.25/+3.23/+1.82$); median bias $+0.0822$ ($+0.0784$ dropping the duplicate-flagged inj2 — same conclusion). The mechanism's directional prediction failed at face value. **And the control failed high and sick**: $\gamma_{\rm rec} = 1.5677 \notin [1.29, 1.43]$, with total resample collapse (1 unique particle of 128, $\sigma_\gamma = 4 \times 10^{-16}$) and the injected source's amplitude railed at $\approx 58\times$ truth. Per the pre-registered honesty clause, a control failing its own window implicates the *injection construction*, not the whitener: both likelihood classes on the same injected field recover γ high by $+0.07$–0.14, the only shared element is the data, and the real residual field carries the real lens's imperfect bright-object scene subtraction. The nuisance readout shows exactly the absorption signature — recovered sources $\times 1.8$–3 larger and $\times 2.4$ brighter than injected truth in all three correlated runs. $n = 3$; no coverage claims.

Two salvage data survive, quoted at their honest strength:

- **The whitener-isolating differential**: on identical data (inj1), $\gamma(\text{corr}) - \gamma(\text{diag}) = -0.0526$ — sign consistent with the mechanism, magnitude $\approx 16\%$ of the real-data 0.330 gap, and *no defensible error bar* (the diagonal leg is a degenerate point mass).
- **What T1.1 could not test by construction**: the injected scene is exactly in-class (truth source = fitted shapelet ridge solve), so the mechanism's lever arm — discounted large-scale structural *misfit* — is largely absent by design. Even a clean exonerate here would not have refuted T0.4-1's direct stationarity rejection, which stands unrefuted.

Verdict: NO-CONFIRM / NO-EXONERATE. The mechanism diagnosis continues to rest on T0.4's direct evidence; T1.1 neither strengthens nor weakens it decisively.

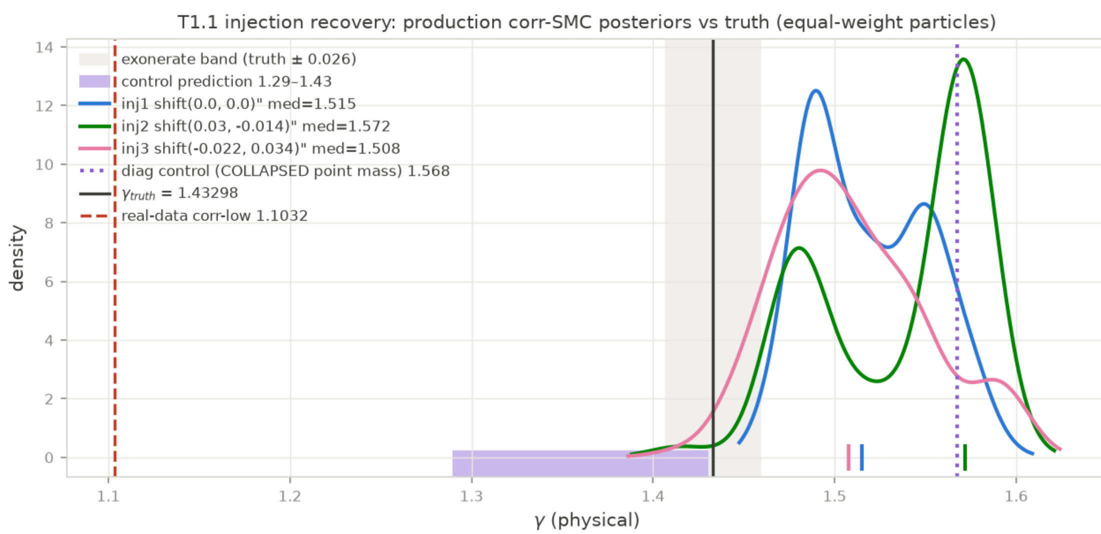

Figure 12: T1.1 injection–recovery: recovered γ posteriors for the three correlated injections and the diagonal control vs. the injected truth 1.4330 and the pre-registered zones. All arms land high (scene residue absorption); the control is both out-of-window and degenerate. Source: `data/t11_gate_eval.json`.

[31] Source artifact: `data/t11_gate_eval.json; research/t11_injection_recovery.md`.

7.6 T1.1b: the fit-side rescue fails structurally — injections must be rebuilt data-side

The obvious rescue — mask the scene-subtraction residue on the fit side (whiten with the production kernel, then drop residue-dominated whitened pixels) — was given a pre-declared entry gate before any GPU spend: the kept-dof loss had to stay $\leq 40\%$. **It failed structurally, and T1.1b stopped at zero cost**: whiten-then-drop loses 85.8% of the whitened dof ($9273 \to 1320$; the region drop alone loses 47.6%), and the arc band — where the slope information lives — retains 108 of 6373 whitened pixels, all outer sky.[32] The 21×21 whitening kernel support smears the excluded regions across the field: fit-side residue masking is structurally incompatible with this injection design, not merely expensive.

The redesign conclusion is data-side: residue-free injection substrates must be built before whitener-bias numbers are quotable — kernel-sampled noise-only fields, sky-set bootstrap, or a deeper multi-start scene subtraction. This was queued as the follow-on experiment and remains unfunded at program close; it is the sharpest single item on the handoff list, because it is what stands between "indicted" and "convicted" for the noise-model mechanism.

7.7 L0-G2: $\gamma = 1.103$ reproduces across stacks and machines

The final mechanism-program result is a positive control on everything above (0.53 A100-h; job 55985451, 31 min). The ported correlated likelihood (§6.4) refit the `v3b` low and steep basins on the scene-API substrate — an independent implementation stack, on a different machine, mirroring the production recipe at seed 2. Against the pre-registered gate ($|\gamma - 1.1032| \leq 0.017$ and $\Delta \log Z(\text{steep} - \text{low}) < 0$), the refit **passes**:[33]

- $\gamma_{\text{low}} = 1.1005$ $[1.0992, 1.1065]$, i.e. $\Delta = 0.0027$ from the certified 1.1032 — well inside both the gate and σ_{seed};
- $\log Z_{\text{low}} = -4770.97$ vs. the old-stack production -4771.08: **0.11 nats of agreement across stacks *and* machines**;
- $\Delta \log Z(\text{steep} - \text{low}) = -29.36$, in sign and magnitude inside the old-stack seed family $(-28.9/-32.3/-31.2)$.

The `CorrelatedImageData` port is thereby licensed at posterior level, and $\gamma = 1.103$ now stands on two independent stacks — the over-correction of §7.4 is a property of the *likelihood class*, not of either implementation. Scope notes (UNCERTIFIED—external; proposed to the upstream team as CGL2-C5): the same frozen whitener bundle feeds both stacks, so this is a stack-implementation cross-check, not an independent noise model; single seed on the new stack, with the old-stack σ_{seed} ($n = 3$, $\pm 46\%$) as context.

8 Results III — Fermat-Potential Sensitivity: a Triple-Disclaimed Teaser

Read the disclaimers first; they are the load-bearing part of this section. (1) DESI–165.4754−06.0423 is **not a time-delay lens** — no source variability, placeholder redshifts; nothing here is a Δt prediction for a real system. (2) The image positions are **synthetic**: four fixed points on a

[32]Source artifact: `data/t11b_residue_mask_report.json`.

[33]Source artifact: `data/results-perlmutter/l0g2_v3b_scene_seed2.json`.

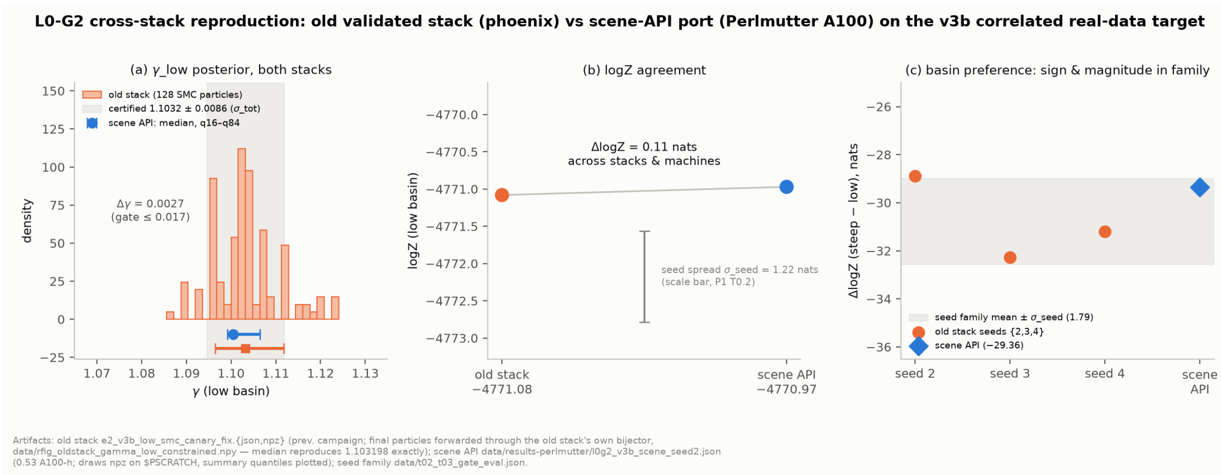

Figure 13: L0-G2 cross-stack reproduction of the correlated `v3b` refit. (a) $\gamma_{\rm low}$: the previous campaign's production SMC ensemble (128 final particles, phoenix, old validated stack; particles forwarded through that stack's own bijector — the median reproduces the ledgered 1.103198 exactly) against the scene-API port's summary quantiles (Perlmutter A100, seed 2): $\Delta\gamma(\text{median}) = 0.0027$, inside both the pre-registered 0.017 gate and the certified 1.1032 ± 0.0086 band. (b) $\log Z$ agreement to 0.11 nats across stacks *and* machines (-4771.08 vs. -4770.97), with the old-stack seed spread of $\log Z_{\rm low}$ ($\sigma_{\rm seed} = 1.22$ nats) as the honest scale bar (per-run bootstrap σ is not recorded in these summary artifacts). (c) $\Delta \log Z(\text{steep} - \text{low}) = -29.36$ on the scene API, inside the old-stack seed family ($-28.88/-32.27/-31.19$). The scene side is plotted from summary quantiles (the posterior-draws NPZ resides on Perlmutter scratch). Sources: `data/results-perlmutter/l0g2_v3b_scene_seed2.json`; `data/rfig_oldstack_gamma_low_constrained.npy`; `data/t02_t03_gate_eval.json`.

$\theta_{\rm E} = 2.655''$ circle about the anchor mass center, the same points for every posterior. (3) The correlated binned-low posterior is the product **known to over-correct** ($\gamma = 1.103$, 17σ below the 1.433 anchor; §7) — so this teaser demonstrates *sensitivity* of Fermat observables to the coadd-covariance modeling choice; it does *not* calibrate which answer is right. Additionally, the primary comparison pair differs by data product (native vs. binned), not only by noise model; the same-product secondary arm isolates the noise model.

With that stated, the number (0 GPU-h; per-sample EPL+shear Fermat potentials through a numpy port of the EPL deflection — Tessore–Metcalf angular series, validated against the vendored jax implementation to max $|\Delta| = 1.3\times10^{-15}$; fractional shifts are distance/redshift-free):[34] swapping the noise model moves the median Fermat-potential differences $\Delta\phi$ between image pairs by

$$\textbf{88\%}\ (\sim 10.7\sigma)\ \ \text{anchor} \to \text{corr}, \qquad \textbf{61\%}\ (\sim 17\sigma)\ \ \text{diag} \to \text{corr, same binned product},$$

against each posterior's own $\Delta\phi$ width of 4–8% — versus the ~1% scale at which such shifts become TDCOSMO-relevant. Representative opposite-pair values (arcsec^2): diagonal-native -0.344 ± 0.028; diagonal-binned-low -0.107 ± 0.004; correlated-binned-low -0.040 ± 0.004. The physical reading is simple: $\Delta\phi$ between images tracks $(\gamma - 1)$, so the correlated likelihood's slope deflation $1.433/1.293 \to 1.103$ propagates near one-to-one into a ~4–9× collapse of $\Delta\phi$ (Fig. 14).

The takeaway is deliberately modest: on a real drizzled *HST* product, the noise-covariance modeling choice moves Fermat observables by tens of percent while typical time-delay error budgets carry no line item for coadd/drizzle noise covariance. The teaser's only claim is that such a line

[34] Source artifact: `data/fermat_dt_teaser.json; research/fermat_dt_teaser.md`.

item is worth pricing on an actual time-delay lens — with a likelihood class that survives §7.4.1's stationarity test, which the present correlated class does not.

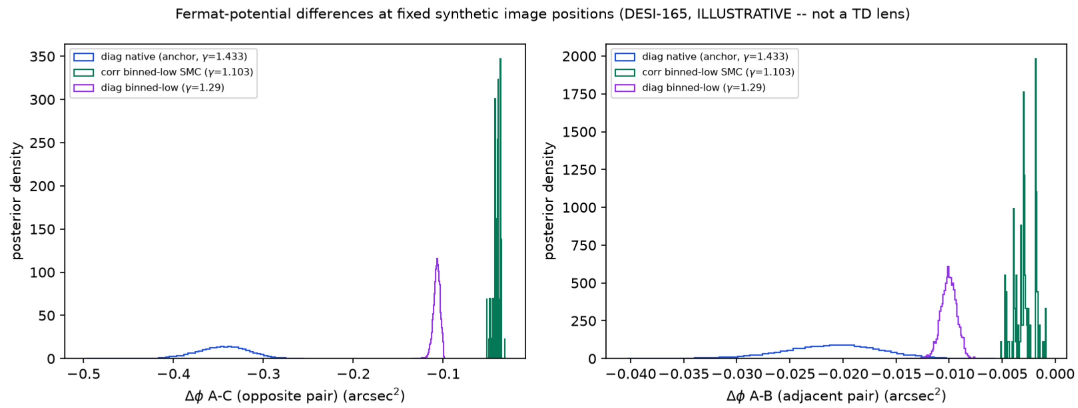

Figure 14: Fermat-potential difference posteriors for an opposite and an adjacent synthetic image pair under the three noise-model posteriors (diagonal-native anchor, diagonal-binned-low, correlated-binned-low). Illustrative only (see the triple disclaimer): the noise-model choice moves $\Delta\phi$ by 61–88% where ~1% is the TDCOSMO-relevant scale. Source: `data/fermat_dt_teaser.json`.

9 The Sampler Decision Matrix

The benchmark's deliverable is not a ranking but a *decision matrix*: for each target class that a lens modeler actually faces, which inference vehicle converges, which yields trustworthy evidence, and at what cost. Four vehicles were exercised under the frozen P0 protocol: **S1**, cold-start prior-seeded MAMS-SMC (tempered SMC with Metropolis-adjusted microcanonical mutations, adaptive λ-schedule, per-stage bootstrap $\log Z$); **warm MAMS**, a MAP-warm-started MAMS-alone arm (no tempering); **MCLMC-mutation SMC**, formally demoted to a diagnostic at the B0 correctness gate (below) and never used as an evidence kernel; and the **classic** MAP→SVI→HMC baseline recipe at frozen single-stage settings. Table 9 is the full matrix; the subsections that follow give each row's evidence and the two cross-cutting findings. Every cell is pre-registered (design checkpoint with derived thresholds logged before submission) and closed in the final gate ledger (§12). The per-fit convergence diagnostics behind every cell — one row for each fit the campaign ever ran, including the failed, rejected, and budget-blocked arms — are tabulated in Appendix A.

9.1 Reading the matrix: measured capability boundaries

Three boundaries are measured, not conjectured. First, **prior-seeded MAMS-SMC works — converges *and* yields evidence — on analytic targets, the DSPL ridge, the 74-d bimodal target (both basins, retention 1.000), and the correlated real-data v3b refit**; it is the only vehicle in the campaign that produced $\log Z$ at all. Second, **when it fails, it fails by cost, never by health**: every non-completion (B1 mocks, B1r real carousel, B3 at N=512, the three arbitration wall-caps) is a wall-clock cap on a sampler whose per-stage diagnostics were clean at the kill point — λ-progress per A100-hour is the binding constraint on scene-substrate real targets. Third, **the classic single-stage recipe produced zero accepted posteriors in this campaign** (B3 0/4, the arbitration arm's NO-VERDICT, and 30/32 unhealthy fits in the X2 SBC arm at

Table 9: The sampler decision matrix. Rows are target classes; columns are vehicles. "—" = not run (pre-registered scope, budget close, or prerequisite unmet; dispositions in `research/gate_ledger_final.md`). All A100 costs are actual ledgered hours. The carousel row uses real MUSE cutouts provided by the team, used with permission; comparison against the team's own runs is deferred pending sign-off. The DSPL and easy-scene target builders come from the upstream development branches (used with permission). Scene-substrate cells are proposed (UNCERTIFIED — external): reproduced/measured on the vendored scene API, with certification belonging to the receiving team.

target class	S1 prior-seeded MAMS-SMC	warm MAMS	MCLMC (demoted)	classic MAP→SVI→HMC	evidence yield
T0 analytic (2-d bimodal; funnel-10; illcond-46) — B0, 0 A100-h	mix2 **PASS** ($\log Z$ err 0.073); funnel10 **FAIL** -0.66 nat (class limitation, reproduced in the old validated stack); illcond46 **PASS** worst-z 1.83 but $\log Z$ err 2.40 nat = $6.6\sigma_{\text{boot}}$	— (an HMC-mutation SMC arm ran as comparison)	**DEMOTED**: $\Delta \log Z$ vs MAMS $+10.79 \pm 0.54$ = 20.1σ on illcond46	n/a	analytic refs; σ_{boot} provably understates $\log Z$ error (finding A)
easy scene 22-d unimodal ×4 — B3, free tier, 0 A100-h	**BLOCKED-BY-BUDGET** 0/4 inside the 5 h/arm L4 fence at N=512; the readable result is the cost row	—	—	4/4 complete but 0/4 pass the frozen acceptance rule ($\hat{R}_{\text{worst}}$ 4.36–7.87) ⇒ **BLOCKED-BY-REFERENCE**	none (no accepted posterior on either arm)
DSPL thin ridge 21-d — B2/B2′, 2.9 A100-h	λ=1 @64 stages, sanity clean; $\hat{m}$ 0/512 vs. imported band ⇒ falsifier control shows the band is realization-specific; B2′ **PASS-UNDERPOWERED**	—	—	—	$\Delta \log Z(\text{orig}-\text{ratio}) = +3.06 \pm 0.35_{\text{boot}}$ where true $\Delta \approx 0$ (finding A, measured)
T2 46-d cond-10^{14} — B4 + arbitration, 4.4 A100-h	healthy but wall-capped ×3 (A100 λ=0.587 @3.5 h; L4 λ=0.418 at a 12.3-h resume leg) ⇒ **budget-infeasible**; B4 itself not evaluable within budget	(the classic arm was warm-started)	—	**NO-VERDICT**: $\hat{R}_{\text{worst}}$ 44.3, all chains railed $\gamma \approx 1.99$ — the known T2 single-stage pathology, reproduced cross-stack	none; premise carried by likelihood parity + the L0-G2 row
T3 74-d bimodal — B5, 1.9 A100-h (+1.7 infra)	**G1 PASS both basins** (λ=1, retention 1.000); **G2 FAIL-as-written** 11.0σ — decomposition indicts our own frozen reference's within-basin coverage (finding B)	—	**G3 AMBIGUOUS**: minor-mode evidence inflated ×56; within-basin γ untouched	—	occupancy robust across all estimators; absolute $\log Z$ NOT robust at the ±5-nat gate scale
carousel 33-d multi-plane, real — B1r, 15.85 A100-h	λ=0.1506 @36 stages after 11.33 h, **0 posterior samples** (sampler healthy; killed by cost)	budget-matched S6br: wall fence in burn round 11/30, **0 draws**, ESS=0 at 41.5% of the grad budget	—	—	none — **NEITHER CONVERGES**; the cost row is the deliverable
v3b correlated, real, 46-d — L0-G2, 0.53 A100-h	**PASS**: γ Δ 0.0027, $\log Z$ Δ 0.11 nat cross-stack; $\Delta \log Z(\text{steep}-\text{low})$ in the seed family	—	—	—	best evidence row of the campaign; correlated port licensed

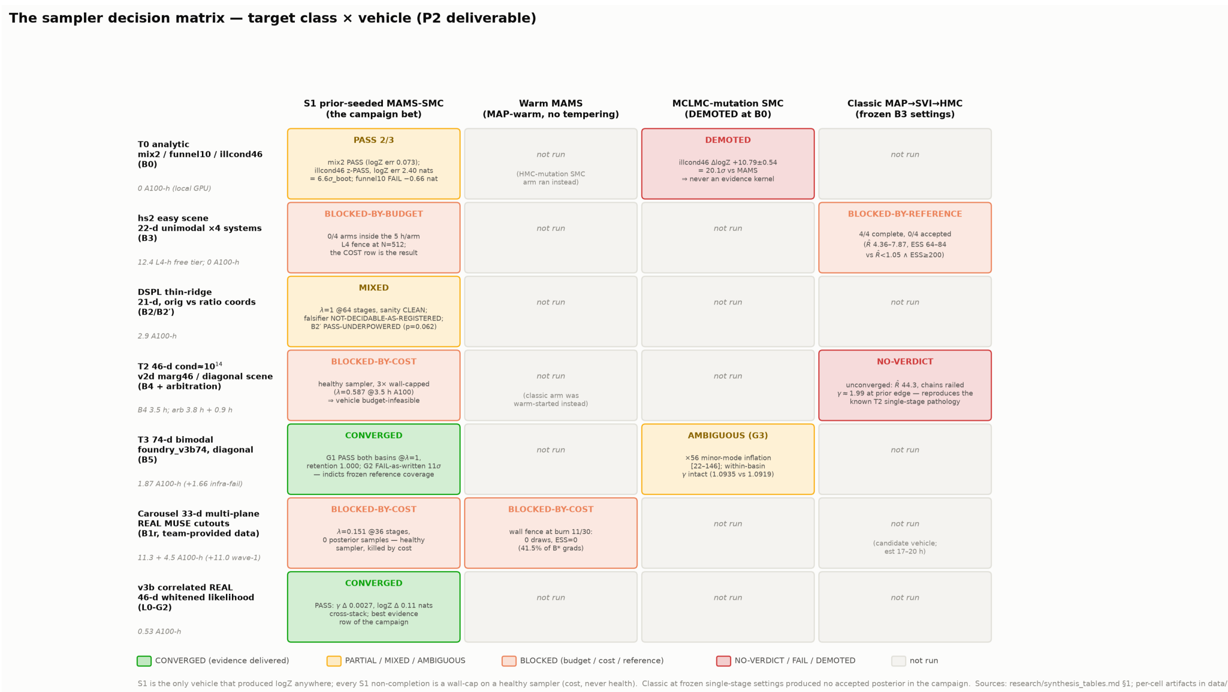

Figure 15: The sampler decision matrix as a capability grid (the graphical form of Table 9): 7 target classes $\times$ 4 vehicles. Cell status is the gate-ledger final disposition (green = converged with evidence delivered; yellow = partial/mixed/ambiguous; salmon = blocked by budget, cost, or reference; red = no-verdict/fail/demoted; gray = not run), with the discriminating measurement and the realized cost in each run cell. The honest reading is baked in: S1 prior-seeded MAMS-SMC is the only vehicle that produced $\log Z$ anywhere; every S1 non-completion is a wall-cap on a healthy sampler; the classic recipe at frozen settings produced no accepted posterior; MCLMC was demoted at B0 ($\Delta \log Z = +10.79 \pm 0.54 = 20.1\sigma$, artifact value) and stayed diagnostic-only. Sources: `research/synthesis_tables.md` §1; `research/gate_ledger_final.md`; per-cell artifacts as cited in Table 9 and the subsections.

reduced budget) — consistent with the prior campaign's finding that this posterior family needs the two-stage re-preconditioned recipe; its unique value here was diagnostic (§9.5).

9.2 Carousel (B1r): neither vehicle converges at campaign budget

The carousel cell is the campaign's designated hard real-data row: a 33-dim multi-plane model on real MUSE cutouts (data provided by the team, used with permission; any comparison against the team's own runs is deferred pending their sign-off). The cold-start arm S1r (N=128, three chained checkpointed legs) spent 11.33 A100-h to reach $\lambda = 0.1506$ at 36 stages — *zero* posterior samples, with the sampler demonstrably healthy at the fence (92–104 unique particles of 128, acceptance 0.871–0.934), so $\lambda = 1$ is infeasible under any campaign-scale budget.[35] The pre-registered close-out framing was written *before* the baseline ran, including the branch that would have scored against our own bet ("any nonzero baseline ESS $\Rightarrow$ decisive cold-start LOSS"). That branch did not fire: the warm MAMS-alone baseline S6br, budget-matched at S1r's realized 3,078,912 gradient evaluations, hit its 4.5-h wall fence in burn round 11 of 30 — **sampling never started** (0 draws, ESS = 0, $\hat{R}$ not computable) at 1,277,632 grads = 41.5% of the budget, with burn itself healthy (acceptance 0.873–0.902).[36] Per-gradient throughput agrees across arms to ∼5% (∼284k vs. 272k grads/h), so the truncation is a *target-cost* fact, not arm inefficiency. The row's honest verdict: **carousel-class real targets are out of reach for both vehicles at these budgets** (an until-converged warm arm is estimated at 17–20 h from the wave-1 anchors, untested). The descoped gates (2-seed self-consistency, $\log Z$ repeatability) remain descoped and are stated as such in the amendment of record.[37]

9.3 T3 bimodal (B5): both basins pass; the gate that fails indicts our own frozen reference

On the 74-d bimodal target, S1 delivers the multimodality certificate **G1** in full: the low basin reaches $\lambda = 1$ in 31 stages (retention 1.000, $\gamma_{\text{eqw}} = 1.0919$, $\log Z = 38376.33 \pm 0.33_{\text{boot}}$) and the steep basin in 22 stages ($\gamma_{\text{eqw}} = 2.5552$, $\log Z = 38518.23 \pm 0.31_{\text{boot}}$).[38]

G2 — basin $\Delta \log Z$ against the frozen per-basin evidence reference from *our own previous campaign* (P2c) — **fails as written**, by a lot: $\Delta(\text{steep} - \text{low}) = +141.90 \pm 0.45$ vs. the reference $+162.2 \pm 1.8$, a -20.30-nat miss against a 5.54-nat band (11.0× the gate σ-scale). We frame this carefully, because the honest decomposition cuts *toward the reference*: the per-basin offsets are +25.16 (low) and +4.86 (steep) — *both* basins carry *more* evidence than the reference, in the same direction — and both $\lambda = 1$ ensembles are γ-*disjoint* from the fixed-step HMC chains the reference was seeded and mutated from (low basin: ours $[1.071, 1.126]$ vs. chains $[1.265, 1.381]$), with two *independent* mutation kernels agreeing on the location (MAMS 1.0919, MCLMC 1.0935). The adaptive anneals reach higher-log-probability shelves that the reference's short fixed-step chains never visited. Comparability was verified line by line before scoring (same target constructor, identical evidence construction, the importance weights cancel; the reference's own on-disk smoke reproduces its $\Delta = 162.7$ under its kernel), so the discrepancy is real — and with n=1 seed on the measuring side, final attribution between reference and vehicle is explicitly undecidable within the closed budget. What the fail *does* establish is cross-cutting finding B (§9.7): frozen MCMC-derived

[35] Source artifact: `data/results-perlmutter/b1r128_carousel33_s1_seed2.PARTIAL.json`.
[36] Source artifact: `data/b1r_decision_matrix.json`.
[37] Source artifact: `research/checkpoint_b1r_close.md`.
[38] Source artifact: `data/b5_gate_eval.json`.

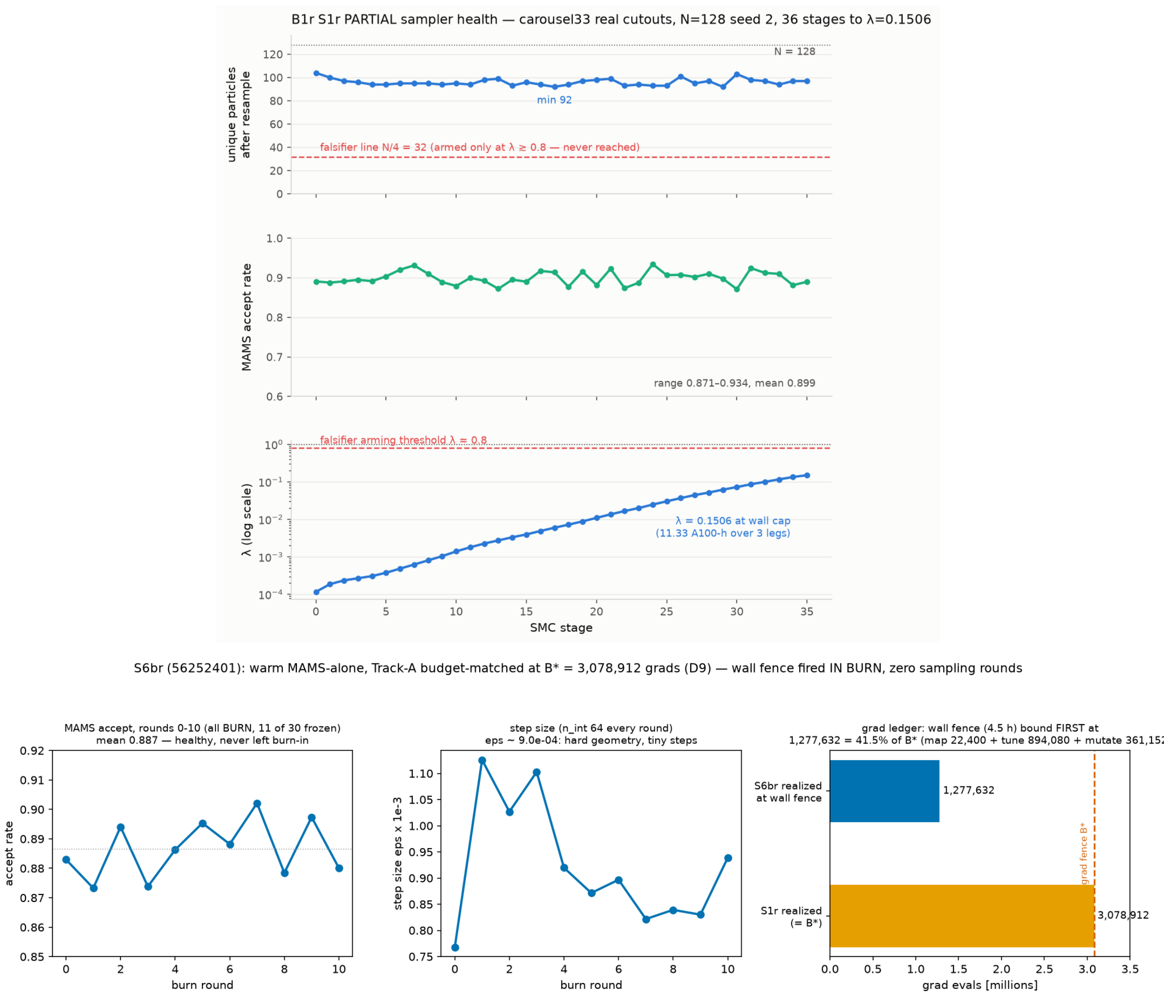

Figure 16: The carousel cell B1r (real MUSE cutouts provided by the team, used with permission; comparison with the team's own runs deferred pending sign-off); both panels plotted before gate text per house rule. Top: the S1r cold-start MC-SMC arm over three chained checkpointed legs to $\lambda = 0.1506$ at 36 stages (11.33 A100-h) — unique-particle counts 92–104 of 128 and acceptance 0.871–0.934 at the fence: a healthy sampler killed by cost. Bottom: the budget-matched warm-MAMS arm S6br (job 56252401) hits its 4.5-h wall fence in burn round 11 of 30 — sampling never started (ESS = 0 at 1,277,632 grads = 41.5% of the matched 3,078,912-gradient budget), with burn itself healthy. Sources: `data/results-perlmutter/b1r128_carousel33_s1_seed2.PARTIAL.json`; `data/b1r_decision_matrix.json`.

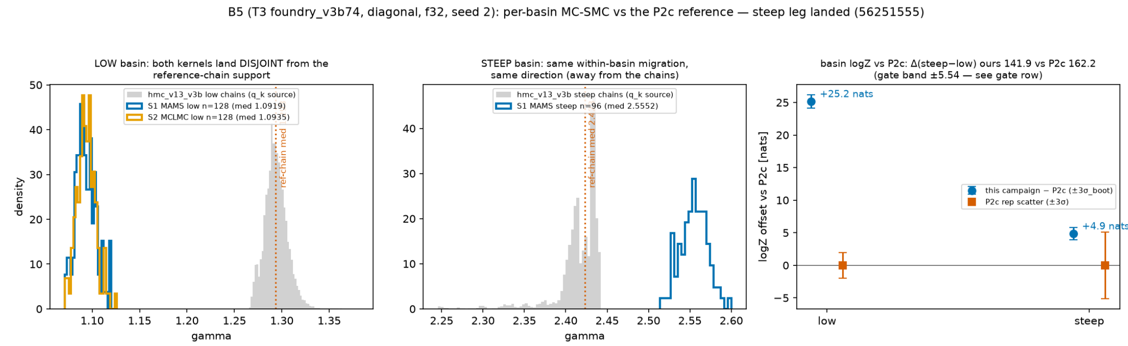

Figure 17: B5 both-basin $\lambda = 1$ posterior overlay on the 74-d bimodal target `foundry_v3b74`, drawn *before* any gate math per house rule (jobs 56006049 / 56251555 / 56006052). The low basin's two independent mutation kernels agree on location (MAMS $\gamma_{\text{eqw}} = 1.0919$, MCLMC 1.0935); the steep basin sits at 2.5552. Source: `data/b5_gate_eval.json`.

evidence references can carry within-basin coverage error far exceeding the gate scale. This is a statement about our own prior artifact, not about any external one.

G3, the pre-registered MCLMC-laundering probe, lands **ambiguous at the frozen thresholds** because its two frozen clauses conflict on the measured value: the diagnostic MCLMC arm inflates minor-mode evidence by $+4.03 \pm 0.49$ nats $= \times 56$ [$\text{CI}_{95,\text{boot}}$ 22–146] — $8.3\sigma_{\text{boot}}$ from zero and $6.0\sigma_{\text{boot}}$ above the pre-registered 1.5–3× ceiling, so SMC resampling does *not* launder the unadjusted-MCLMC bias at point estimate — yet the shift sits inside the imported 5.56-nat σ_{seed} repeatability band, so the laundering falsifier technically fires. Within-basin location is untouched (γ 1.0935 vs. 1.0919): the bias lands in the evidence, not the posterior. Either way the physical conclusion is robust across every estimator: the minor mode is negligible ($w_{\text{low}} \leq 1.3 \times 10^{-60}$ in all constructions), while *absolute* evidence calibration at the ±5-nat gate scale is not.

9.4 DSPL (B2/B2′): the realization-calibration lesson

The DSPL thin-ridge cell was scored twice, and the sequence is the lesson. As written, B2 asked S1 (in the original, deliberately pathological coordinates) to recover a pre-registered minor-arm mass band 0.103 ± 0.045, imported at design time. The measured $\hat{m} = 0/512$ fails the band — but the pre-registered falsifier control, an *exact* reparameterization of the same posterior on the same data (which cannot suffer coordinate mode-death), measures this realization's true arm mass at $5/512 = 0.0098 \pm 0.0043$, itself far outside the band.[39] The band does not transfer across data realizations; the gate was **not decidable as registered**, and the pre-registered branch mandated re-registration rather than silent re-scoring. The re-registered B2′ (rule and attainable-significance clause declared before scoring) compares the arms directly: two-sided Fisher $p = 0.0619 \geq \alpha = 0.01$ — **PASS-UNDERPOWERED (consistent)**, with the honest caveat that with only five minor-arm particles in total no outcome could have failed at α; minor-arm under-coverage up to $\sim \times 3$ is not excluded (the one-sided signal, $p = 0.031$, is $\sim 2\sigma$ and suggestive at most). The dominant arm is statistically identical across coordinates ($\Omega_{m,0}$ median 0.4701 vs. 0.4702). The headline stands at "no detected coordinate pathology" strength — and the cell's most valuable by-product is an evidence-error datum: $\Delta \log Z(\text{orig} - \text{ratio}) = +3.06 \pm 0.35_{\text{boot}}$ nats on identical data where the exact reparameterization forces the true $\Delta \approx 0$ (§9.7).

[39] Source artifact: `data/b2_gate_eval.json`.

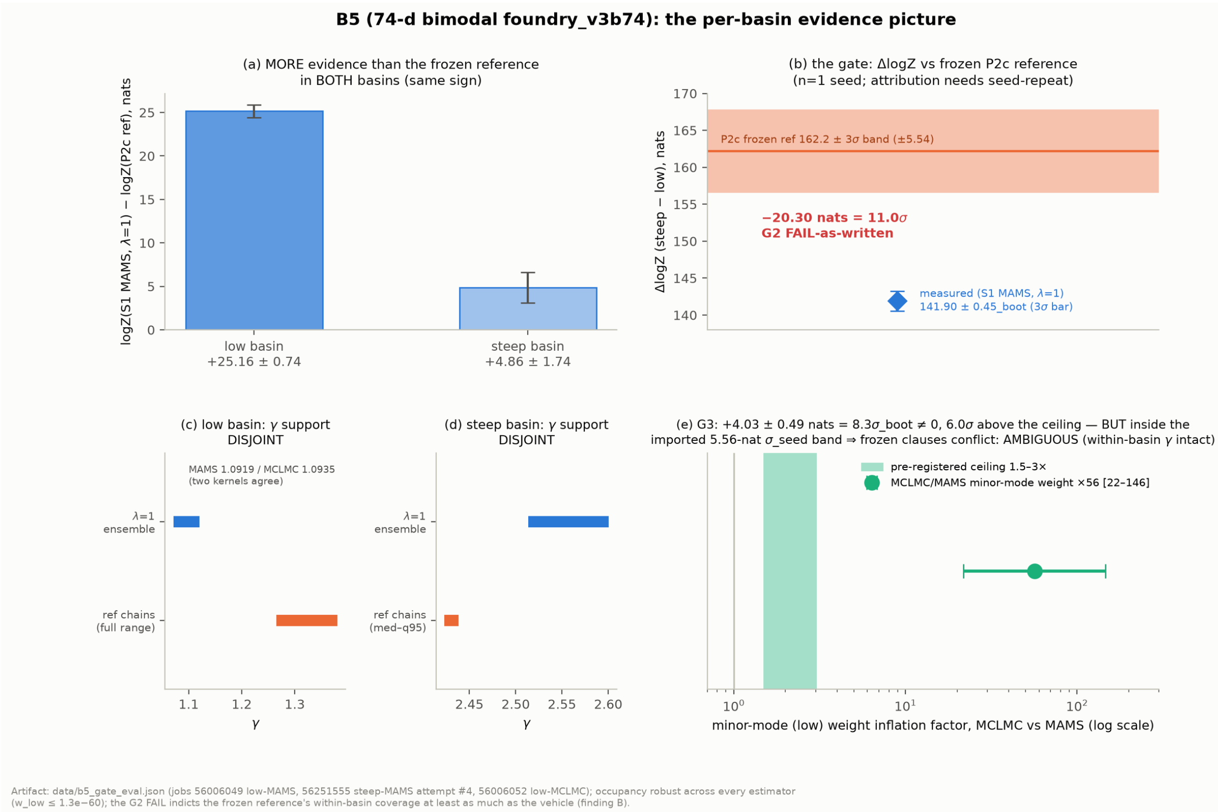

Figure 18: B5 per-basin evidence picture on the 74-d bimodal target. (a) The S1 $\lambda = 1$ ensembles find *more* evidence than the frozen P2c reference in *both* basins ($+25.16 \pm 0.74$ low, $+4.86 \pm 1.74$ steep — same sign). (b) The G2 gate as written therefore fails at 11.0σ (measured $\Delta = 141.90 \pm 0.45_{\text{boot}}$ vs. reference 162.2 ± 1.8, band ± 5.54 nats) — but the decomposition in (a) plus (c,d) indicts the frozen reference's within-basin coverage at least as much as the vehicle: both $\lambda = 1$ ensembles are γ-disjoint from the reference-chain support that the reference was seeded *and* mutated from, while two independent kernels (MAMS 1.0919, MCLMC 1.0935) agree on the low-basin location (cross-cutting finding B; n=1 seed, attribution undecidable within the closed budget). (e) The MCLMC diagnostic arm inflates minor-mode evidence $\times 56$ [$\text{CI}_{95,\text{boot}}$ 22–146] $= +4.03 \pm 0.49$ nats — $6.0\sigma_{\text{boot}}$ above the pre-registered 1.5–3$\times$ ceiling yet inside the imported 5.56-nat σ_{seed} repeatability band, so the two frozen G3 clauses conflict: AMBIGUOUS, and the B0 demotion stands. Minor-mode occupancy is robust across every estimator ($w_{\text{low}} \leq 1.3 \times 10^{-60}$). Source: `data/b5_gate_eval.json` (jobs 56006049, 56251555 attempt #4 chunk-32, 56006052; frozen P2c reference constants therein).

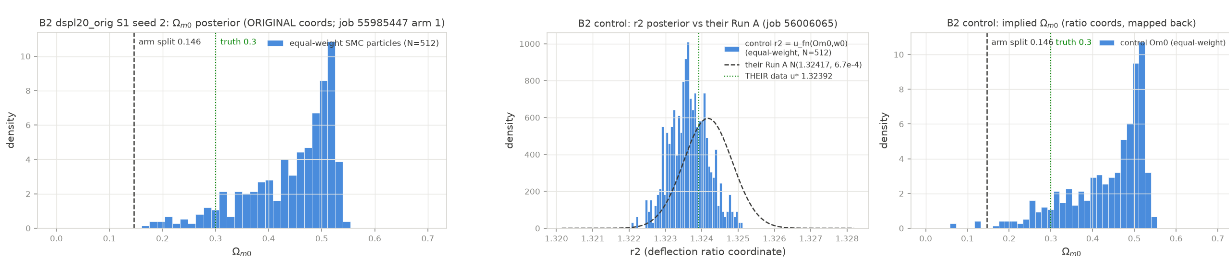

Figure 19: DSPL thin-ridge cell (B2/B2′). Left: the dominant arm is statistically identical across coordinates ($\Omega_{m,0}$ median 0.4701 vs. 0.4702). Right: the exact-reparameterization ratio-coordinates falsifier control that proved the imported minor-arm band realization-specific (this realization's true arm mass $5/512 = 0.0098 \pm 0.0043$ vs. the imported band 0.103 ± 0.045), forcing the pre-registered re-registration to B2′ (PASS-UNDERPOWERED). Source: `data/b2_gate_eval.json`.

9.5 The classic arm: a NO-VERDICT that reproduces the T2 pathology cross-stack

The anchor-arbitration classic arm — warm-started MAP→SVI→HMC at the frozen B3 reference settings on the parity-certified 46-d cond-10^{14} v2d target — is a **NO-VERDICT (unconverged)** under the pre-registered worst-parameter rule: $\hat{R}_{\mathrm{worst}} = 44.3$ (11.7 at the escalated stage, vs. the 1.05 gate), $\mathrm{ESS}_{\mathrm{min}} = 63$ at the escalated stage (53.6 at the initial 50-chain stage — both quoted, per the flagged-record rule), with every chain railed at the prior edge, $\gamma \approx 1.987$ $[1.983, 1.991]$ — so γ is not quotable.[40] The scientific content is what this reproduces: the same single-stage-HMC pathology this posterior exhibits on the old validated stack (single-stage $\hat{R} = 3.1$; only the two-stage re-preconditioned recipe converges), now demonstrated on the scene substrate at 0.9 A100-h. The difficulty is target-intrinsic, not stack-specific — exactly the premise-check the arbitration needed once every SMC vehicle had wall-capped, and the cheapest useful thing a non-converging run can prove.

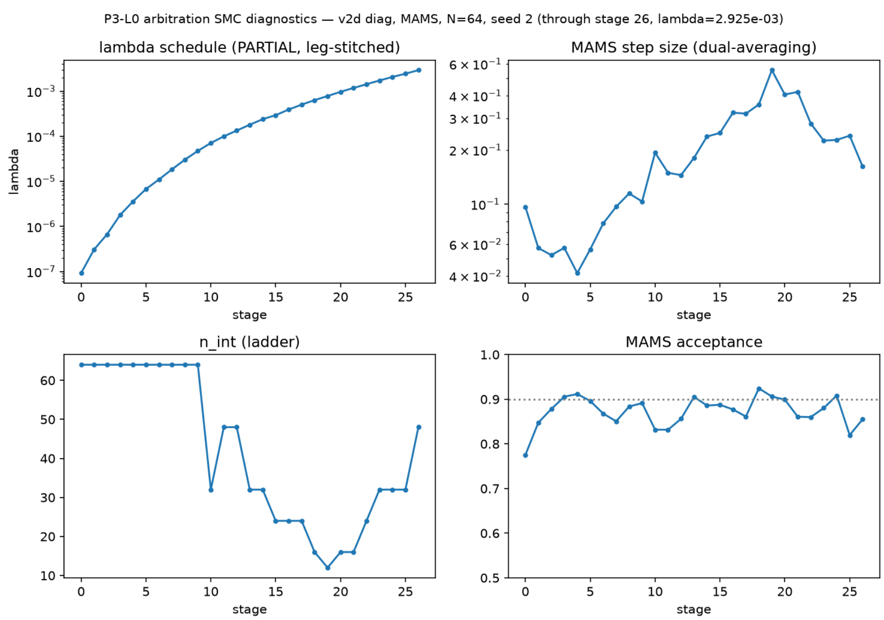

Figure 20: The arbitration MC-SMC arm on the 46-d cond-10^{14} `v2d` target (job 56168446): healthy per-stage diagnostics to $\lambda = 0.587$ at the 3.5-h A100 fence — budget-infeasible, not unhealthy. Together with the L4 resume leg ($\lambda = 0.4183$ at a 12.26-h leg, on-disk record) and the classic arm's NO-VERDICT, this is the vehicle-exhaustion evidence behind the anchor arbitration's PARTIAL-BY-VEHICLE-EXHAUSTION close. Disposition: `research/gate_ledger_final.md`.

9.6 Easy scene (B3): double-blocked, and honest about it

The nominally easiest row — four 22-d unimodal systems from the hundred-systems class — returned no evaluable accuracy cell, for two independent reasons that we report as such. The SMC arms are **blocked by budget**: no arm finished inside the pre-registered 5-h/arm free-tier L4 fence at N=512 (12.38 L4 GPU-h consumed; the readable result is the cost row: scene-substrate MC-SMC at N=512 costs > 5 h/target on an L4).[41] The classic reference arms are **blocked by reference**:

[40] Source artifact: `data/results-perlmutter/l0_arbitration_classic.json`.

[41] Source artifact: `data/b3_cells.json`.

all four posteriors completed but all four fail the frozen reference-acceptance rule ($\hat{R}_{\text{worst}}$ 4.36–7.87 vs. < 1.05; ESS_{min} 63.7–84 vs. ≥ 200), so "4/4 references complete" must not be read as "references usable" — each per-cell artifact records `reference_accepted: false`. The double block is itself a datum: it is additional, independent evidence for the single-stage-recipe convergence finding above, on targets designed to be easy.

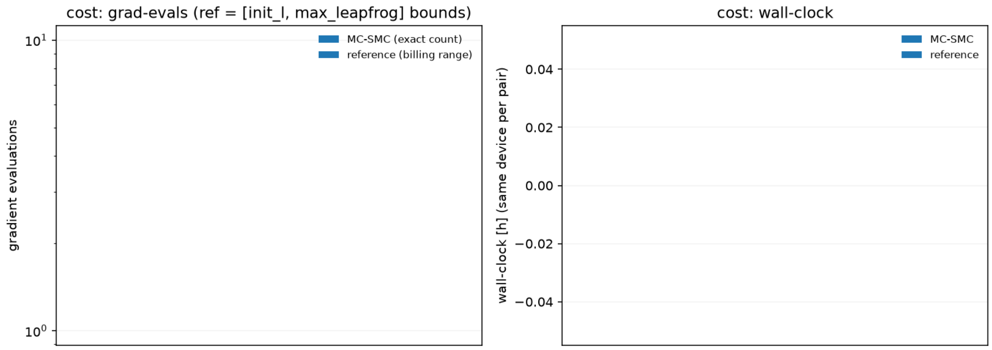

Figure 21: B3's readable result is its cost row: scene-substrate MC-SMC at N=512 costs > 5 h per 22-d target on the free-tier L4 (12.38 L4 GPU-h consumed, 0/4 arms complete inside the pre-registered 5-h/arm fence). Source: `data/b3_cells.json`.

9.7 Two cross-cutting findings

(A) Bootstrap error bars understate evidence error. Measured twice, in different ways. At B0, MAMS-SMC on the analytic ill-conditioned target returns $\log Z$ -145.80 vs. the analytic -148.20: a 2.40-nat error at $\sigma_{\text{boot}} = 0.37$, i.e. $6.6\sigma_{\text{boot}}$ (the HMC-mutation arm errs by 21.6 nats at $\sigma_{\text{boot}} = 0.41$).[42] On DSPL, on *identical data* where an exact reparameterization forces the true difference to zero, $\Delta \log Z(\text{orig} - \text{ratio}) = +3.06 \pm 0.35_{\text{boot}}$. The practical rule the campaign adopted, and that we recommend: never gate a $\Delta \log Z$ claim on σ_{boot} alone; the seed-repeat scale ($\sigma_{\text{seed}} = 1.79$ nats here, n=3, ±46% sampling error on that estimate) is the load-bearing error bar for every evidence gate.

(B) Frozen-reference coverage risk. The B5-G2 fail decomposes into both-basin positive evidence offsets with both ensembles γ-disjoint from the reference chains and two independent kernels agreeing on the new location (§9.3): frozen references built from short fixed-step MCMC chains can carry within-basin coverage error far exceeding the σ-scale of any gate built on them. The X2 frozen-set validity question (§10.1) is the same genus on the SBC side. The lesson we apply to ourselves first: an evidence reference needs a coverage certificate before it is frozen into a gate.

Finally, the demotion that held all campaign: **unadjusted-MCLMC mutations are not an evidence kernel**. The B0 bias screen measured $\Delta \log Z = +10.79 \pm 0.54$ vs. MAMS on the ill-conditioned target — 20.1σ by the artifact of record (the ledger prose says "30σ"; the flagged, binding artifact value is 20.1)[43] — and B5-G3's $\times 56$ minor-mode inflation confirms the channel on a real-lens-derived target. Within-basin locations stay correct in both measurements: MCLMC remains useful precisely as the cheap *diagnostic* it was demoted to.

[42] Source artifact: `data/smc_b0_report.json`.

[43] Source artifact: `data/smc_b0_report.json`.

10 Calibration Gifts and the Handoff Package

10.1 X2: the first formal SBC of the pipeline class

The X2 cross-pollination arm ran the first formal simulation-based calibration (SBC; rank-statistic) test of this pipeline class: N=32 prior-matched mock systems through the certified port of the classic MAP→SVI→HMC pipeline at reduced free-tier budgets (∼29 A16-h; N=32 of the assigned ≥ 64 — an under-delivery selected by the pre-stated budget rule and reported as such). Every verdict is *proposed (UNCERTIFIED — external)*: the ranks are the deliverable; interpretation and certification belong to the team.[44]

Three results, in decreasing order of confidence. **(i) A validity question about the frozen mock set**, posed as a question and not a finding of fault: the pre-existing frozen SBC set carries, per its own provenance in the upstream development branches, a generation prior narrower than the current fitting prior — under which SBC fails by construction (truth draws must follow the fitting prior). Such a mismatch is a legitimate robustness-testing design if intentional; the question for the team is whether a matched-prior set should exist. Our arm therefore ran on prior-matched *regenerated* mocks. (Detail is internal-memo material, publication-gated on the team's sign-off.) **(ii) Severe lens-light rank miscalibration**: the lens-light photometric block shows one-sided, sign-coherent rank pile-ups — I_e rank-location $z = -5.27$ ($p = 7.6 \times 10^{-11}$; 18/32 systems in the bottom bin), $R_{\text{sérsic}}$ $z = +5.76$ ($p = 1.5 \times 10^{-10}$), $n_{\text{sérsic}}$ $z = +4.75$ — which is not an under-mixing shape; three candidate channels (genuine pipeline miscalibration, reduced MAP/SVI budget, plug-in error map) are not separable from this run and are said so in the claims register. The mass block shows no location bias (worst $|z| = 2.25$) but 6 of 8 parameters fail uniformity via U-shapes, the classic under-mixing signature, with only 2 of 32 fits passing the health rule at this budget. **(iii) The glass-house disclosure**, included in the same harness by pre-registered rule: our own previous campaign's E1c SBC *failed* the same γ-rank gate ($p = 6.5 \times 10^{-5}$, 68% coverage 0.34) and *flipped to PASS* ($p = 0.53$) when unhealthy fits were rerun deep and re-read healthy-only — sampler pathology can fake rank failures, and that amendment path is exactly what $n_{\text{healthy}} = 2$ blocks here. The turnkey harness ships in the handoff package for an $N \geq 64$ rerun at reference settings.[45]

10.2 The handoff package

The handoff directory (`papers/handoff/`) contains the engagement memo (with its close-of-campaign addendum) and `CLAIMS.md`: fourteen claims in the receiving team's own claims-register format, every verdict *proposed (UNCERTIFIED — external)*, with the sign-off-gated claims (those touching data or targets from the upstream development branches) explicitly marked, and a mandatory doubt report attached to each. That file — not this report — is the interface of record for the team: it names the artifact behind every number and reserves "validated" for the old certified stack throughout.

The gap list `research/handoff_gap_list.md` inventories the package honestly against what the plan promised: delivered in substance are the claims register, the certified correlated-noise likelihood port (`cgl2/correlated.py`, `cgl2/marg.py`, with its exact-limit validation record), the tempered-SMC implementation with checkpoint/resume (`cgl2/samplers/`, 58-test suite), and the turnkey SBC harness (`30_sbc_gift.py`). Promised but *not* packaged at close: the SMC-stage adapter specification, the correlated-likelihood handoff document and upstream-shaped patch files, the DSPL note, and the Hessian-stage restore (never started). These remain owed offers, listed as

[44]Source artifact: `data/x2_sbc.json`.
[45]Source artifact: `research/x2_sbc_gift.md`.

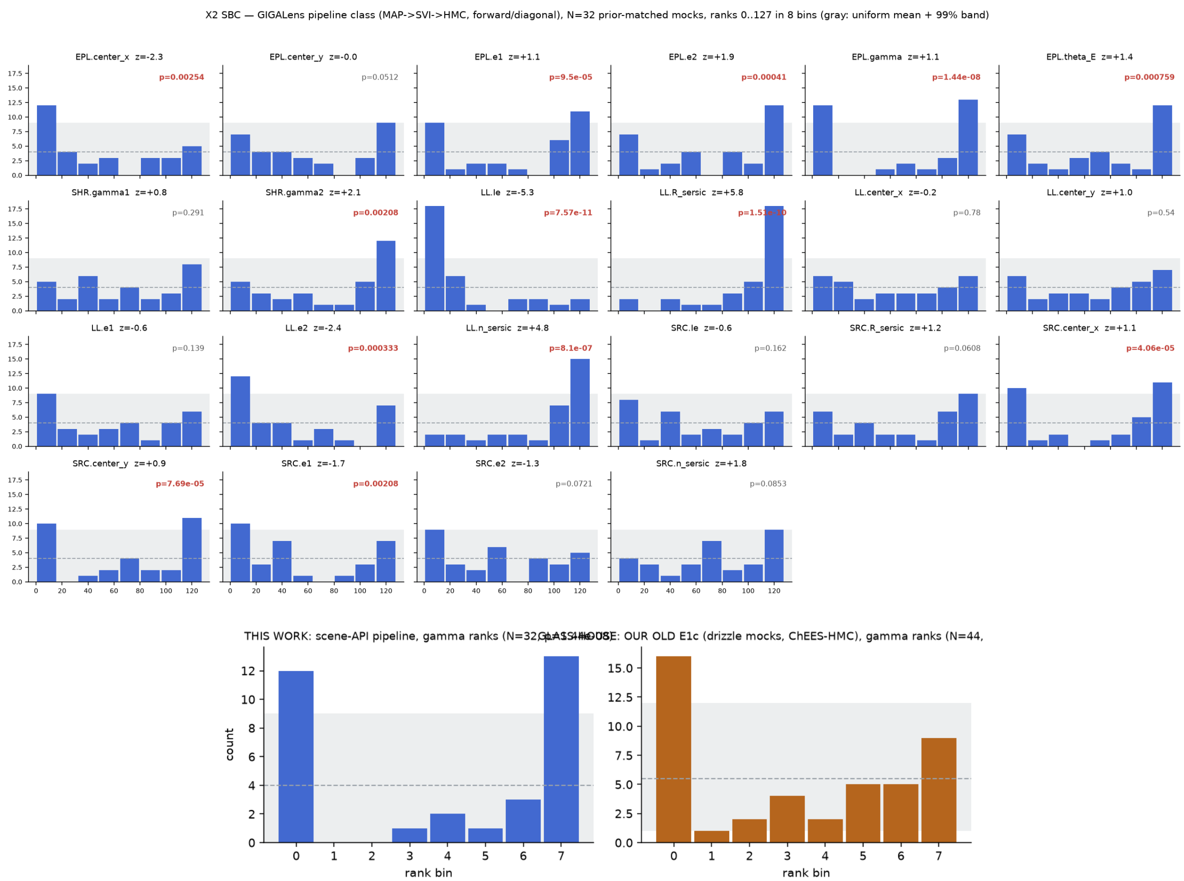

Figure 22: X2: the first formal SBC of the pipeline class (N=32 prior-matched mocks, reduced free-tier budget; all verdicts proposed, UNCERTIFIED — external; the pipeline runs on the upstream development branches, whose unpublished results are not characterized here). Top: rank histograms — the lens-light photometric block shows one-sided, sign-coherent pile-ups ($|z|$ up to 5.76), which is not an under-mixing shape; the mass block shows under-mixing U-shapes without location bias (worst $|z| = 2.25$). Bottom: the glass-house disclosure — our own previous campaign's E1c γ-rank gate failed ($p = 6.5 \times 10^{-5}$) and flipped to PASS ($p = 0.53$) when unhealthy fits were rerun deep and re-read healthy-only: sampler pathology can fake rank failures, and $n_{\mathrm{healthy}} = 2$ blocks that amendment path here. Source: `data/x2_sbc.json`.

such in the memo addendum, not silently dropped.

11 Discussion

11.1 What the campaign settles

Five things are now measured that were conjectures at the start. (1) **The bridge holds**: the scene-substrate forward model is certified against the validated old stack at machine precision for our config class, and the correlated-noise likelihood ported onto it reproduces the certified real-lens result on an independent stack and machine — γ to $\Delta = 0.0027$ and $\log Z$ to 0.11 nats (L0-G2). The $\gamma = 1.103$ result now stands on two stacks. (2) **The mechanism narrows**: stationarity of the noise-model class behind that number is rejected on the real field ($p = 0.010$, replicated spatial pattern), the companion and spectrum-flooring suspects are eliminated, and the profile-curvature fork died at its zero-cost entry gate — noise-model *class* misspecification is the standing working hypothesis for the over-correction, with source/PSF representation the surviving alternative. (3) **prior-seeded MAMS-SMC is the campaign's only evidence-producing vehicle**, and its failure mode is purely economic: cost walls on healthy samplers, never divergence or mode loss. (4) **Unadjusted MCLMC is not an evidence kernel** (20.1σ screen; $\times 56$ minor-mode inflation; locations intact) — but is a cheap, useful diagnostic. (5) **The classic single-stage recipe does not converge on this posterior family**, now shown on both stacks and three independent cells (B3, arbitration, X2 health rates); the two-stage re-preconditioned recipe remains the only known converging direct-sampler path.

A post-campaign epilogue (E1, 2026-07-22; PI-requested MCLMC fits of both diagonal targets, §1) sharpened two of these after close. On (5), the direct-sampler gap is narrower than the campaign body suggests: the team's own vendored MCLMC, warm-started and run with its windowed mass-matrix adaptation and regularization option, delivered the first fully convergence-certified posterior on the `v2d` native diagonal target ($\hat{R}_{\mathrm{worst}} = 1.0031$, $\mathrm{ESS}_{\mathrm{min}} = 30{,}334$; $\gamma = 1.4683\,[1.4343, 1.5048]$, $0.72\sigma_{\mathrm{comb}}$ from the anchor) at 1.49 free-tier L4-hours — so the convergence failures above are specific to the classic single-stage recipe, not to direct samplers per se. On (2), the mechanism gains its sampling-side face: on the binned product the same vehicle under the same *diagonal* likelihood migrated every one of 64 chains, in all four seed groups, out of the physical basin it was warm-started in and down to $\gamma \approx 1.08$–1.15 (kept-draw pooled median 1.1047, 0.0015 from the correlated fit's 1.1032; NO-QUOTE — the ensemble fails every gate, and the one allowed escalation was declined because migration is structural, not a depth problem). The binned data's preference for the unphysical low-γ shelf is therefore a property of the *product*, not of any single likelihood or sampler: it is reached by tempered SMC under the correlated likelihood and, from a physical-basin warm start, by MCLMC under the plain diagonal one — which is the T0.4 mechanism narrative restated from the sampling side, and one more reason the mechanism's confirmatory data-side injection test (above) is worth running.[46]

11.2 What it opens

The honest open list is long, and most of it is deliberate non-spend recorded in the gate ledger rather than silent omission. **Data-side injections**: the T1.1 injection-recovery test came back confounded — the control arm failed high *and* sick, implicating the injection construction (scene-subtraction residue) under the pre-registered honesty clause — and the fit-side rescue (T1.1b) failed its entry gate with an 85.8% whitened-dof loss. Residue-free injections must be built on the data

[46] Source artifact: `data/e1_harvest.json`.

side (kernel-sampled noise-only fields, sky-patch bootstrap, or deeper multi-start subtraction); the mechanism's confirmatory test is therefore still unrun, with one directional datum in its favor (same-data corr−diag differential −0.0526, no defensible error bar). **The two-stage port**: a converged scene-substrate posterior on the cond-10^{14} target needs the two-stage re-preconditioning recipe ported (∼0.5 day + 2–4 A100-h, estimated, not spent); until then "the 1.433 anchor reproduces at posterior level on the new substrate" is untested. **The locally-stationary whitener class**: the rejected stationary class needs a successor; the ported likelihood deliberately keeps a pluggable whitener seam for it, and the kernel-scan arm (E3) that would inform it was deferred by plan and never funded — it lapses at program close as the standing caveat on absolute-γ claims. **L1/L2 never ran**: there is *no* production-SMC coverage certificate (L1) and no supersampling/sub-pixel-PSF decomposition (L2); nothing in this report should be read as implying either exists. **PSF marginalization was promoted and then never built** (its pool was reallocated by a later decision); the D7 promotion language should not be read as delivered work. The B4 preconditioning question (does per-λ ensemble preconditioning replace the two-stage recipe at cond-10^{14}?) closed not-evaluable-within-budget and is genuinely open.

11.3 Guidance for practitioners

The decision matrix compresses to four rules. (1) If you need **evidence or mode structure** — $\log Z$, basin weights, cold-start robustness — on targets up to the 74-d bimodal class, prior-seeded MAMS-SMC is the vehicle, at single-digit A100-hours per basin; take the evidence error from seed repeats, not from the per-run bootstrap (finding A). (2) If you need a **posterior on a carousel-class real multi-plane system**, budget an order of magnitude more than we did or use a warm-started method and accept the cold-start blind spot: at $\lesssim$16 A100-h *neither* family converges (B1r). (3) Use **MCLMC as a diagnostic**, freely — and never as the evidence kernel without Metropolis adjustment or a measured bias budget. (4) **Certify your references**: before freezing an MCMC-derived posterior or evidence reference into an acceptance gate, demand a coverage certificate at the gate's own σ-scale (findings B; B3's blocked-by-reference row; the X2 frozen-set question). Our own frozen reference failed this standard at 11σ-equivalent scale — the most expensive single lesson of the campaign, and one we incurred against ourselves.

11.4 Operational lessons

Four ops findings each cost real budget once and are now protocol. **(1) Pin the high-memory GPUs and re-measure after remat changes**: two independent OOM failures (an injection arm on a 40-GB node, 1.49 h burned; the arbitration's 21-GiB single mutate allocation) traced to memory sizings measured *with* rematerialization active and applied to runs without it — memory figures do not transfer across remat settings; the fix is a hardware constraint pin plus fresh sizing. **(2) The scheduler snapshots scripts at submission**: a fix "resubmitted" after editing the batch script re-ran the old code (two failed attempts); protocol now verifies the *submitted* snapshot's checksum against the intended file before trusting any resubmission. **(3) One writer per manifest**: concurrent session fronts clobbered the deploy checksum manifest; the rule is pull-remote-first, union-merge, single writer per wave. **(4) Checkpoint everything**: the first production wave burned 94% of its hours with zero artifacts (18.7 h spent, 1.1 h readable) — jobs died at walls holding all state in memory. The mid-campaign checkpoint retrofit (per-stage full-state checkpoints, a graceful wall-cap exit protocol distinguishing PARTIAL from crash, bit-identical resume including the full $\log Z$ bootstrap state, gated by a 58-test suite) converted that failure mode into three-of-four readable recoveries, and is the only reason the 12-hour carousel chain and every λ-progress readout in Table 9 exist at all. Smaller pins that will save a future campaign a day each: chunked mutation sizes must divide *both* the particle count and the pilot subset; a watchdog state for "completed without artifact" doubles as the harvest reminder; BLAS thread caps matter on aarch64 login nodes; and the lensing likelihood is never run on CPU.

12 Reproducibility

Ledgers and provenance. The authoritative record is `CAMPAIGN.md`: nine locked decisions (D1–D9), the A100-hour ledger (rows appended *before* results are read, failed and zero-artifact jobs counted in full), the gate record, and a dated stage log. Every gate ever registered — including the six that ended without a run and one promoted-then-unbuilt item — has a final disposition in `research/gate_ledger_final.md`; nothing lapses silently. Among the never-run items, two of our own programmed arms are named explicitly: the S7 flow-preconditioned-MAMS arm (prerequisite unmet — the B1 λ=1 ensemble it required never came to exist) and the X3 evidence-scored Vela-ladder prototype (never mobilized). Two additional internal-reference benchmark arms were omitted under the campaign's pre-declared external-collaboration bright line and are dispositioned in the same ledger. Every number in this report traces to an on-disk artifact via `research/synthesis_tables.md`, whose seven discrepancy flags (artifact vs. ledger prose) are binding on this text: wherever the two disagree the artifact value is quoted (e.g. the MCLMC demotion screen is 20.1σ by `data/smc_b0_report.json`, not the ledger's "30σ"; the L4 arbitration leg's on-disk record is $\lambda = 0.4183$ at a 12.26-h resume leg, with the ledger's $\lambda \approx 0.45$/18-h figure including earlier checkpointed legs). The external claims register is `papers/handoff/CLAIMS.md` (§10.2). External release of this report is gated on the sign-off checklist `papers/handoff/SIGNOFF_CHECKLIST.md`, which lists the six passages characterizing unpublished upstream code state that require the team's sign-off (internal use is unaffected; the sixth passage is the E1 epilogue's description of the vendored experimental MCLMC driver, §1 item 3).

Pre-registration discipline. Every experimental cell was pre-registered: a design checkpoint with hypothesis, predicted direction and magnitude, falsifier, and derived thresholds, logged before submission; figures plotted before gate text was written; amendments (B1-reduced descope, the S6br budget-match, the B2′ re-registration) ledgered as dated decisions with checksummed checkpoints, never edited in place. Proposer and grader are separated throughout: every externally relevant verdict in this report is *proposed (UNCERTIFIED — external)*, with certification belonging to the receiving team, and scene-substrate results are labeled "reproduced/measured, UNCERTIFIED (external)" — "validated" is reserved for the old certified stack.

Environment freezes. The campaign venv pins python 3.13, `jax/jaxlib` 0.6.2, `blackjax` 1.3, `tfp` 0.25.0, `numpy` 2.4.6 (a declared deviation from the upstream 2.1.3 pin, covered by the gate battery), no `tensorflow`; all installs pass `-c constraints.txt`. The substrate is a vendored, *unpatched* snapshot of the upstream development branch (full SHA in `VENDORED_REF.txt`; the three convention reconciliations live on our side of the seam), and the old validated stack is imported by path at its certified ref. The Perlmutter-native environment re-passed the full parity battery (F8, job 55980038) with the phoenix-side F1 value bit-consistent. Production deploys are non-git rsync copies audited file-by-file against a checksum manifest (225 files at close) before every submission.

Compute. Table 10 is the phase budget against actuals: **53.51 of 100 A100-h** at program close, nothing in flight (per-row `sacct` rounding makes the component sum 53.52; flagged, not smoothed).

Table 10: A100-hour budget by phase: caps vs. ledgered actuals (`CAMPAIGN.md` ledger; per-row `sacct`, Elapsed × 1 GPU, shared QOS). T1.1 was funded from the pool freed by D7's retirement of P4; P2b by the D8 reallocation (10 h freed-P4 + 8 h stretch). The 100-h hard stop was never approached.

phase	scope	cap [h]	actual [h]
P0	scaffold, parity gates F1–F8, SMC correctness (B0)	1	0.00
P1	certification slice (T0.2/T0.3/T0.4)	10	2.21
T1.1	injection-recovery on real drizzle noise (D7 pool)	10	7.80
P2	sampler benchmark (B1–B5 waves + recovery lane)	24	22.45
P2b	carousel B1-reduced (S1r + S6br; D8/D9)	18	15.85
P3	scene-substrate payload (L0, arbitration, L0-G2)	17	5.21
P4	profile-class fork (retired at entry gate X1-G0)	10	0.00
campaign total (hard stop 100)		100	**53.51**

Free-tier phoenix spend (not A100): 12.38 L4 GPU-h (B3), ∼29 A16-h (X2), and — after program close — the E1 epilogue's 9.56 L4-h (§1). Zero-artifact and failed jobs are counted at full cost in their phases; the wave-1 post-mortem and the recovery-lane economics are part of the record, not an embarrassment to be netted out.

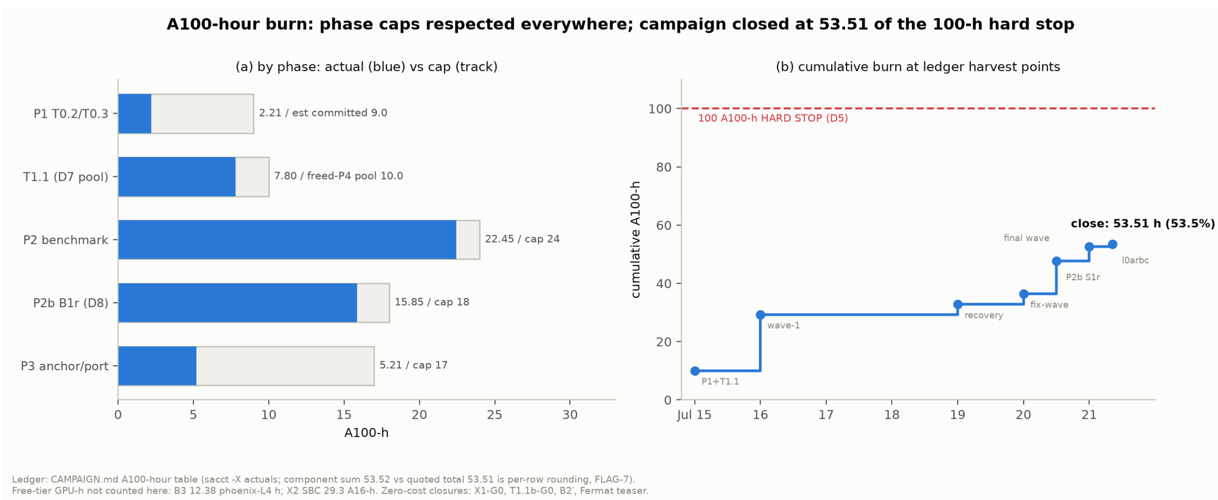

Figure 23: A100-hour burn. (a) Actual vs. cap by phase — every fence respected (P1 2.21, T1.1 7.80 of the D7-freed 10-h pool, P2 22.45/24, P2b 15.85/18, P3 5.21/17). (b) Cumulative burn at the ledger's harvest points (events in ledger order; sub-day x-offsets are display-only), closing at **53.51** of the 100-h hard stop; the 53.52 component sum is per-row `sacct` rounding (flagged, not smoothed). Free-tier GPU-h (B3: 12.38 L4-h; X2: 29.3 A16-h) are additional and uncounted here. Zero-cost closures: X1-G0, T1.1b, B2′, the Fermat teaser. Source: `CAMPAIGN.md` A100-hour ledger (append-before-read protocol).

Rebuilding the results. The parity battery is `01_parity_scene.py` (gates frozen in the README at P0); the SMC correctness gates are `04_smc_micro_validation.py`; each benchmark cell names its runner, slurm script, seed, and artifact in its ledger row; and each gate evaluation is a standalone JSON under `data/` (Perlmutter production artifacts under `data/results-perlmutter/`). The report builds with `make -C papers pdf` against the shared `reproductions/tech-report.sty` preamble.

A Convergence Diagnostics for Every Fit

Table 11 is the complete per-fit convergence ledger: one row for every fit this campaign ever ran, including the failed, rejected, collapsed, budget-blocked, and zero-artifact arms that the headline tables summarize or omit. It extends the consolidated ledger of Table 2 (§1, item 4) from "every fit quoted in this report" to "every fit executed," and backs each cell of the decision matrix (Table 9, §9).

Two provenance rules govern the table. First, the two inherited companion-campaign diagonal fits (the anchor $\gamma = 1.433 \pm 0.034$ and the diagonal-low $\gamma = 1.293 \pm 0.012$) appear as provenance rows only.[47] Second, arms that ran the sampler but left no readable ensemble on disk (the zero-artifact wave-1 losses) are still given rows: their diagnostics cells quote the postmortem and ledger record ("no artifacts — TIMEOUT") because that is all that exists to quote. Only the X2 free-tier SBC census (N=32 classic-recipe fits, summarized with the gift) and pure infrastructure failures that never reached a sampler are ledgered in the coverage notes after the table rather than rowed.

Conventions. For SMC fits the diagnostics are: tempering endpoint (λ reached; λ=1 is completion), stage count, final-stage weighted ESS (wESS; the adaptive schedule holds per-stage wESS at $0.7N$ by construction), unique particles after the final resample, MAMS acceptance against its 0.90 target, and σ_{seed} where independent-seed repeats exist. For MCMC arms: $\hat{R}_{\text{worst}}$ and ESS_{min} across parameters, against the frozen acceptance rule $\hat{R} < 1.05 \wedge \text{ESS} \geq 200$; the post-campaign E1 MCLMC fits (group M — lettered for the vehicle, and to avoid collision with the campaign's L0/L1/L2 lane names) carry their own, stricter frozen gate, $\hat{R} < 1.01 \wedge \text{ESS} \geq 1000$ pooled over 64 chains plus a $\gamma \in [1.15, 2.0]$ containment band. Verdicts use a fixed vocabulary: **CONVERGED** (sampler completed with healthy diagnostics; any science-gate outcome is annotated separately), **PARTIAL-BY-BUDGET** (healthy sampler stopped by a pre-registered wall cap short of λ=1), **REJECTED** (completed but fails the frozen acceptance rule), **COLLAPSED** (degenerate terminal population), and **BLOCKED** (no readable ensemble — killed by a budget fence, a walltime TIMEOUT, or a since-fixed artifact-writer defect; the cause is named in each row). Every number is quoted from the artifact named in the last column; the spine `research/synthesis_tables.md` binds, and its discrepancy flags are honored here (the MCLMC bias screen is quoted at its artifact value 20.1σ, not the ledger's "30σ"; the classic-arm ESS_{min} is quoted per stage; the L4 arbitration λ is quoted from the on-disk resume-leg summary with the ledger's multi-leg figure noted).

[47] Provenance footnote for the companion diagonal fits: those posteriors were produced by the companion campaign's two-stage re-preconditioned recipe, and their per-fit $\hat{R}$/ESS convergence diagnostics live in the companion report's campaign ledger (Benson and Huang, 2026). They are cited here, never re-derived and never invented; this campaign consumed the fits only as inherited draws (e.g. the L0-G2 q-fit construction and the head-to-head model points of Table 1).

Table 11: Convergence diagnostics for every fit the campaign ran. wESS = final-stage weighted ESS; "uniq" = unique particles after the final resample (for T0.2-family runs the stored-draw unique-row count is also given where it differs); accept = MAMS Metropolis acceptance (target 0.90; unadjusted MCLMC accepts by construction and its acceptance is not quoted). σ_{boot} = per-stage bootstrap $\log Z$ error. Verdict vocabulary as defined in the text.

fit / arm	data + likelihood	vehicle	N or ch×draws	diagnostics	verdict	artifact
A. SMC correctness gate (B0) — analytic targets with exact references; local GPU, 0 A100-h; seed 0						
B0 mix2, MAMS	analytic 2-d bimodal mixture (no image data)	tempered SMC, MAMS mutation	512	λ=1 @6 stages; wESS 464/512; uniq 440; accept 0.89–0.93; $\log Z$ err 0.073 (σ_{boot}=0.068); $\hat{w}_{\text{minor}}$ 0.219 vs 0.2	**CONVERGED** (gate PASS)	`data/smc_b0_report.json`
B0 mix2, HMC-mutation	same	tempered SMC, HMC mutation	512	λ=1 @6 stages; wESS 486/512; uniq 463; accept 0.75–0.95; $\log Z$ err 0.003; $\hat{w}_{\text{minor}}$ 0.191	**CONVERGED** (gate PASS)	same
B0 mix2, MCLMC	same	tempered SMC, unadjusted-MCLMC mutation	512	λ=1 @6 stages; wESS 483/512; uniq 471; $\log Z$ err 0.009; $\hat{w}_{\text{minor}}$ 0.205	**CONVERGED** (gate PASS)	same
B0 funnel10, MAMS	analytic 10-d Neal funnel	tempered SMC, MAMS	512	λ=1 @7 stages; wESS 508/512; uniq 500; accept 0.88–0.94; $\log Z$ -0.664 vs 0 (σ_{boot}=0.070)	**CONVERGED** ($\log Z$ gate FAIL — sampler-class limitation, reproduced in the old validated stack)	same
B0 funnel10, HMC-mutation	same	tempered SMC, HMC	512	λ=1 @8 stages; wESS 512/512; uniq 498; accept 0.76–0.86; $\log Z$ -0.545 vs 0	**CONVERGED** ($\log Z$ gate FAIL)	same
B0 funnel10, MCLMC	same	tempered SMC, MCLMC	512	λ=1 @7 stages; wESS 461/512; uniq 446; $\log Z$ -0.383 vs 0	**CONVERGED** ($\log Z$ gate FAIL)	same
B0 illcond46, MAMS	analytic 46-d ill-conditioned Gaussian	tempered SMC, MAMS	512	λ=1 @187 stages; wESS 407/512; uniq 415; accept 0.89–1.00; worst-param $\|z\|$ 1.83 (rule $<$ 3); $\log Z$ -145.80 vs ref -148.20: err 2.40 nats at σ_{boot}=0.37 (6.6σ_{boot} — finding A)	**CONVERGED** (location gate PASS; $\log Z$ error noted)	same
B0 illcond46, HMC-mutation	same	tempered SMC, HMC	512	λ=1 @195 stages; wESS 417/512; uniq 417; accept 0.71–0.98; worst-$\|z\|$ 1.87; $\log Z$ err 21.6 nats at σ_{boot}=0.41	**CONVERGED** (location PASS; $\log Z$ error noted)	same
B0 illcond46, MCLMC	same	tempered SMC, MCLMC	512	λ=1 @185 stages; wESS 445/512; uniq 425; worst-$\|z\|$ 2.93; $\log Z$ err 13.2 nats; $\Delta \log Z$ vs MAMS $+10.79 \pm 0.54 = 20.1\sigma$	**CONVERGED**; kernel **DEMOTED** to diagnostic ($> 3\sigma$ rule)	same
B. T0.2 seed certification + T0.3 companion mask — the production correlated fit family (old validated stack; seed-2 legs inherited from the companion campaign)						

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Table 11, continued.

fit / arm	data + likelihood	vehicle	N or ch×draws	diagnostics	verdict	artifact
`v3b` corr low, seed 2 (**money fit**)	real HST/WFC3-IR `v3b` binned drizzle; whitened-correlated likelihood ($n_{\mathrm{keep,w}} = 9273$)	prior-seeded MAMS-SMC	128	λ=1 @28 stages; wESS 118.1/128; uniq 77 (37 stored rows); γ 1.1032; $\log Z$ −4771.08	**CONVERGED**	companion `e2_v3b_low_smc_canary_fix.json`; `data/t02_t03_gate_eval.json`
`v3b` corr low, seed 3	same	same	128	λ=1 @28; wESS 127.7/128; uniq 82 (14 stored rows — dup-cluster, reported not sick); γ 1.0967; $\log Z$ −4769.12	**CONVERGED**	`e2_v3b_low_smc_seed3.json`
`v3b` corr low, seed 4	same	same	128	λ=1 @28; wESS 127.5/128; uniq 79 (27 stored rows); γ 1.1005; $\log Z$ −4771.37. Across seeds 2–4: $\sigma_{\mathrm{seed}}(\gamma) = 0.00325$ (kill $\sigma > 0.024$ not tripped), $\sigma_{\mathrm{seed}}(\log Z) = 1.22$ nats	**CONVERGED** + seed-certified (n=3: ±46% on σ_{seed} itself)	`e2_v3b_low_smc_seed4.json`
`v3b` corr steep, seed 2	same	same	96	λ=1 @21; wESS 36.0/96 (final resample low); uniq 50; γ 2.6393	**CONVERGED**	companion `e2_v3b_steep_smc_canary_p96.json`; `data/t02_t03_gate_eval.json`
`v3b` corr steep, seed 3	same	same	96	λ=1 @21; wESS 96.0/96; uniq 59; γ 2.6522	**CONVERGED**	`e2_v3b_steep_smc_seed3.json`
`v3b` corr steep, seed 4	same	same	96	λ=1 @22; wESS 90.3/96; uniq 58; γ 2.6485. Across seeds: $\sigma_{\mathrm{seed}}(\gamma) = 0.00664$; $\Delta \log Z(\mathrm{steep} - \mathrm{low}) = -28.9/-32.3/-31.2$ per matched seed, $\sigma_{\mathrm{seed}}(\Delta \log Z) = 1.79$ nats	**CONVERGED**; low-basin preference seed-stable	`e2_v3b_steep_smc_seed4.json`
T0.3 companion-masked refit (seed 2)	same data, whiten-then-drop companion mask ($n_{\mathrm{keep,w}}$ 9273 → 8247)	same	128	λ=1 @28; wESS 127.4/128; uniq 84 (17 stored rows); γ 1.1011 (shift −0.0021 vs production, $0.26\sigma_{\mathrm{stat}}$)	**CONVERGED** (companion exonerated)	`e2_v3b_low_smc_compmask.json`; `data/t02_t03_gate_eval.json`
C. T1.1 injection recovery — $\gamma_{\mathrm{truth}} = 1.433$ arcs injected into the real `v3b` drizzle residual field (Perlmutter A100)						
T1.1 inj1, corr	injected scene 1; whitened-correlated ($n_{\mathrm{keep,w}} = 9273$)	prior-seeded MAMS-SMC	128	λ=1 @27 stages; wESS 125.6/128; uniq 76 (52 stored rows); γ_{rec} 1.5151 ± 0.036 ($z = +2.25$)	**CONVERGED** (bias +0.082, outside both pre-registered zones)	`t11_inj1_smc.json`; `data/t11_gate_eval.json`
T1.1 inj2, corr	injected scene 2 (dup-flagged shift)	same	128	λ=1 @34; wESS 124.4/128; uniq 85 (18 stored rows — dup-cluster flag); γ_{rec} 1.5719 ± 0.043 ($z = +3.23$)	**CONVERGED** (flagged; dropping it leaves the median bias +0.078, same zone)	`t11_inj2_smc.json`
T1.1 inj3, corr	injected scene 3	same	128	λ=1 @20; wESS 107.9/128; uniq 81; γ_{rec} 1.5076 ± 0.041 ($z = +1.82$)	**CONVERGED**	`t11_inj3_smc.json`

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Table 11, continued.

fit / arm	data + likelihood	vehicle	N or ch×draws	diagnostics	verdict	artifact
T1.1 inj1, diag control	same injected scene 1; delta-diagonal likelihood (16,653 px)	same	128	λ=1 @37; wESS 128.0/128 (degenerate); **1/128 unique particle rows**; $\gamma_\sigma \approx 4 \times 10^{-16}$; srcS.$I_e$ railed ∼58× truth; γ 1.5677 $\notin [1.29, 1.43]$	**COLLAPSED** (total resample collapse; per the honesty clause the injection construction is implicated)	`t11_inj1_diag_smc.json`; `data/t11_gate_eval.json`
D. L0-G2 cross-stack reproduction — the correlated fit re-run on the scene API port, Perlmutter A100 (proposed, UNCERTIFIED — external)						
L0-G2 low leg, seed 2	`v3b` binned; ported `CorrelatedImageData` whitened-correlated likelihood	prior-seeded MAMS-SMC on the scene API	128	λ=1 @27 stages; wESS 123.5/128; uniq 81; γ 1.1005 [1.0992, 1.1065]; $\log Z$ −4770.97	**CONVERGED** (PASS: $\Delta\gamma$ = 0.0027; $\log Z$ agrees with the old stack to 0.11 nats)	`l0g2_v3b_scene_seed2.json`
L0-G2 steep leg, seed 2	same	same	96	λ=1 @22; wESS 47.6/96; uniq 54; $\log Z$ −4800.33; $\Delta \log Z(\text{steep} - \text{low}) = -29.36$, in the P1 seed family	**CONVERGED** ($\log Z$ agreement 0.37 nats)	same
E. B5 — T3 74-d bimodal (`foundry_v3b74`, diagonal likelihood, f32; Perlmutter A100)						
B5 low basin, MAMS	`v3b` binned, diagonal likelihood (old-stack target)	prior-seeded MAMS-SMC	128	λ=1 @31 stages; basin retention 1.000; min uniq 88; accept 0.862–0.998; $\log Z$ 38376.33 $\pm$ 0.33_{boot}; γ_{eqw} 1.0919	**CONVERGED** (G1 PASS; the 11σ G2 FAIL indicts the frozen reference's within-basin coverage, §9.3)	`b5_v3b74_low_mams_seed2.json`; `data/b5_gate_eval.json`
B5 steep basin, MAMS	same	same	96	λ=1 @22 stages; retention 1.000 (frac-low 0.0); min uniq 70 $\geq N/4$; accept 0.886–0.998; $\log Z$ 38518.23 $\pm$ 0.31_{boot}; γ_{eqw} 2.5552	**CONVERGED** (G1 PASS)	`b5_v3b74_steep_mams_seed2.json`
B5 low basin, MCLMC (diagnostic arm)	same	tempered SMC, unadjusted-MCLMC mutation (demoted kernel)	128	λ=1 @36 stages; $\log Z$ 38380.36 $\pm$ 0.36_{boot}; γ_{eqw} 1.0935 (within-basin $\Delta\gamma = 0.0016$ vs MAMS); minor-mode weight inflated ×56 [CI_{95} 22–146]	**CONVERGED** (diagnostic only; G3 ambiguous at frozen thresholds — never an evidence kernel)	`b5_v3b74_low_mclmc_seed2.json`; `data/b5_gate_eval.json`

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Table 11, continued.

fit / arm	data + likelihood	vehicle	N or ch×draws	diagnostics	verdict	artifact
B5 S1 low leg, wave-1 attempt (job 55985449)	same	prior-seeded MAMS-SMC	128	no artifacts: the RC3 artifact-writer defect (`TypeError` after the driver returned) destroyed the write; λ=1 was reached (structural: the driver raises if λ<1), diagnostics lost; `set -e` killed the job before the steep leg; 1.02 A100-h. Fix landed + regression-locked; the converged rows above are the reruns	**BLOCKED** (writer defect — infra, not science)	`research/p2_wave1_postmortem_redesign.md`
B5 S2 MCLMC low leg, wave-1 attempt (55985450)	same	tempered SMC, MCLMC mutation	128	no artifacts: same RC3 defect at the write (∼36 min in); λ=1 reached (structural), diagnostics lost; 0.64 A100-h	**BLOCKED** (writer defect)	same
F. B2/B2′ — DSPL thin-ridge cosmology (21-d, seeded mock per the team's generation recipe; Perlmutter A100)						
B2 `dspl20_orig`, seed 2	DSPL mock, diagonal likelihood, original (Ω_{m0}, w_0) coords	prior-seeded MAMS-SMC	512	λ=1 @64 stages; min uniq 358 ($\geq N/4$), final 468; min incremental ESS 358.4; $\log Z$ 17294.01±0.25$_{\text{boot}}$; Ω_{m0} med 0.4701	**CONVERGED** (falsifier NOT-DECIDABLE-AS-REGISTERED: band uncalibrated for this realization, §9.4)	`b2_dspl20_orig_s1_seed2.json`; `data/b2_gate_eval.json`
B2′ `dspl20_ratio` control, seed 2	identical data, exact ratio-coordinate reparameterization	same	512	λ=1 @64 stages; min uniq 360, final 424; accept 0.822–0.927; $\log Z$ 17290.95 ± 0.24$_{\text{boot}}$; r_2 shape PASS (width ratio 0.79 ∈ [0.67, 1.5]); $\Delta \log Z(\text{orig}-\text{ratio}) = +3.06 \pm 0.35_{\text{boot}}$ where true $\Delta \approx 0$ (finding A, measured)	**CONVERGED** (PASS-UNDER-POWERED)	`b2_dspl20_ratio_s1_seed2.json`; `data/b2_gate_eval.json`
G. B3 — easy scene, 22-d unimodal ×4 (`hs2` seeded mocks; free-tier L4, 0 A100-h)						
B3 `hs2_sys0` classic reference	`hs2_sys0` mock, diagonal (ForwardProbModel conventions)	classic MAP→SVI→HMC, frozen settings	50ch×750 (250 burn); esc. 50×1500 (500)	$\hat{R}_{\text{worst}}$ 7.65 (esc. 7.57) vs rule < 1.05; ESS$_{\text{min}}$ 62.9 (esc. 64.0) vs ≥ 200	**REJECTED** (blocked-by-reference)	`data/b3_run_ref_sys0_l4-8.json`
B3 `hs2_sys1` classic reference	`hs2_sys1` mock, same	same	same	$\hat{R}_{\text{worst}}$ 8.81 (esc. 7.87); ESS$_{\text{min}}$ 65.0 (esc. 63.7)	**REJECTED**	`data/b3_run_ref_sys1_l4-8.json`
B3 `hs2_sys2` classic reference	`hs2_sys2` mock, same	same	same	$\hat{R}_{\text{worst}}$ 4.90 (esc. 5.66); ESS$_{\text{min}}$ 74.5 (esc. 74.5)	**REJECTED**	`data/b3_run_ref_sys2_l4-8.json`
B3 `hs2_sys3` classic reference	`hs2_sys3` mock, same	same	same	$\hat{R}_{\text{worst}}$ 4.44 (esc. 4.36); ESS$_{\text{min}}$ 82.5 (esc. 83.9)	**REJECTED**	`data/b3_run_ref_sys3_l4-8.json`
B3 `hs2_sys0` SMC arm	same data	prior-seeded MAMS-SMC	512	none written: killed at the pre-registered 18,000-s (5-h) per-arm L4 fence before completion; no readable ensemble	**BLOCKED** (by budget)	`data/b3_cells.json`; `data/b3_lane2_sys0_smc.log`

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Table 11, continued.

fit / arm	data + likelihood	vehicle	N or ch×draws	diagnostics	verdict	artifact
B3 `hs2_sys1` SMC arm	same	same	512	same (5-h fence)	**BLOCKED** (by budget)	`data/b3_lane2_sys1_smc.log`
B3 `hs2_sys2` SMC arm	same	same	512	same (5-h fence)	**BLOCKED** (by budget)	`data/b3_lane2_sys2_smc.log`
B3 `hs2_sys3` SMC arm	same	same	512	same (5-h fence)	**BLOCKED** (by budget)	`data/b3_lane2_sys3_smc.log`
H. B4 — T2 `foundry_marg46` (`v2d` 46-d marg-ridge, cond$\sim 10^{14}$; Perlmutter A100, wave-1)						
B4 marg46 S1, seed 2 (job 55985448)	`v2d` native, marg46 ridge-marg likelihood (old validated stack)	prior-seeded MAMS-SMC	256	no artifacts — TIMEOUT 3:30 (3.51 A100-h): run log shows build (78 s) + warmup log-prob only; zero stage observability (no per-stage checkpointing, RC1), and the since-fixed RC3 writer defect sat on the same code path; measured cost lower bound 3.5 h at N=256, never re-funded	**BLOCKED** (TIMEOUT; ledger disposition: proposed NOT-EVALUABLE-WITHIN-BUDGET)	`b4_marg46_s1_seed2_run.log`; `research/gate_ledger_final.md` §B.1
I. B1/B1r — carousel 33-d multi-plane (Perlmutter A100): wave-1 mock arms, then B1r on REAL MUSE cutouts (sign-off-gated)						
B1 S1 cold-start SMC, seed 2 (mock; job 55985444)	carousel33 seeded-mock stand-in (real cutouts not yet landed), diagonal	prior-seeded MAMS-SMC	512	no artifacts — TIMEOUT 4:00 (4.01 A100-h), zero per-stage observability (RC1); postmortem estimate: the 4-h wall bought $\sim$5–8 early stages, $\lambda \lesssim 10^{-4}$	**BLOCKED** (TIMEOUT; mock arms retired by the B1-REAL-DATA-LANDED amendment, superseded by B1r)	`b1_carousel33_s1_seed2_run.log`; `research/p2_wave1_postmortem_redesign.md`
B1 S1 cold-start SMC, seed 3 (mock; 55985445)	same	same	512	no artifacts — TIMEOUT 4:00 (4.01 A100-h), same failure mode	**BLOCKED** (TIMEOUT; retired by amendment)	`b1_carousel33_s1_seed3_run.log`
B1 S6b warm MAMS-alone, seed 2 (mock; 55985446)	same	MAP warm start → MAMS-alone	64 chains	no artifacts — TIMEOUT 3:00 (3.01 A100-h); run log: MAP done (302 s, 22,400 grads), burn round 0 only ($\sim$25 min/round incl. compile)	**BLOCKED** (TIMEOUT; retired by amendment; motivated the checkpoint retrofit that made the B1r rows below readable)	`b1_carousel33_s6b_seed2_run.log`

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Table 11, continued.

fit / arm	data + likelihood	vehicle	N or ch×draws	diagnostics	verdict	artifact
B1r S1r cold-start SMC, seed 2	real MUSE cutouts (`source4-5.fits`, `source9.fits`), diagonal	prior-seeded MAMS-SMC, 3 chained checkpoint legs (11.33 A100-h)	128	λ = 0.1506 @36 stages (λ=1 structurally out of reach); uniq 92–104/128 every stage; accept 0.871–0.934 (mean 0.899 vs 0.90); stage wESS pinned at $0.7N$; **0 posterior samples**	**PARTIAL-BY-BUDGET** (sampler healthy — a cost verdict, §9.2)	`b1r128_carousel33_s1_seed2.PARTIAL.json`; `data/b1r_decision_matrix.json`
B1r S6br warm MAMS-alone, seed 2	same	MAP (64×350) warm start → MAMS-alone, budget-matched at $B^* = 3{,}078{,}912$ grads	64 chains, 0 draws	4.5-h wall fence fired at burn round 11/30 — **sampling never started**: draws $(0, 64, 32)$, ESS = 0, $\hat{R}$ not computable; burn healthy (accept 0.873–0.902); realized 1,277,632 grads = 41.5% of B^*; throughput within ∼5% of S1r	**BLOCKED** (burn-fenced; a target-cost fact, not arm pathology)	`b1r128_carousel33_s6b_seed2.json`; `data/b1r_decision_matrix.json`
J. L0 anchor arbitration — `v2d` native diagonal, parity-certified scene API target (T2 class, cond∼10^{14})						
L0arb MC-SMC, L4 leg	`v2d` native, diagonal masked-err ridge-marg	prior-seeded MAMS-SMC (phoenix L4, checkpoint-chained)	64	resume leg from stage 26: summary λ_{reached} = 0.4183 (stage-66 checkpoint; the trace's last row is stage 67 at $\lambda = 0.4501$), 12.26-h leg; accept 0.853–0.946 (the ledger's $\lambda \approx 0.45$ @∼18 h presumably folds in the earlier leg's wall; the on-disk summary binds)	**PARTIAL-BY-BUDGET** (healthy; wall-capped)	`data/l0_arbitration_smc.json`
L0arb MC-SMC, A100 hbm80g (job 56168446)	same	same (Perlmutter A100 80 GB)	128	$\lambda = 0.587$ @stage 76 at the 3.5-h cap; MAMS healthy (accept ∼0.88); peak 22.7 GB; third consecutive wall-capped MC-SMC attempt ⇒ vehicle amendment	**PARTIAL-BY-BUDGET** (vehicle budget-infeasible on this target)	`CAMPAIGN.md` ledger row 56168446 (ckpts stage_000–075 on CFS)
L0arb classic warm arm, initial	same	warm MAP→SVI→HMC, B3-frozen settings (seeds 0/1/2)	50ch×750 (250 burn)	$\hat{R}_{\text{worst}}$ = 44.3; ESS_{min} = 53.6; all chains railed $\gamma \approx 1.987$ $[1.983, 1.991]$ at the prior edge	**REJECTED** (NO-VERDICT: γ not quotable; reproduces the known T2 single-stage pathology on the scene API, §9.5)	`l0_arbitration_classic.json`
L0arb classic warm arm, escalated	same	same	50ch×1500 (500 burn)	$\hat{R}_{\text{worst}} = 11.7$; $\text{ESS}_{\text{min}} = 63.0$ (escalated stage — the initial-stage value is 53.6; quoted per stage); chains still railed	**REJECTED** (NO-VERDICT)	same
K. Companion-campaign diagonal fits — provenance rows (cited, not re-derived; see the provenance footnote in the appendix text)						
anchor $\gamma = 1.433 \pm 0.034$	`v2d` native, diagonal	companion two-stage re-preconditioned recipe	see companion ledger	per-fit $\hat{R}$/ESS ledgered in the companion report	converged there; **cited**	Benson and Huang (2026)

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Table 11, continued.

fit / arm	data + likelihood	vehicle	N or ch×draws	diagnostics	verdict	artifact
diag-low $\gamma = 1.293 \pm 0.012$	`v3b` binned, diagonal	same	see companion ledger	per-fit $\hat{R}$/ESS ledgered in the companion report	converged there; **cited**	Benson and Huang (2026)
M. E1 post-campaign epilogue (2026-07-22) — PI-requested MCLMC diagonal fits (vendored windowed-adaptation driver; phoenix L4 free tier, 9.56 L4 GPU-h, 0 A100-h; frozen gates $\hat{R} < 1.01 \wedge$ ESS ≥ 1000 pooled 64 chains + containment $\gamma \in [1.15, 2.0]$)						
E1 `v2d` native MCLMC	`v2d` native (80^2@0.13″), parity-certified diagonal masked-err ridge-marg (scene API)	vendored MCLMC (windowed mass matrix, regularization ON; warm start = mapped healthy anchor chains)	2 grp × 32ch × 4000 (2000 burn)	$\hat{R}_{\rm worst}$ 1.0031 (LL2 $n_{\rm sérsic}$); ESS$_{\rm min}$ 30,334; per-group $\hat{R}_{\rm worst}$ 1.0034/1.0031; γ: $\hat{R}$ 1.0017, ESS 39,794; containment 0/64 violations (all draws in $[1.324, 1.637]$); kernel-nonan 1.0; 1.49 h, peak 9.1 GB. $\gamma = 1.4683\,[1.4343, 1.5048] = 0.72\sigma_{\rm comb}$ from the anchor; seed-group medians Δ 0.0002	**CONVERGED** (PASS E1-G1+G2, not escalated; γ quotable)	**`mclmc_diag_v2d.{npz,json}`; `data/e1_harvest.json`**
E1 `v3b` binned MCLMC, seed group 0	`v3b` binned (130^2@0.08″), same diagonal construction	same (warm start = the 1.2941 $\gamma < 1.7$ conditional cloud, init ONLY)	16ch × 4000 (2000 burn)	$\hat{R}_{\rm worst}$ 1.271 (LL2 $n_{\rm sérsic}$), ESS$_{\rm min}$ 45.4; init med 1.2914 → kept med 1.1066; 16/16 chains below the band	**REJECTED** (E1 gate)	**`mclmc_diag_v3b.{npz,json}`; `data/e1_harvest.json`**
E1 `v3b` MCLMC, seed group 1	same	same	same	$\hat{R}_{\rm worst}$ 1.635 (LL3 $R_{\rm sérsic}$), ESS$_{\rm min}$ 26.2; init med 1.2940 → kept med 1.0981; 16/16 below the band	**REJECTED**	same
E1 `v3b` MCLMC, seed group 2	same	same	same	$\hat{R}_{\rm worst}$ 1.420 (LL1 I_e), ESS$_{\rm min}$ 33.2; init med 1.2900 → kept med 1.1049; 16/16 below the band	**REJECTED**	same
E1 `v3b` MCLMC, seed group 3	same	same	same	$\hat{R}_{\rm worst}$ 1.166 (LL2 $R_{\rm sérsic}$), ESS$_{\rm min}$ 67.8; init med 1.2937 → kept med 1.1074; 16/16 below the band	**REJECTED**	same
E1 `v3b` MCLMC, pooled verdict (the 4 groups above)	same	same	pooled 64ch × 4000	$\hat{R}_{\rm worst}$ 1.379 / ESS$_{\rm min}$ 139 (both LL1 I_e); 46/46 scene params over the $\hat{R}$ gate; γ $\hat{R}$ 1.261 (ESS 181); 64/64 chains out of band (frac outside min 0.94, median 1.00; all reported, none dropped); kept-draw pooled med 1.1047 (descriptive only, 0.0015 from the corr-low 1.1032); wall 8.07 h	**REJECTED** — NO-QUOTE + FINDING (the binned-product migration, §1); the one allowed escalation **declined**: structural migration, not sampling depth	same; **`figs/e1_v3b_migration.png`**
E1 `v3b` launch attempt 1 (PID 184374)	same	same (original 32ch × 2-group geometry)	2 grp × 32ch	no artifacts: OOM in the main scan (`RESOURCE_EXHAUSTED`, a 10.1 GB allocation atop the working set — the 130^2 lstsq-marg grad tape is $\sim 4\times$ `v2d`'s) ⇒ ledgered amendment 3: 16ch × 4 groups, pooled 64 unchanged, no gate or threshold change	**BLOCKED** (OOM → geometry amendment)	**`data/mclmc_lane_v3b.log`; `CAMPAIGN.md`** E1 amendment 3

continued on next page

Table 11, continued.

fit / arm	data + likelihood	vehicle	N or ch×draws	diagnostics	verdict	artifact
E1 `v3b` launch attempt 2 (PID 189281)	same	same (amendment-3 geometry)	4 grp × 16ch	no artifacts: silent kill — the relaunched lane process vanished within minutes of launch with no traceback and no log output flushed; the only trace is the watchdog's `VANISHED_NO_ARTIFACT` record. The subsequent relaunch is the completed run rowed above	**BLOCKED** (silent kill — infra, cause unattributed)	`data/watchdog_state.json` (`phoenix:189281`)

Coverage notes (what is summarized rather than rowed). For completeness of the "every fit ever ran" claim: (i) the **X2** SBC lane ran N=32 classic-recipe fits on the free tier; they are gift deliverables scored as a set, so their health census (2/32 healthy, 30/32 unhealthy at that budget) is reported with the SBC gift rather than as 32 rows (§10.1, `data/x2_sbc.json`) — consistent with the B3 and arbitration classic-arm rows here. (ii) The q-fit proposal constructions (SVI-cov canaries, e.g. `t11_inj*_canary_svicov.json`) are prep stages of the prior-seeded-SMC vehicle, not posterior fits, and are subsumed in their fit's row. (iii) Pure infrastructure failures that never yielded a sampler state — the T1.1 inj3 and l0arb OOMs, the vendor-guard path-identity trip, the B5-steep snapshot-ordering reruns, and the classic-arm attempt #1 `IndexError` at MAP — are ledgered per-job in `CAMPAIGN.md`; their successful reruns are the corresponding rows above. (iv) After the two dead E1 `v3b` launches (group M BLOCKED rows), a second concurrent launch of the *same* production lane — identical config and seeds (`data/mclmc_lane_v3b_r3.log`) — ran alongside the lane that completed and was killed unread at harvest open; it wrote no artifact and none of its draws were read or used, so the completed lane's `npz/json` is the sole `v3b` E1 record. It is noted here rather than rowed because it is a duplicate of a rowed fit, not a distinct one.

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